Conditioned States as Toric Kähler Data

A companion note to "Probability, Context and Differentiation: The Layers"
(working draft)

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Abstract

The central point of this note is that *leading-order balance identifies candidates, not the selected state*. On a compact Delzant polytope with potential $W_T = \sum_k T^{-\lambda_k} y^{v_k}$, the unique positive critical point (Galkin's lemma) has a valuation u^* at which the leading terms balance with positive weights: Birch at target 0. The converse fails; on F_1 a whole segment balances and higher order selects. A selection theorem settles which point is chosen: u^* is the lexicographic max–min point, with error $O(1/\log(1/T))$. The positive orthant is a Lefschetz thimble, and its law $e^{-W/\hbar}dy/y$ has a tropical limit: uniform on shrunk polytopes, then concentrated on the candidates, level by level. Around this sits the toric Kähler dictionary in probabilistic terms: Fisher metric, relative entropy, Duistermaat–Heckman. Soft evidence of size n shifts $\lambda \mapsto \lambda - (n/N)\rho$, and acts only through circuits; F_1 has one, with a wall inside the Kähler cone where the leading set jumps.

Part II reads this on contingency tables. The margins are a point of the base and the Kähler class of the fibre, the coupling polytope. On the fibre, Tier A is Fisher's noncentral hypergeometric family, the best smallest cell is $\min(f_i/m, g_j/n)$, and the selected coupling is the facet-leximin one. One count of data can cross a leximin wall and change the Floer fibres without changing the surface.

Part III turns to morphisms. By the cited theorems, distinct objects are orthogonal, each has a Clifford endomorphism algebra (for the probabilistic object, of its Fisher information), and no higher products survive. Across an outcome wall the category over Λ is unchanged up to A_∞ equivalence. But the probabilistic object's Clifford form, reduced mod T, drops rank, and the Stokes walls, where thimble bases mutate, crowd into an $O(1/\log(1/T))$ window. Floer statements are cited, not proved. A ledger records every operation.

Part IV goes to 2×4, where the fibre is three-dimensional. Its 148 Kähler chambers are classified exactly, with ampleness certified by integer intersection numbers on invariant curves (Nakai–Moishezon, in its toric form): 14 are Fano and 134 weak Fano, and coupled three-step selection occurs only in the weak Fano ones. In three examples, one per role, the Clifford form of the positive object has one scale per dimension that selection removes, including a double scale where one level removes two; we state this as a conjecture. A path through the column-binding example (Stage IV) crosses two leximin walls in one Kähler chamber: the example itself sits on a level-two wall, which forces a collision of critical points, while the level-one wall is crossed without one. Composing couplings by Markov gluing is associative; composing them by selection is not, and the associahedron K_4 then carries an exact configuration of eleven distinct composites. An A_∞ structure on chains of coupling polytopes, with point classes composing and the diagonal coupling as unit, is trivial up to quasi-equivalence; so the information lies in selection itself. On binary chains selection is Fréchet-midpoint gluing. On uniform margins it composes correlations by a signed minimum where Markov gluing multiplies

them, and it is associative; among the other margins tested it is associative everywhere for none, but on full-dimensional cells for some, which we certify exactly. On binary chains the selected composite is the marginal of the unique Floer-nontrivial torus fibre of the reduced space, by Cho–Oh and by Alston–Amorim's pairing with the real Lagrangian, while Markov gluing takes the Birch point of the same space. In the dagger category of couplings, selection keeps units, reversal and copied factors, but not independent factors or coarse-graining. For binary ends it is the centre of the interval the data determine, which makes it harder to flip by poisoning, about 13% of the records against 2.6% for Markov gluing in one example, but sensitive to how the variables are recorded. Both rules are sections of the projection from joint laws to marginal data. On the 4-cycle (CHSH) cover, and on every n-cycle, the noncontextuality facets are simplices: at every point of the contextuality boundary the data determine the joint law and every rule agrees, while structural zeros leave a family of dimension $3 \cdot 2^{n-2} - 2n$. Simpson reversals arise either from the rule (selection sensitivity) or from coarsening the middle (information loss). The cheapest undetectable poisoning of such data is a conformal sum of a few circuits of the consistent-edit space, usually one circuit of four cells whatever the number of variables. The number of undetectable placements grows monotonically with the hidden support, whose freedom splits into room to move the data and room for the rule. Removing batches or contexts is a contravariant functor, and it explains a decision's change over time; conformal sums explain it cell by cell. Every decision gets a certified poisoning radius, the gap divided by a Lipschitz constant computed from dual vertices and circuits; the fibre's dimension can jump under poisoning while its width cannot. Bayesian inversion is the dagger of the coupling category, a symmetry of the marginal projection rather than a section, and composition commutes with it for reversal-equivariant rules. Local data determine the value of information exactly when its value functionals are sums over the contexts, independently of whether they determine the law or the decision, and the worst-case change in robust decision value is twice the Hausdorff distance between fibres. Value of information is a geometry of consequences: resolving a fibre is worth a minimax regret computed by linear programs and bounded by its width along utility differences, the cheapest poisoning on a rigid facet raises no value of information at all, dilation occurs on marginal data, and Das's commuting square fails for re-selection, so commitment is the stable policy. Rational data can give an irrational maximum-entropy law, the number of integer explanations grows like N to the fibre dimension, and only selection rules that are not concave jump. The viability share, the least share of the data that any disturbance needs to tip a decision, is one linear program and is the only quantity that warns of on-facet poisoning; in open loop decisions are eventually constant, with finite memory and feedback eventually periodic. Along a history of data snapshots, the selected law can depend on the path only within one fibre, and the decision only when that fibre straddles the threshold. I-projection updating over growing sets of contexts is flat, giving maximum entropy at every step, while forgetting and minimal-change updating are path dependent. Local tables always glue until the cycle of contexts closes, and only then can contextuality block the gluing. Fisher geometry finds the fibre as the null space of the data's information, but statistical distinguishability is not decision relevance: decision value is Lipschitz in Fisher–Rao distance and not conversely, phases are decision-null, and volume coefficients depend on coordinates. Whether maximum entropy is the only associative section is left open. 120 numerical, 7 certificate and 3 ledger checks corroborate the statements.

Part I
The geometry

1 What this note does and does not do

The layers paper ends its toric section with an outlook remark: the precise dictionary is between conditioned states and moment fibres, not between sources and Lagrangians [28, Remark "outlook: a symplectic reading"]. This note makes that remark exact in two places and leaves the rest open. Its components have different status.

component	status	source
toric Kähler structure, moment map, Fisher metric, symplectic potential, Duistermaat–Heckman	Used	[54, 4, 41, 7]
Birch's theorem; relative interiors of convex hulls	Used	[29, 71]
one positive critical point (theorem 3(1)); the positive locus as a Lefschetz thimble	Used	Galkin's lemma [49]; [50, 56]
the valuation u^* maximises $\min_k \ell_k$; a recursive construction of u^*	Used	[57, Thm. 1.2]; [46]
the bridge (theorem 3(2)–(4)): leading-order balance, candidates versus selection, the sign split	Derived	this note
the orthant is the ascending thimble J_+; Laplace concentration (proposition 7)	Adapted	[49, 50]
the tropical limit of the thimble law (theorem 8) and its F_1 levels (example 9)	New	this note
linear-programming duality, strict complementarity; regular subdivisions and secondary fans	Used	[72, 52, 60]
soft evidence as $\lambda - s\rho$; the balance programme; candidates = relative interior of argmax (definition 10 and proposition 11)	Derived	this note
where data can act: circuits, least evidence, the F_1 wall (corollary 12 and example 14)	New	this note
coupling polytopes are Delzant; Kähler class = product coupling; chambers and blow-ups	Used	[12]
Tier A on the fibre is Fisher's noncentral hypergeometric family (proposition 22)	Adapted	[45, 30]
the max–min coupling $\min(f_i/m, g_j/n)$; leximin walls; the one-count example (theorem 23 and example 25)	New	this note
selection: the positive valuation is the lexicographic max–min point (theorem 27)	Adapted	[46, 57, 37]; proof here
morphisms: orthogonality, Clifford endomorphisms, summands (proposition 31)	Cited	[32, 33, 77]
no higher products; A_∞ equivalence across the F_1 wall (proposition 32 and theorem 34)	Derived (on cited m_2)	this note
the Clifford shadow of selection; Stokes walls crowd the outcome wall (theorem 33 and proposition 35)	New (35: computed)	this note
the column-binding example classified; the reduction of the Clifford form; its Hom table (propositions 36 and 39 and theorem 38)	New (Hom: on cited theorems)	this note
Stage IV: a path across two leximin walls; the collision at the level-two wall (propositions 40 and 41)	New (exact tropical; collision numerical)	this note
Markov gluing of couplings is associative; maximum entropy is Markov (proposition 43)	Used	[47]
composition by selection: not associative; the exact K_4 configuration (proposition 44 and theorem 45)	New (exact computation)	this note

component	status	source
no-go for contractible morphism spaces; Fréchet-midpoint gluing; signed-minimum correlations; cells of associativity (propositions 47 and 50, theorem 48, and corollary 49)	New (cells exact; census on grids)	this note
binary selection picks the Floer-nontrivial fibre of the reduced space (proposition 52)	Cited (exact checks)	[32, 8, 46]
couplings form a dagger symmetric monoidal category under Markov gluing (definition 54 and proposition 55)	Used	[47, 34]
what selection keeps and breaks: units, reversal, copied factors; not independent factors or coarse-graining (propositions 57 and 58)	New (proofs; exact counterexamples)	this note
the identified interval of a binary composite; selection is its centre; poisoning and zero-cost changes (propositions 60 and 61 and example 62)	Adapted (identified sets); New	[62]; this note
rules as sections of the marginal projection; fibre dimension over faces; from sections to a category (theorems 64 to 66)	New (proofs)	[83, 16]; this note
maximum entropy and its consistency axioms (remark 69)	Cited	[75, 80]
associativity as a Beck–Chevalley-type condition (Markov property); symmetry as a hyperconnected quotient (remark 68)	Cited (dictionary)	[47, 27]
gluing and locality; signed extensions always exist; the fibre is $\mathcal{A}_d \cap \mathcal{J}$; contextual fraction (remark 70)	Cited	[2, 6, 1]
what the entropy cocycle gives: with $c > 0$ system independence, with $c < 0$ no section; the power family (remark 70 and proposition 71)	New (proof; family numerical)	[14, 81, 18]; this note
the 4-cycle: $\mathcal{D} = \mathcal{L} \cap \mathrm{CH}$; CH facets are simplices, positivity facets leave fibre 4 (propositions 72 and 73)	Adapted (Fine's theorem, exact); New (fibre table)	[26, 17, 2]
sections at the CHSH boundary; Simpson reversals made by a rule or by the middle resolution (propositions 75 and 76)	New (exact; paths numerical)	[76]; this note
every n-cycle: noncontextuality facets are simplices, so the contextuality boundary is rigid (corollary 74 and theorem 77)	Adapted (facet list cited); New (identification reading; exact for $n \leq 8$)	[9]; this note
how little poisoning in many dimensions: two thresholds, circuit sparsity, placement monotone in the hidden support, removal as a functor; changes read cell by cell (propositions 79 to 82 and remark 84)	New (proofs; censuses exact; streams numerical)	[69, 67, 64]; this note
the geometry of poisoning: certified radii, deformation, crossing, continuity of rules (proposition 85 and example 86)	New (proofs; example numerical)	[83]; this note
Bayesian inversion is the dagger; reversal-equivariant sections commute with it; order drift as Bayesian inconsistency (proposition 87 and remark 88)	Adapted (dagger); New (sections)	[34, 68, 63]; this note
decisions and the value of information over fibres; when local data determine it; value-based distance = Hausdorff distance (propositions 89, 90 and 93 and theorem 91)	New (proofs; computations numerical)	[20, 61, 79, 53]; this note
Fisher geometry against decision relevance: phases decision-null; fibre = Fisher-null space of the data; one-sided Fisher–Rao bound; volume coefficients coordinate-dependent (proposition 94)	Adapted (Fisher–Rao, Fubini–Study); New (reading of theorem 91)	[66]; this note

component	status	source
phases of the objects: A is three Bloch spheres (orthonormal product basis); holonomies over each fibre form finite groups in B and C; link to probabilistic pseudo knots (proposition 117, section 14.5)	Proved (A); Numerical (groups); Outlook (graded category)	[24]; this note
trajectories of decision viability: viability shares under any disturbance; open and closed loops; ranking radius; topology caveats (propositions 97 to 99 and remark 101)	New (proofs; streams numerical)	[5, 24]; this note
histories and holonomy: holonomy bounded by the fibre; KL updating flat over growing contexts, forgetting and minimal change curved; gluing fails only when the context cycle closes (propositions 103 and 104 and example 105)	Adapted (I-projection, Jeffrey); New (holonomy bounds, example)	[19, 39, 79]; this note
algebra of the operations: removable singularities, irrational maximum entropy, counting explanations, Maxwell jumps (proposition 95)	Adapted (Birch, Ehrhart, Markov bases); New (examples)	[25, 15, 40]; this note
Floer cohomology of toric fibres and the potential W	Cited, not proved	[32, 46]
ampleness by invariant curves: Nakai–Moishezon, toric Kleiman, wall relations (proposition 107)	Used	[58, 36]
the 148 chambers of 2×4; selection depth in the Fano chambers; examples A, B, C (theorem 108 and propositions 109 and 111)	New (exact computation)	this note
weak Fano corrections $W^{\mathrm{LF}} = \sum(1+\delta_k)T^{-\lambda_k}y^{v_k}$ (remark 113)	Cited	[35]
one scale per removed dimension (Conjecture 115); proved at leading order in A, B, C (proposition 116)	Conjecture; New	this note
Novikov weights against the environment orders of the layers paper; topological early warning of decision instability	Open	sections 13.23 and 15

On decisions, the note does not claim to connect contextuality with decision-making, which Tohmé, Caterina and Gangle and Abramsky, Barbosa and Mansfield have done in other forms. Its claim is a joint quantitative treatment over marginal fibres: which actions the data identify, the regret from ambiguity that cannot be resolved, what structural and instance information are worth and when local data determine that worth, and how all of these respond to undetectable poisoning (sections 13.20, 13.21 and 13.23).

Two probability structures must be kept apart. The measures in this note live on *conditioned states*, points of the polytope. They are not the source weights π_l of the layers paper. The uniform measure of proposition 1(5) is a prior on states, not on sources.

2 A reading guide: information states

This section restates the note in one vocabulary. It adds no results; every claim below points to where it is proved, checked or left open. The common object is an *information state*

$$\mathcal{I} = (S, d, J_d), \qquad J_d = \pi_S^{-1}(d) \cap \Delta(\Omega),$$

where S is the collection of observed contexts, d the observed marginal tables, π_S the linear map from a joint law to its marginals on S, and J_d the *fibre*: the convex set of joint laws compatible with the data. Almost everything in the note is either an operation on information states or a statement about what a fibre does or does not determine.

Operations on information states. The operations are of different kinds and obey different laws, and much of the note consists in keeping them apart.

operation	what it does to (S, d, J_d)	where
marginalisation	forgets distinctions: projects J_d	theorem 64 and section 13.15
conditioning	restricts to the laws compatible with an observation; soft evidence moves the dual coordinates	sections 7 and 10
Markov gluing	composes couplings; associative; the composition of a category	propositions 43 and 55
Bayesian inversion	the dagger of that category	proposition 55 and section 13.22
independent product	the tensor (with the copied-factor caveats)	section 13.13
selection	chooses one law $s(d) \in J_d$; not a composition: in general not associative	section 13.10 and theorems 27 and 65
poisoning	perturbs d; asks the least edit that moves a decision	sections 13.14, 13.20 and 13.21
contextuality test	asks whether $J_d \neq \varnothing$	sections 13.16 and 13.19
value of information	measures decision-relevant variation inside J_d	section 13.23
history and transport	carries a selection along a sequence of information states	section 13.26
viability	asks whether acceptable decisions survive every admissible disturbance	section 13.25

Three of these are compositions in the categorical sense (gluing, inversion as dagger, product as tensor) and are treated as such in section 13.13. Selection is not: it is a section of the marginal projection, and theorem 65 says exactly when a family of sections composes associatively (interface consistency; maximum entropy satisfies it, leximin does not). The note does not force every operation into one category.

Geometry of the fibre. The fibre carries three geometric structures with different jobs.

- *Convex and toric.* J_d is a polytope; with its log-linear (maximum-entropy) parametrisation it is the moment polytope of a toric Kähler structure whose metric is the Fisher information (proposition 1). Its faces, walls and chambers organise selection (theorem 27) and, for 2×4 couplings, the toric manifolds of Part IV (theorem 108).
- *Statistical.* At an interior law the observed data have Fisher information I_{obs}, and its null space is exactly the direction space of the fibre (proposition 94(2)):

  $$\ker I_{\text{obs}}(P) = T_P J_d.$$

  The directions the data cannot distinguish are exactly the directions in which compatible laws vary.
- *Decision-theoretic.* A decision problem cuts the fibre into policy cells, on each of which the value of information is linear (section 13.23).

The spine: what the data determine for a decision. The strongest single connection in the note joins the second and third structures. By theorem 91, the value of information is constant on J_d exactly when every full-dimensional policy cell's value functional annihilates the fibre's directions; by proposition 94(2) this reads

$$\text{VOI determined by the data} \iff \text{every relevant value functional is Fisher-visible.}$$

This separates several kinds of uncertainty that are easily run together:

- *statistical*: sampling variation of the observed tables (not modelled here beyond section 13.14);
- *observational*: the Fisher-null directions, the fibre itself;
- *contextual*: whether any compatible law exists at all ($J_d = \varnothing$ for contextual data, section 13.16);
- *decision*: whether the action changes over the fibre ($R > 0$, proposition 89);
- *value*: whether the value of information changes over the fibre (theorem 91).

They are not interchangeable. Laws far apart in Fisher–Rao distance can have the same value of information, and laws arbitrarily close can lie on opposite sides of a policy wall (proposition 94(3)–(4)). Decision value is Lipschitz in Fisher–Rao distance; the converse fails.

Selection, transport and stability. Selection turns an information state into a law, and the note compares rules by what they preserve. Maximum entropy is the I-projection, composes associatively and is flat under updating along growing context sets (proposition 104); leximin is the tropical limit of the positive critical point (theorems 8 and 27) and is not associative. Along a history of information states a rule may depend on the path: the holonomy $\mathrm{TV}(\sigma(H), \sigma(H'))$ is bounded by the fibre (proposition 103), vanishes on rigid faces and for identified decisions, and the path-independence condition $\sigma = s \circ d$ is the selection-level sheaf condition of [79]. Gluing of local tables can fail only when the cover has a cycle, which is where contextuality enters (example 105). Poisoning, viability and value-based stability measure how far the data must move before a decision does (sections 13.21 and 13.25 and proposition 93).

The toric and Floer layer. For coupling fibres the polytope is the moment polytope of a toric manifold, the potential W has critical points (Floer objects: a torus fibre with a flat connection), and the positive critical point selects the leximin coupling in the tropical limit (theorems 3 and 27). Part IV tests this beyond two-dimensional fibres. In the three 2×4 examples the valuations of the eigenvalues at the positive object match the dimensions removed at successive selection levels (proposition 116); the general statement remains Conjecture 115. The objects also carry phases. They are decision-null (proposition 94(1)), and they form finite groups over each fibre; only in the product case do they become a canonical Hilbert-space basis (section 14.5). This layer enriches the information state; it does not replace its probabilistic meaning.

What is not claimed. Three tempting unifications are left as outlook, not results.

- *An indexed category of information states*: contexts and refinement as a base, fibres J_d over it, kernels and sections as the transformations, assembled by a Grothendieck construction. The pieces exist (sections 13.13, 13.15 and 13.26); the construction and its compatibility laws for products, forgetting, conditioning and mixtures are not proved here. In particular the fibre of a product of data is in general larger than the product of fibres, since correlations across the two systems are unconstrained.
- *Beyond finite outcome spaces*: theorem 91 and proposition 94 are finite-dimensional; an infinite-dimensional version is not attempted.
- *Phase-sensitive decisions*: a value of information that sees the phases would need non-commuting measurements, which the note never uses.

How to read the note. Part I builds the toric Kähler structure, the bridge to the Floer side and the thimble. Part II applies it to contingency tables and coupling fibres. Part III (section 13 onwards) treats the operations: morphisms, gluing, sections, contextuality, poisoning, inversion, decisions, value of information, algebra, viability and histories, roughly in the order of the table above. Part IV is the three-dimensional test. The status table and the ledger (appendix A) give the status of every statement: proved, adapted, exact, numerical or open.

3 Tier A: the toric Kähler structure of conditioned states

Setting. Let $S = \{S_i\} \subset \mathbb{Z}^d$ be a finite support whose affine span is $\mathbb{R}^d$, with coefficients $c_i > 0$. In log coordinates $u = \log x$ put

$$\varphi(u) = \log \sum_i c_i e^{\langle S_i, u\rangle}, \qquad q_i(u) = \frac{c_i e^{\langle S_i, u\rangle}}{\sum_j c_j e^{\langle S_j, u\rangle}},$$

so $q(u)$ is the monomial law of the layers paper and $\mu(u) = \sum_i q_i S_i$ is its moment map. On the torus $(\mathbb{C}^*)^d$, write $z_j = e^{(u_j + i\theta_j)/2}$.

Proposition 1 (the toric Kähler data)**.** *(1) $F(z) = \varphi(\log|z|^2)$ is a Kähler potential for a torus-invariant Kähler form ω on $(\mathbb{C}^*)^d$, the pullback of the Fubini–Study form by the monomial map $z \mapsto [\sqrt{c_i}\, z^{S_i}]$. In the coordinates (u, θ), up to the normalising constant of one's convention, $\omega = \sum_{j,k} \varphi_{jk}(u)\, du_j \wedge d\theta_k$, and the moment map of the rotation action is $\nabla\varphi = \mu$.*

(2) Fisher information is the metric. *The Hessian $\varphi_{jk} = \mathrm{Cov}_{q(u)}(S)_{jk}$ is the Fisher information of the exponential family $q(u)$ in its natural parameter, and the Kähler metric is $\sum \varphi_{jk}(du_j du_k + d\theta_j d\theta_k)$.*

(3) The positive real locus is a Lagrangian section. *$\{\theta = 0\}$ is Lagrangian, and it meets each moment fibre $\mu^{-1}(p)$, p in the interior of $Q = \mathrm{conv}(S)$, in exactly one point (Birch).*

(4) The symplectic potential is a relative entropy. *The Legendre dual $\psi(p) = \sup_u(\langle p, u\rangle - \varphi(u))$ equals $\sum_i q_i \log(q_i/c_i)$ evaluated at the exponential-family law $q = q(u(p))$. That law is the unique minimiser of $D(q\|c) = \sum_i q_i \log(q_i/c_i)$ among the laws on S with mean p; many other laws have mean p. So $\psi(p) = \min\{D(q\|c) : \sum_i q_i S_i = p\}$, the entropy of the minimiser relative to the weights c, with the sign reversed. Moreover $\nabla\psi(p) = u(p)$ and $\mathrm{Hess}\,\psi = (\mathrm{Hess}\,\varphi)^{-1}$, the Fisher information in the expectation parameter.*

(5) Duistermaat–Heckman. *The pushforward of the Liouville measure $\omega^d/d!$ by μ is $(2\pi)^d$ times Lebesgue measure on the interior of Q. In natural parameters it has density* $\det \mathrm{Hess}\,\varphi$.

Proof. These are classical. For (1), see [54, 4] and [48, Section 4.2]. (2) holds because φ is the log-partition function of the exponential family [7]. (3): on $\{\theta = 0\}$ every $d\theta_k$ vanishes, and a moment fibre is $\{u(p)\} \times T^d$, which meets $\theta = 0$ once because $u(p)$ is unique [29]. (4) is Legendre duality for a strictly convex function, and $\langle p, u\rangle - \varphi(u) = \sum_i q_i \log(q_i/c_i)$ at $\mu(u) = p$. That $q(u(p))$ minimises $D(\cdot\|c)$ under the mean constraint is the Lagrange (Gibbs) characterisation of exponential families [7]. (5) is the change of variables $p = \nabla\varphi(u)$, whose Jacobian is $\mathrm{Hess}\,\varphi$ [41]; the θ-integral gives $(2\pi)^d$. Checks: KB1 checks (1)–(2) on 30 random supports, against finite differences. KB2 checks (4) on 20 supports. KB3 checks (5) by comparing $\int \det \mathrm{Hess}\,\varphi\, du$ over boxes with the areas of their moment images. □

In general this structure lives on the open orbit $(\mathbb{C}^*)^d$ only. A Delzant polytope Q determines a compact symplectic toric manifold X_Q [38], but a metric built from an arbitrary support need not extend smoothly to it. It does extend in one case. Suppose S is the set of lattice points of a Delzant lattice polytope Q. On a smooth projective toric variety an ample line bundle is very ample [48, Section 3.4]. So the monomial map extends to an embedding of X_Q in $\mathbb{P}^{|S|-1}$, and ω is the restriction of the Fubini–Study form; the weights c_i act by a diagonal automorphism. The polytope of the layers paper's blueprint is not Delzant: its vertex at $(1,1)$ has determinant 3. So the blueprint gives an orbifold, and the examples below are $\mathbb{CP}^2$ and F_1.

4 The bridge

Setting. Let $Q = \{p : \ell_k(p) = \langle v_k, p\rangle - \lambda_k \geq 0,\ k = 1, \dots, m\}$ be a compact Delzant polytope, with primitive inward facet normals v_k. Its potential, in the convention of Hori–Vafa and Cho–Oh [55, 32], is

$$W_T(y) = \sum_k T^{-\lambda_k} y^{v_k}, \qquad y \in (\mathbb{C}^*)^n,\ 0 < T < 1.$$

A family of critical points y_T has *valuation* $u = \lim_{T\to 0} \log|y_T| / \log T$ when the limit exists. At u, the term k has order $\ell_k(u)$, and the *leading terms* are $K(u) = \{k : \ell_k(u) \text{ minimal}\}$. Say the leading terms *balance* at u if $0 = \sum_{k\in K(u)} a_k v_k$ for some $a_k > 0$.

Lemma 2 (balance is Birch at target zero)**.** *For vectors v_k, $k \in K$, the following are equivalent:*

- *there are weights $a_k > 0$ with $\sum_k a_k v_k = 0$;*
- *there is a law q with $q_k > 0$ for every $k \in K$ and mean $\sum_k q_k v_k = 0$;*
- *0 lies in the relative interior of $\operatorname{conv}\{v_k : k \in K\}$.*

When they hold, the positive monomial law with support $\{v_k\}$ and mean 0 is unique, up to the torus directions orthogonal to the span of the differences (Birch).

Proof. Normalising the weights gives the first equivalence. The relative interior of the convex hull of finitely many points is the set of their convex combinations with all coefficients positive [71, Theorem 6.9]. Uniqueness is Birch's theorem [29], which is [28, Theorem "the toric realization"(1)] with target 0. □

Theorem 3 (the Birch bridge)**.** *Let the λ_k be rational.*

(1) One positive critical point *(Galkin's lemma [49]; Used). For each $T \in (0,1)$, W_T has exactly one critical point y_T in the positive torus $(0,\infty)^n$. A positive y is critical iff the law $w_k \propto T^{-\lambda_k} y^{v_k}$ of the terms of W_T has mean exponent $\sum_k w_k v_k = 0$: Birch at target 0.*

(2) Its valuation balances. *The valuation u^* of y_T exists. As $T \to 0$ the law w tends to a law supported exactly on $K(u^*)$, with positive weights and mean 0. So the leading terms balance at u^*.*

(3) Candidates versus selection. *The converse of (2) fails: the set of points where the leading terms balance can be larger than $\{u^*\}$, and higher-order terms select u^* among them.*

(4) The sign split. *At the valuation of a critical point that is not positive, the leading coefficients ρ_k of the surviving terms give a linear dependence $0 = \sum_K \rho_k v_k$ with every $\rho_k \neq 0$, real of either sign or complex. Unlike the positive case, this dependence does not define a probability law, and the leading terms there need not balance.*

Proof. (1) In log coordinates $b = \log y$, $\log W_T(e^b) = \log \sum_k T^{-\lambda_k} e^{\langle v_k, b\rangle}$ is strictly convex. Its Hessian is the covariance of the v_k under w, which is positive definite because the v_k *affinely* span $\mathbb{R}^n$. Indeed the v_k of a compact polytope positively span $\mathbb{R}^n$, so 0 lies in the interior of $\operatorname{conv}\{v_k\}$, which is therefore full-dimensional. Positive spanning also makes the function coercive: every direction increases some term. It therefore has exactly one critical point. Since $W_T > 0$ there, a critical point of $\log W_T$ is a critical point of W_T, and its gradient is the mean of w. (2) Write $T = \tau^N$, with N a common denominator of the λ_k. The positive critical point is a semialgebraic function of τ, so it has a Puiseux expansion with positive leading coefficients [31], and its valuation exists. The weights w_k then have orders $\ell_k(u^*)$ in T, up to a common shift, so they concentrate on $K(u^*)$. The limits are strictly positive on every $k \in K(u^*)$, namely $w_k \to \rho_k / \sum_{j \in K(u^*)} \rho_j$, where the $\rho_k > 0$ are the leading coefficients of the positive Puiseux series $T^{-\lambda_k} y_T^{v_k}$. That strict positivity is what places 0 in the relative interior. This is the arc-limit theorem of the layers paper. The mean stays 0 in the limit; apply lemma 2. Judd and Rietsch prove that the valuation satisfies tropical critical conditions of this kind for complete positive Laurent polynomials over generalised Puiseux series [57]. (3) is example 6(b). (4) The same expansion, with complex leading coefficients ρ_k, gives $0 = \sum_K \rho_k v_k$ with every $\rho_k \neq 0$. Example 6(a) shows a real critical point with one negative coordinate, whose leading terms admit no positive balance. The checks below corroborate these statements; they are not proofs. KB4 checks (1)–(2) on $\mathbb{CP}^2$ and on F_1 at five blow-up sizes, from four starts each. KB5 checks (3) at exact rational points. KB6 finds all four critical points of F_1 in 80-digit arithmetic and checks (4). □

5 The Floer reading

For a compact toric Fano manifold X_Q, Cho and Oh show that the potential of the Lagrangian torus fibres is exactly W, with no corrections, and that a fibre $L(u)$ has non-vanishing Floer cohomology for some local system iff a critical point of W has valuation u [32]. Fukaya, Oh, Ohta and Ono extend the picture to all compact toric manifolds, where W acquires corrections [46]. We use only the Fano statement, and we prove nothing about Floer cohomology. Both examples below are Fano: $\mathbb{CP}^2$, and F_1, the blow-up of $\mathbb{CP}^2$ at a point, which is a del Pezzo surface. The Hirzebruch surfaces F_n with $n \geq 2$ are not Fano.

Corollary 4 (the probabilistic fibre)**.** *On a compact toric Fano manifold, exactly one critical point of W is positive. Its valuation determines* one *torus fibre $L(u^*)$ with non-vanishing Floer cohomology, not the only one. At u^*, the terms of W of least order form a conditioned law with mean exponent* 0*. Other critical points, including non-positive ones, may determine further fibres with non-vanishing Floer cohomology, but those are not represented by the positive probabilistic law. The fibre $L(u^*)$, a Lagrangian torus, is a different object from the positive real Lagrangian section of proposition 1(3), which meets it once.*

Proof. Combine theorem 3(1)–(2) with the cited theorem. □

Example 5 ($\mathbb{CP}^2$)**.** Here $v = (1,0), (0,1), (-1,-1)$ and $\lambda = (0,0,-1)$, so $W_T = y_1 + y_2 + T/(y_1 y_2)$. The critical points are $y_1 = y_2$ with $y_1^3 = T$: three of them, one positive, all with valuation $(\frac{1}{3}, \frac{1}{3})$,

the barycentre. There all three terms tie, and $\frac{1}{3}\big((1,0)+(0,1)+(-1,-1)\big)=0$. The barycentre is also the only point where the leading terms balance. Its fibre is the Clifford torus.

Example 6 (the Hirzebruch surface F_1)**.** Take $Q=\{p_1\geq 0,\ p_2\geq 0,\ p_1+p_2\leq 1,\ p_2\leq 1-\alpha\}$, the triangle of $\mathbb{CP}^2$ with the corner $(0,1)$ cut by a facet of length $\alpha\in(0,1)$. So $v=(1,0),(0,1),(-1,-1),(0,-1)$ and $W_T=y_1+y_2+T/(y_1y_2)+T^{1-\alpha}/y_2$. The critical equations are $y_1^2y_2=T$ and $y_2^2=T/y_1+T^{1-\alpha}$. With $y_2=w^2$ and $y_1=T^{1/2}/w$, they become $w^4-T^{1/2}w-T^{1-\alpha}=0$, so there are four critical points. The Newton polygon of this quartic gives their valuations.

(a) $\alpha<\frac{1}{3}$. Three critical points lie over $(\frac{1}{3},\frac{1}{3})$, one of them positive, and there the first three terms balance. The fourth lies over $(\alpha,1-2\alpha)$. Its leading coefficient is $w\approx -T^{1/2-\alpha}$, so $y_1<0$. Its leading terms are $(1,0),(-1,-1),(0,-1)$. They admit the signed balance $(1,0)+(-1,-1)-(0,-1)=0$ (in general $a_1=a_3=-a_4$), but no positive one. By corollary 4, the probabilistic fibre is over the barycentre, and the second fibre that the cited theorem provides is outside the positive theory.

(b) $\alpha>\frac{1}{3}$. All four critical points lie over $u^*=(\frac{1+\alpha}{4},\frac{1-\alpha}{2})$, one of them positive. But the leading terms $(0,1)$ and $(0,-1)$ balance at every point of the open segment $u_2=\frac{1-\alpha}{2}$, $\frac{1-\alpha}{2}<u_1<\alpha$. The endpoints are excluded. There a third term joins the leading set, which is $\{v_1,v_2,v_4\}$ at $u_1=\frac{1-\alpha}{2}$ and $\{v_2,v_3,v_4\}$ at $u_1=\alpha$. Neither set balances positively, because the first coordinates of its vectors are $1,0,0$, or $0,-1,0$. Leading order gives a segment of candidates, and the next order selects u^*: the same mechanism by which the layers paper's ties at the limit are broken at higher order.

At $\alpha=\frac{1}{3}$ all four facets tie at the barycentre. KB5 and KB6 corroborate every statement here at $\alpha=\frac{1}{5}$ and $\frac{3}{5}$; KB4 also uses $\frac{1}{4},\frac{2}{5},\frac{4}{5}$ (fig. 1).

6 The positive thimble and its tropical limit

The positive critical point has a second, analytic role: it is the critical point of an oscillatory integral. In mirror symmetry the integral $\int e^{-W/\hbar}\,dy/y$ over the positive real cycle is the one tied to Galkin's conifold point and to Gamma conjecture I; under homological mirror symmetry that cycle is expected to correspond to the structure sheaf [49, 50, 56]. Here we only need its elementary properties, and what the tropical limit does to the law it carries. Give $(\mathbb{C}^*)^n$ the flat metric $\sum_j|d\log y_j|^2$, and let $\hbar>0$.

Proposition 7 (the positive thimble)**.** *(1) The positive orthant $(0,\infty)^n$ is the ascending Lefschetz thimble J_+ of y_T for $\mathrm{Re}(W_T/\hbar)$, the set of points whose downward gradient flow tends to y_T. In $b=\log y$, the Hessian of $\mathrm{Re}\,W_T(e^b)$ at $b_T=\log y_T$, in the coordinates $(\mathrm{Re}\,b,\mathrm{Im}\,b)$, is $\mathrm{diag}(H,-H)$ with $H=\mathrm{Hess}(W_T\circ\exp)(b_T)$ positive definite. So the real directions ascend, and $\mathrm{Im}\,W_T=0$ on J_+.*

(2) $\mu_\hbar=Z^{-1}e^{-W_T(y)/\hbar}\,dy/y$ on J_+ is a probability law, and $Z<\infty$.

(3) Laplace. *As $\hbar/W_T(y_T)\to 0$, the law of $b=\log y$ under $\mu_\hbar$ is asymptotically normal, with mean b_T and covariance $\hbar H^{-1}$.*

Proof. (1) $W_T\circ\exp(b)=\sum_k T^{-\lambda_k}e^{\langle v_k,b\rangle}$ is holomorphic, and its restriction to real b is strictly convex and coercive (theorem 3(1)). For a holomorphic f, the Hessian of $\mathrm{Re}\,f$ in $(\mathrm{Re}\,b,\mathrm{Im}\,b)$ is

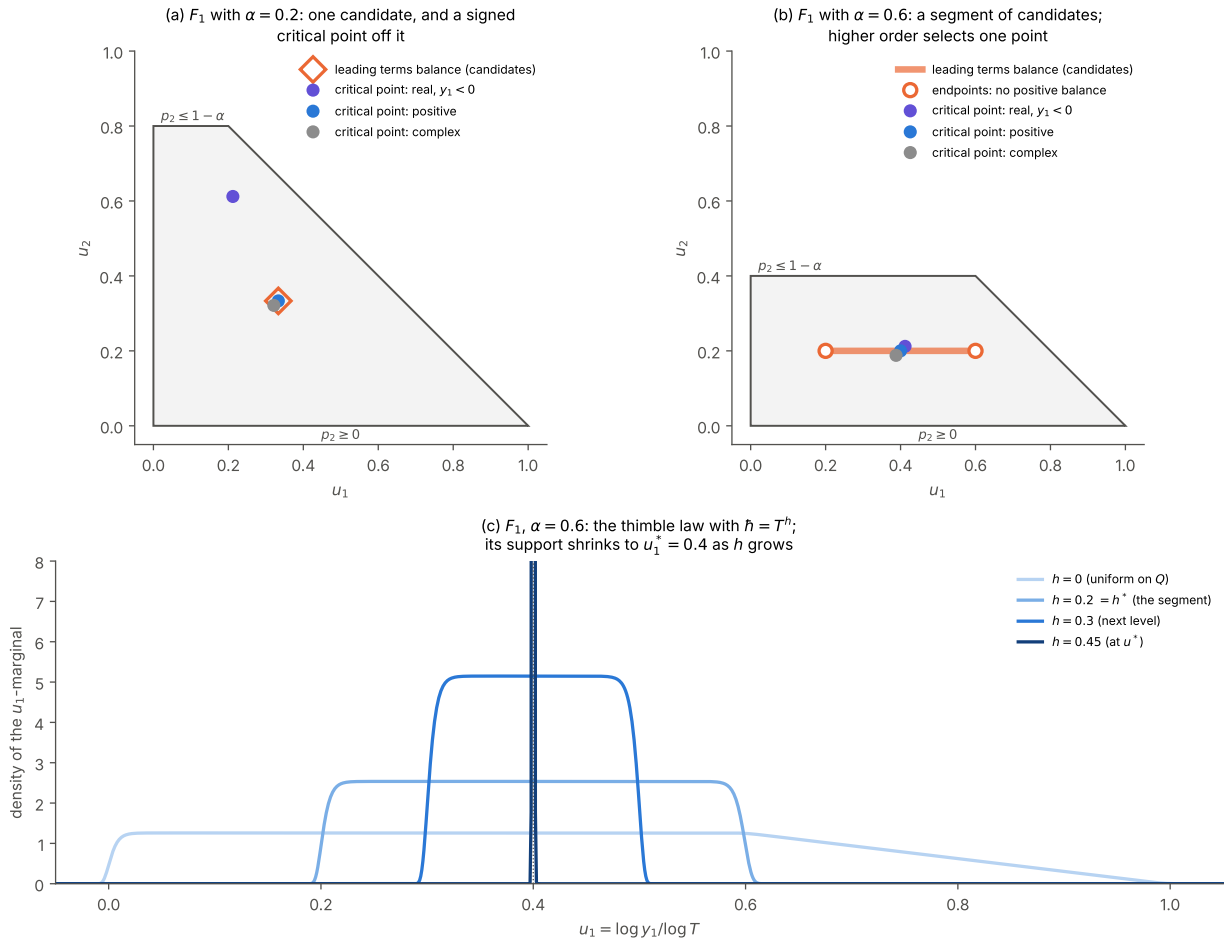


Figure 1: F_1 at two blow-up sizes. (a) $\alpha = 0.2$: the only point where the leading terms balance is the barycentre, which carries the positive critical point; a real critical point with $y_1 < 0$ sits over $(\alpha, 1 - 2\alpha)$, outside the positive theory. (b) $\alpha = 0.6$: the leading terms balance along an open segment, whose endpoints (open circles) have no positive balance, and the critical points all sit over one point of it. (c) $\alpha = 0.6$: the u_1-marginal of the thimble law with $\hbar = T^h$ (theorem 8 and example 9). At $h = 0$ it is the marginal of the uniform law on Q; at $h = h^* = 0.2$ it is uniform on the segment; at $h = 0.3$ it is uniform on $[0.3, 0.5]$; above 0.4 it is concentrated at $u_1^* = 0.4$. Here $\log T = -200$, and -100 for $h = 0.45$.

$\operatorname{diag}(H,-H)$ with H the real Hessian, by the Cauchy–Riemann equations. The metric is real, so the gradient flow of $\operatorname{Re} W_T$ preserves the real locus. There the flow is the gradient flow of a strictly convex, coercive function, and every orbit tends to b_T. So the orthant lies in the stable manifold J_+, which is an n-ball. The orthant is an n-manifold, so it is open in J_+; it is closed in $(\mathbb{C}^*)^n$, so it is closed in J_+. Since J_+ is connected, the two are equal. (2) $W_T > 0$ on J_+, and $W_T(e^b) \geq c|b| - C$ by positive spanning, so $Z < \infty$. (3) This is Laplace's method at the non-degenerate minimum b_T of $W_T \circ \exp$; the relative corrections are $O(\hbar/W_T(y_T))$. KB7 corroborates (1) and (3), at $\hbar/W_T(y_T) = 10^{-2}, 10^{-3}, 10^{-4}$. □

Laplace describes the limit $\hbar \to 0$ at fixed T. The tropical limit couples the two, $\hbar = T^h$, and reads y on the scale of the valuation, $u = \log y / \log T$. Since $T^{-\lambda_k} y^{v_k} = T^{\ell_k(u)}$, the pushforward $\nu_{T,h}$ of μ_{T^h} to u has density proportional to

$$f_{T,h}(u) = \exp\Big(-\sum_k T^{\ell_k(u)-h}\Big).$$

Let $g(u) = \min_k \ell_k(u)$, which is the tropicalisation of W in the min convention. Let $h^* = \max_u g$, and let $Q_h = \{u : g(u) \geq h\} = \{\langle v_k, u\rangle \geq \lambda_k + h\}$, so that $Q_0 = Q$.

Theorem 8 (the tropical limit of the thimble law)**.** *(1) If $h < h^*$, then $\nu_{T,h}$ tends, in total variation, to the uniform law on the polytope Q_h. At $h = 0$ this is the normalised Duistermaat–Heckman measure of X_Q.*

(2) If $h \geq h^*$, then $\nu_{T,h}$ concentrates on $Q_{h^*} = \operatorname{argmax} g$: $\nu_{T,h}(U) \to 1$ for every open $U \supseteq Q_{h^*}$.

(3) $\operatorname{argmax} g = \{u : 0 \in \operatorname{conv}\{v_k : k \in K(u)\}\}$. It is a polytope. It contains every candidate, every u with $0 \in \operatorname{relint}\operatorname{conv}\{v_k : k \in K(u)\}$, and in particular u^*; it can also contain non-candidates. Every u outside it has $g(u) < h^*$. This applies to the valuation of a signed critical point of example 6(a) as well.

Proof. Write $L = \log(1/T)$. Since the v_k positively span, $g(u) \leq C - c|u|$, so h^* is finite and attained, and the level sets of g are compact. (1) For u with $g(u) > h$, every exponent is positive and $f_{T,h}(u) \to 1$. Off Q_h, some exponent is negative and $f_{T,h}(u) \to 0$, and ∂Q_h is null. For $T \leq T_0 < 1$, $f_{T,h} \leq \min(1, \exp(-T_0^{g-h}))$, which is integrable. By dominated convergence, $\int f_{T,h} \to \operatorname{vol} Q_h > 0$, since Q_h has interior when $h < h^*$. Scheffé's lemma then gives convergence in total variation. (2) Choose $\delta > 0$ with $\{g \geq h^* - 2\delta\} \subseteq U$. Put $s = h - h^* \geq 0$. Off U, $T^{g-h} \geq T^{-(s+2\delta)}$, so $\int_{U^c} f_{T,h} \leq C \exp(-\frac{1}{2} T^{-(s+2\delta)})$. Near a maximiser u_0, on the ball of radius $1/(ML)$, where $M = \max|v_k|$, every $\ell_k \geq h^* - 1/L$. So $f_{T,h} \geq \exp(-me\,T^{-s})$ there, and the mass of the ball is at least $cL^{-n}\exp(-me\,T^{-s})$. The ratio of the two bounds tends to 0, because $T^{-2\delta} \to \infty$. (3) g is concave and piecewise linear, and its superdifferential at u is $\operatorname{conv}\{v_k : k \in K(u)\}$ [71, Theorem 23.8]. A point maximises a concave function iff 0 lies in its superdifferential. Candidates satisfy this, and so does u^* by theorem 3(2). This recovers, in our setting, the theorem of Judd and Rietsch that the tropical critical point maximises the tropicalisation [57, Thm. 1.2]. KB8 corroborates (1)–(3) on $\mathbb{CP}^2$ and on F_1 with $\alpha = \frac{1}{5}$ and $\frac{3}{5}$. □

So the first level of the tropical limit sees exactly the max–min set. That set is the first step of the iterated max–min by which Fukaya, Oh, Ohta and Ono locate a fibre [46], and of the recursive construction of Judd and Rietsch [57]. The selection of u^* inside it is a second-level question.

Example 9 (the levels on F_1)**.** For $\mathbb{CP}^2$, and for F_1 with $\alpha < \frac{1}{3}$, $h^* = \frac{1}{3}$ and $\operatorname{argmax} g$ is the barycentre. The signed fibre of example 6(a) has $g = \alpha < h^*$. For F_1 with $\alpha > \frac{1}{3}$, $h^* = \frac{1-\alpha}{2}$, and

argmax g is the closed segment $S = \{u_2 = \frac{1-\alpha}{2},\ \frac{1-\alpha}{2} \le u_1 \le \alpha\}$. Its relative interior consists of the candidates, and its endpoints are maximisers that are not candidates.

(a) At $h = h^*$ the u_1-marginal of $\nu_{T,h}$ tends to the *uniform* law on $[\frac{1-\alpha}{2}, \alpha]$. Leading order gives the selected point no extra mass.

(b) On S, $\ell_1 = u_1$ and $\ell_3 = \frac{1+\alpha}{2} - u_1$. The next max–min level is $h_2 = \max_S \min(\ell_1, \ell_3) = \frac{1+\alpha}{4}$, attained at $u_1 = \frac{1+\alpha}{4} = u_1^*$. For $h^* < h < h_2$, the u_1-marginal tends to the uniform law on $[h, \frac{1+\alpha}{2} - h]$. For $h > h_2$, it concentrates at u_1^*.

Sketch. Near S, the transverse factor $\exp(-T^{u_2-h} - T^{1-\alpha-u_2-h})$ does not depend on u_1. The coupling of u_2 to ℓ_3 matters only in boundary layers of width $O(1/L)$. So the u_1-marginal behaves as in theorem 8, now for $\min(\ell_1, \ell_3)$ on S. KB8 corroborates (a) and (b) for $\alpha = \frac{3}{5}$, at $h = \frac{1}{5}, \frac{3}{10}, \frac{9}{20}$ (fig. 1(c)). That the mean at level h^* equals u_1^* here is a symmetry of this example: u_1^* is the midpoint of S. The mechanism of selection is the second level, not the mean.

7 Soft evidence moves λ

This section starts the programme of conditioning existing structures on data and recording what changes. It treats one kind of data, soft evidence whose strength grows with the amount of data. Hard evidence (zero cells) and forgetting are not treated here.

Definition 10 (the data action)**.** Fix a reference scale N and put $T = e^{-N}$. Evidence that multiplies term k of W_T by $e^{-n\rho_k}$ after n observations, with rates $\rho_k \ge 0$, multiplies it by $T^{(n/N)\rho_k}$. So it replaces λ by

$$\lambda^{\mathrm{eff}} = \lambda - s\rho, \qquad s = n/N.$$

Which term each observation weighs, and the scale N, are modelling choices. A likelihood of fixed size has $s \to 0$ as $N \to \infty$. It changes the coefficients smoothly, as in proposition 1, and changes no valuation.

Let Δ_m be the laws on the m facets, and let $B = \{q \in \Delta_m : \sum_k q_k v_k = 0\}$ be the *balance polytope*, the laws with mean exponent 0. It has dimension $m - n - 1$, because 0 lies in the interior of $\mathrm{conv}\{v_k\}$. Write $g_\lambda = \min_k \ell_k$, $h^*(\lambda) = \max g_\lambda$, and $F(\lambda)$ for the face of B on which $\langle q, \lambda\rangle$ is maximal.

Proposition 11 (the max–min level is a balance programme)**.** *(1)* $h^*(\lambda) = \min_{q\in B}\big(-\langle q, \lambda\rangle\big)$. *So h^* is concave and piecewise linear in λ.*

(2) Let K_0 be the union of the supports of the laws in $F(\lambda)$, and $\sigma_0 = \mathrm{conv}\{v_k : k \in K_0\}$. Then $K(u) = K_0$ on the relative interior of argmax g_λ*, and $K(u) \supsetneq K_0$ on its relative boundary. The candidates (theorem 8(3)) are exactly the relative interior, and* $\dim \mathrm{argmax}\, g_\lambda = n - \dim \sigma_0$.

(3) The limit law of the positive critical point (theorem 3(2)) lies in $F(\lambda)$.

Proof. (1) $\max_u \min_k \ell_k = \max_u \min_{q\in\Delta_m} \sum_k q_k \ell_k(u)$, and the minimax theorem exchanges the two. For fixed q, $\sup_u \sum_k q_k(\langle v_k, u\rangle - \lambda_k)$ is $-\langle q, \lambda\rangle$ if $\sum_k q_k v_k = 0$ and $+\infty$ otherwise. (2) By complementary slackness, $q_k > 0$ for an optimal q forces $\ell_k(u) = h^*$ at every maximiser u, so $K(u) \supseteq K_0$ on argmax g_λ. By strict complementarity [72], some maximiser has $\ell_j > h^*$ for every $j \notin K_0$. The same then holds on the relative interior of argmax g_λ, which is $\{u : \ell_k(u) = h^*\ (k \in$

$K_0)$, $\ell_j(u) > h^*$ $(j \notin K_0)\}$. A law in the relative interior of $F(\lambda)$ is positive on K_0 and has mean 0, so those points are candidates (lemma 2). At a boundary point, suppose 0 were in the relative interior of $\operatorname{conv}\{v_k : k \in K(u)\}$. Then there would be a balanced law positive on all of $K(u)$. That law would be optimal, since $\sum_k q_k \ell_k(u) = h^*$, so its support would lie in K_0, a contradiction. The linear span of $\{v_k : k \in K_0\}$ is its affine span, since it contains 0, so the equations $\ell_k = h^*$ on K_0 cut out dimension $n - \dim \sigma_0$. (3) The limit law q is balanced and supported on $K(u^*)$, where every $\ell_k(u^*) = h^*$. So $-\langle q, \lambda\rangle = \sum_k q_k \ell_k(u^*) = h^*$, and q is optimal. DW2 and DW3 corroborate (1)–(3) on 40 and 23 configurations. □

So candidates against selection appears twice, in dual spaces. In u-space, u^* is selected inside the relative interior of $\operatorname{argmax} g_\lambda$, which has dimension $n - \dim \sigma_0$. In law space, the limit law is selected inside $F(\lambda)$. The two can be non-trivial at different parameters (example 14).

Corollary 12 (where data can act)**.** *(1) The leading set K_0, the candidate dimension and the face F change only on the walls of the normal fan of B, the* outcome walls*. Within a chamber, h^* is affine in λ, and so is u^* when σ_0 is an n-simplex.*

(2) Evidence along a translation, $\rho_k = \langle v_k, t\rangle$, changes nothing but $u^* \mapsto u^* - st$. Evidence along the constant, $\rho = \mathbf{1}$, changes nothing but $h^* \mapsto h^* + s$. Only the quotient of $\mathbb{R}^m$ by these $n+1$ directions can cross a wall. It has dimension $m - n - 1$, the number of independent affine dependences (circuits) of $\{v_k\}$.

(3) Suppose there is one circuit a, normalised to be primitive and integral. The wall is $c(\lambda) = -\langle a, \lambda\rangle = 0$, and along $\lambda_0 - s\rho$ it is reached at $s^* = -c(\lambda_0)/\langle a, \rho\rangle$ when this is positive, never otherwise. Over rates with $\max_k \rho_k - \min_k \rho_k \le 1$, the least such s is $|c(\lambda_0)|/(\frac{1}{2}\|a\|_1)$.

Proof. (1) $F(\lambda)$ is constant on the relative interior of each cone of the normal fan of B; apply proposition 11. On a chamber, $h^* = -\langle q_0, \lambda\rangle$ for a fixed vertex q_0. When σ_0 is an n-simplex, u^* is the unique solution of $\ell_k(u) = h^*$, $k \in K_0$, which is linear in λ. (2) Under a translation, ℓ_k becomes $\ell_k(u + st)$. Under the constant, every ℓ_k shifts by s. Both are orthogonal to $B - B$, which consists of affine dependences. (3) B is a segment, and its normal fan has one wall, $\langle q_A - q_B, \lambda\rangle = 0$, where $q_A - q_B$ is a multiple of a. Since $\sum_k a_k = 0$, adding a constant to ρ leaves $\langle a, \rho\rangle$ unchanged. So the best value over range at most 1 is $\sum_k a_k^+ = \frac{1}{2}\|a\|_1$. □

Example 13 ($\mathbb{CP}^2$: no wall)**.** Here $m = n + 1$, so there is no circuit, and B is the single law $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. No soft evidence changes the leading set of $\mathbb{CP}^2$. Data only move the selected point and the level (DW4, DW7).

Example 14 (F_1 across its wall)**.** For general λ, put $L = -(\lambda_1 + \lambda_2 + \lambda_3)$ and $\alpha = -\lambda_1 - \lambda_3 + \lambda_4$. These are the size of the triangle and the length of the blown-up facet, and both are invariant under translation. The Kähler cone is $0 < \alpha < L$. Its walls are viability walls, where the polytope loses facet 4 or collapses, and X_Q changes. The one circuit is $a = (2, -1, 2, -3)$, with $c(\lambda) = 3\alpha - L$. The balance polytope is the segment from $q_A = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0)$ to $q_B = (0, \frac{1}{2}, 0, \frac{1}{2})$, and $h^* = \min\big(L/3, (L - \alpha)/2\big)$.

(a) The outcome wall $\alpha = L/3$ lies *inside* the Kähler cone: the manifold does not change there. Below it, $K_0 = \{1, 2, 3\}$ and the candidate is a point. Above it, $K_0 = \{2, 4\}$, the candidates form an open segment, and u^* is selected at the second level (example 9). The lower diagonal of the subdivision flips from v_1v_3 to v_2v_4.

(b) u^* is continuous and piecewise affine across the wall (fig. 2(a)). What jumps is the leading set, the candidate dimension ($0 \to 1$) and the mechanism of selection (leading order $\to$ second level). The signed fibre over $(\alpha, 1-2\alpha)$ reaches the barycentre at the wall. By the cited theorem [32], the number of torus fibres with non-vanishing Floer cohomology drops from two to one there. We claim nothing about the Fukaya category.

(c) *At the wall* ($\alpha = 1/3$, $L = 1$), the substitution $w = T^{1/6}x$ turns the quartic of example 6 into $x^4 - x - 1 = 0$ for every T. So the positive law is $(1/x, x^2, 1/x, 1/x^2)/\Sigma \approx (0.2156, 0.3922, 0.2156, 0.1766)$ for every T, where $x \approx 1.2207$. It is the interior point $(1-t)q_A + tq_B$ of $F = B$, with $t \approx 0.353$. Here the candidate in u-space is a single point, but the law is selected inside a segment.

(d) The discriminant of $w^4 - T^{1/2}w - T^{1-\alpha}$ is $-27T^2 - 256T^{3-3\alpha}$, which never vanishes: no two critical points collide for $T > 0$. The outcome wall is where its two terms have equal order. It is a wall of the tropical discriminant, not of the discriminant itself.

(e) *Data.* Start at $\alpha = 1/5$, $L = 1$, so $c = -2/5$. Evidence against term 1, $\rho = e_1$, has $\langle a, \rho\rangle = 2$ and reaches the wall at $s^* = 1/5$, that is $n^* = N/5$ (fig. 2(b)). Along the way $\alpha/L = (1/5+s)/(1+s)$ stays in the Kähler cone. Evidence against term 4 ($\langle a, \rho\rangle = -3$) never reaches the wall. The most efficient rates with range at most 1 are $\rho = (1, 0, 1, 0)$, against terms 1 and 3, with $s = 1/10$.

DW4–DW7 corroborate (a)–(e).

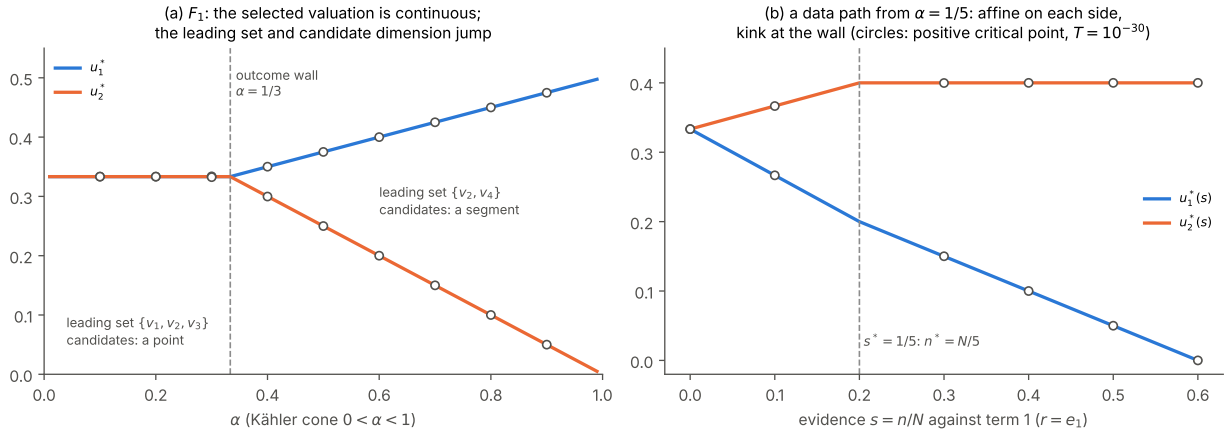


Figure 2: Soft evidence on F_1. (a) The selected valuation u^* across the Kähler cone. The curves are the chamber formulas and the circles the positive critical point at $T = 10^{-30}$. The outcome wall at $\alpha = 1/3$ changes the leading set and the candidate dimension, not the position of u^*; here $h^* = u_2^*$. (b) A data path from $\alpha = 1/5$ with evidence against term 1: $u^*(s)$ is affine on each side of $s^* = 1/5$, with a kink at the wall.

On F_1, then, the least data that changes the perspective changes *which terms carry the limit law*, not where the selected state sits. A decision that reads the leading set jumps at n^*. A decision that reads u^* continuously does not. Decision, contextuality and value-of-information walls need their own definitions, a decision model, a named complex and obstruction class, and an information-acquisition model, before they can be compared with the outcome walls. We record them separately and have not computed them.

Part II
Back to contingency tables

Part I went from toric geometry to conditioned probability. Part II runs the dictionary backwards, from a table of counts to the potential, and asks what the Floer-selected fibre means for the table. It works the independence model exactly. Every operation it uses, and every operation it does not yet treat, is recorded in the ledger (appendix A).

8 Two configurations of one model

An $r \times c$ table of counts t_{ij} has row margins t_{i+} and column margins t_{+j}. The independence model is the log-linear model whose design matrix A has one column per cell, the row indicator stacked on the column indicator. It is toric [40, 51]: its sufficient statistic is the vector of margins, and by Birch the MLE is the unique law in the model with the observed margins, $\hat{p}_{ij} = t_{i+}t_{+j}/n^2$ [29, 30]. The moment polytope of this model is $\Delta_{r-1} \times \Delta_{c-1}$, so its manifold is $\mathbb{CP}^{r-1} \times \mathbb{CP}^{c-1}$, which is Delzant and Fano.

The model carries two different point configurations, and they must not be confused.

Proposition 15 (cells and facets)**.** *(1)* Cells. *The rc columns of A satisfy $(r-1)(c-1)$ independent linear relations. These are spanned by the basic moves $+1$ at $(i,j),(k,l)$ and -1 at $(i,l),(k,j)$, which change a table without changing its margins. This is the Tier A support of proposition 1.*

(2) Facets. *The $r+c$ inward facet normals of $\Delta_{r-1}(L_1) \times \Delta_{c-1}(L_2)$ have exactly one circuit, $a = (c\,\mathbf{1}_r, -r\,\mathbf{1}_c)/\gcd(r,c)$. This is the configuration W is built from.*

Proof. (1) A has rank $r+c-1$, and the moves lie in its kernel and have rank $(r-1)(c-1)$ [40]. (2) $m-n-1 = (r+c)-(r+c-2)-1 = 1$. The row normals sum to 0, and so do the column normals. So $\sum_k a_k v_k = 0$, and $\sum_k a_k = rc - cr = 0$. CT1 checks both for $r, c \in \{2,3,4\}$. □

So the 2×3 model has two moves but one circuit. The moves act on tables with fixed margins, within a fibre. The circuit acts on λ, across Kähler classes. The second statement is a special case of a general count.

Proposition 16 (products add one circuit)**.** *If $v^{(1)}$ and $v^{(2)}$ both have 0 in the interior of their convex hulls, the product configuration has $(m_1 - n_1 - 1) + (m_2 - n_2 - 1) + 1$ independent circuits. The extra one is $(q^{(1)}, -q^{(2)})$ for balanced laws $q^{(i)}$ of the factors. Its wall is $h_1^* = h_2^*$ when each factor has a single balanced law.*

Proof. The count is $(m_1+m_2)-(n_1+n_2)-1$. The vector $(q^{(1)}, -q^{(2)})$ sums to $1-1=0$, and both halves balance. It is independent of the circuits of the factors, which sum to zero on each factor separately. When $B_i = \{q^{(i)}\}$, the balance programme gives $h^* = \min(h_1^*, h_2^*)$ (proposition 11), with a wall where they are equal. □

9 The independence model: its wall and its selection

In the coordinates $b =$ the first $r-1$ row margins and the first $c-1$ column margins, the facets of $Q = \Delta_{r-1}(L_1) \times \Delta_{c-1}(L_2)$ have $\ell_k(b)$ equal to the k-th margin. Here L_1 and L_2 are the totals of the

row and column margins. So $W = W_{\text{row}}(y) + W_{\text{col}}(z)$ with $W_{\text{row}} = y_1 + \cdots + y_{r-1} + T^{L_1}/(y_1 \cdots y_{r-1})$, and similarly for columns.

Theorem 17 (the independence model). *(1) $h^* = \min(L_1/r,\, L_2/c)$. The leading set is the row facets if $cL_1 < rL_2$, the column facets if $cL_1 > rL_2$, and all $r+c$ facets on the wall $cL_1 = rL_2$. The wall is $c(\lambda) = -\langle a, \lambda\rangle = (cL_1 - rL_2)/\gcd(r,c) = 0$.*

(2) The critical points of W are $y_i = \zeta T^{L_1/r}$, $z_j = \eta T^{L_2/c}$ with $\zeta^r = \eta^c = 1$, so there are rc of them. All have the valuation $u^* = (L_1/r, \ldots; L_2/c, \ldots)$, the uniform margins, for every L_1, L_2. The positive law is exactly proportional to $T^{L_1/r}$ on each row facet and $T^{L_2/c}$ on each column facet.

(3) Off the wall the candidates have dimension $c-1$ (if $cL_1 < rL_2$) or $r-1$, and the uniform margins are selected at the second level. On the wall the candidate is the single point u^*.

(4) By the cited theorem [32], exactly one torus fibre has non-vanishing Floer cohomology, for every L_1, L_2. So the outcome wall of the independence model changes no Floer data. On F_1 it does (example 14(b)).

Proof. (1) Each factor has the single balanced law, uniform on its facets, with values L_1/r and L_2/c. Apply proposition 16. (2) The variables separate. $\partial W_{\text{row}}/\partial y_i = 0$ gives $y_i \prod_j y_j = T^{L_1}$ for every i, so all y_i are equal and $y^r = T^{L_1}$. The term values are then $T^{L_1/r}$ times a root of unity. (3) Off the wall the leading set is one factor's facets, so σ_0 has dimension $r-1$ (or $c-1$); apply proposition 11(2). (4) All critical points share one valuation. CT2 and CT3 check (1)–(3), with exact rationals and gradients. □

10 Margins against evidence

Data enter in two ways, and the note keeps them apart.

Proposition 18 (two roles of data). *(1)* Observed margins *b pick a point of Q, the fibre $\mu^{-1}(b)$ and the MLE (Birch). For the independence model, $\ell_k(b)$ is exactly the k-th margin, the facet $\{\ell_k = 0\}$ is a zero margin, and the MLE lies in the open model iff every margin is positive [30]. For general hierarchical models the facets correspond to facial sets of cells, and the MLE fails to exist exactly when the margins lie on a proper face [43, 44]. We use only the independence case.*

(2) Evidence on the terms of W *moves λ (definition 10), and with it u^*, h^* and the walls. With λ fixed, u^* does not depend on b.*

(3) Both b and u^* are points of Q, in moment coordinates. The valuation of a critical point is the moment coordinate of its torus fibre $L(u^*) = \mu^{-1}(u^*)$ [32]. So the MLE fibre and the Floer-selected fibre can be compared in one space. For the independence model we use the Fisher–Rao distance within the model, between $\hat{p}$ and the law at u^* (uniform margins). With surveys of sizes n_1 for rows and n_2 for columns,

$$d(\hat{p}, p^*)^2 = n_1\, d_\Delta(\hat{p}_{\text{row}}, \mathbf{1}/r)^2 + n_2\, d_\Delta(\hat{p}_{\text{col}}, \mathbf{1}/c)^2, \qquad d_\Delta(p,q) = 2\arccos \sum_i \sqrt{p_i q_i}.$$

Joint sampling of n cells is the case $n_1 = n_2 = n$. A coarser index is the centrality *$g(b)/h^*$: the smallest margin divided by its largest possible value.*

Proof. (1) The first statements are direct, and the MLE formula shows positivity. (2) is corollary 12 together with the fact that W does not involve b. (3) The Fisher information of the independence family is block diagonal, $n_1 \operatorname{Cov}(\text{row}) \oplus n_2 \operatorname{Cov}(\text{col})$, because the log-likelihood separates into a row part and a column part. So the model is a Riemannian product of two scaled simplices, and on each simplex the Fisher–Rao distance is $2 \arccos$ of the Bhattacharyya coefficient [7]. CT4 and CT5 check (1) and the block structure. □

The MLE fibre and the Floer-selected fibre coincide exactly when the margins are uniform. That the Floer fibre is the uniform-margin table, the one farthest from every zero-margin degeneracy in lattice distance, is a reading of the cited theorem, not a new statistical result. The polytope used here is the *base*, the space of margins. Section 12 turns to the *fibre* over a point of it, the coupling polytope, where the same margins become the Kähler class.

11 Data and walls in the independence model

Example 19 (2×2 and 2×3)**.** Under joint sampling, $L_1 = L_2 = n$. The 2×2 table therefore sits on its wall for every n, which is the monotone $\mathbb{CP}^1 \times \mathbb{CP}^1$. For 2×3, $cL_1 - rL_2 = n > 0$, so $h^* = n/3$, the column facets lead, and the candidates are the open segment of row margins $b_1 \in (n/3, 2n/3)$ with uniform column margins. Its endpoints are maximisers but not candidates. The selected point is $b_1 = n/2$. To reach the wall the column survey must grow to $n_2 = 3n/2$: $n/2$ more column observations. In the terms of corollary 12(3), this is ρ on a column facet with $\langle a, \rho \rangle = -r/\gcd$, so $s^* = n/2$. Crossing it changes the leading set and the candidate dimension. By theorem 17(4) it changes no Floer data. CT6 checks every number.

Example 20 (hard evidence: a structural zero)**.** Declare cell $(2, 2)$ of the 2×2 table structurally zero. This is the quasi-independence model on three cells. The three remaining cells span a unimodular triangle, which is *not* a face of the square: hard evidence need not restrict to a toric divisor. Its facet configuration is that of $\mathbb{CP}^2$, which has no circuit, so the one wall of the 2×2 model is gone. CT7 checks this. The general case, deleting cells of larger tables, is open.

12 One coupling polytope through every layer

What this section shows, in plain words. Fix two margins, a row law f and a column law g. The tables (couplings) with those margins form a polygon, the *fibre*. That polygon is the picture of a toric surface, and everything in Part I can be drawn on it.

- The usual conditional law of a table given its margins, the one behind Fisher's exact test, *is* the toric Kähler structure (fig. 4).
- The positive critical point of W picks the *leximin* coupling: the one whose smallest cell is as large as possible (fig. 3).
- The positive thimble is a family of laws on couplings. It runs from "every coupling equally likely" to "only the leximin coupling" (fig. 5).
- Two kinds of wall cut the space of margins. Blow-up walls change the surface; leximin walls change which cells lead. Here one count of data crosses a leximin wall and changes the Floer fibres, while the surface stays the same (fig. 6).

The example is the one followed in the architecture note [13]: $f = (2/5, 3/5)$, $g = (3/10, 43/100, 27/100)$.

12.1 Base and fibre

A pair of margins is a point of the *base*, $\Delta_{n-1} \times \Delta_{m-1}$ (section 9). Over it sits the *fibre*, the coupling polytope $\mathrm{Cpl}(f, g)$ of tables with those margins. For nondegenerate margins the fibre is Delzant, its facets are cells $\{\alpha_c = 0\}$, and the Kähler class of its toric manifold is the product coupling, $[\omega_{f,g}] = \sum_c f_i g_j [D_c]$ [12]. So the same data play two roles:

	base (sections 8 to 10)	fibre (this section)
points	margins (f, g)	couplings α with those margins
facets	a zero margin	a zero cell
the margins are	a point (the Birch/MLE fibre)	the Kähler class: λ_c is read off (f, g)
W is a sum over	margin facets	cell facets
Tier A support	cells, with $(r-1)(c-1)$ moves	integer tables with these margins

For 2×3, write a coupling by $x = \alpha_{11}$, $y = \alpha_{12}$. Then $\alpha_{13} = f_1 - x - y$, $\alpha_{21} = g_1 - x$, $\alpha_{22} = g_2 - y$ and $\alpha_{23} = x + y - (f_1 + g_1 + g_2 - 1)$, so cell c has $\ell_c = \langle v_c, (x, y) \rangle - \lambda_c$ with $v_c \in \{(1,0), (0,1), (-1,-1), (-1,0), (0,-1), (1,1)\}$.

Example 21 (the fibre)**.** For $f = (2/5, 3/5)$, $g = (3/10, 43/100, 27/100)$ the fibre is the pentagon with vertices $(13/100, 0)$, $(3/10, 0)$, $(3/10, 1/10)$, $(0, 2/5)$, $(0, 13/100)$ and area $1331/20000$, so $X = \mathrm{Bl}_2\mathbb{CP}^2$. Its facets are the cells $11, 12, 13, 21, 23$. Cell 22 never vanishes, since $\alpha_{22} \geq f_2 + g_2 - 1 = 3/100$. Scaled by $n = 100$, its lattice points are the 715 integer tables with margins $(40, 60)$ and $(30, 43, 27)$. By Pick's theorem, $715 = 665.5 + 97/2 + 1$. CF1 and CF2 check this exactly.

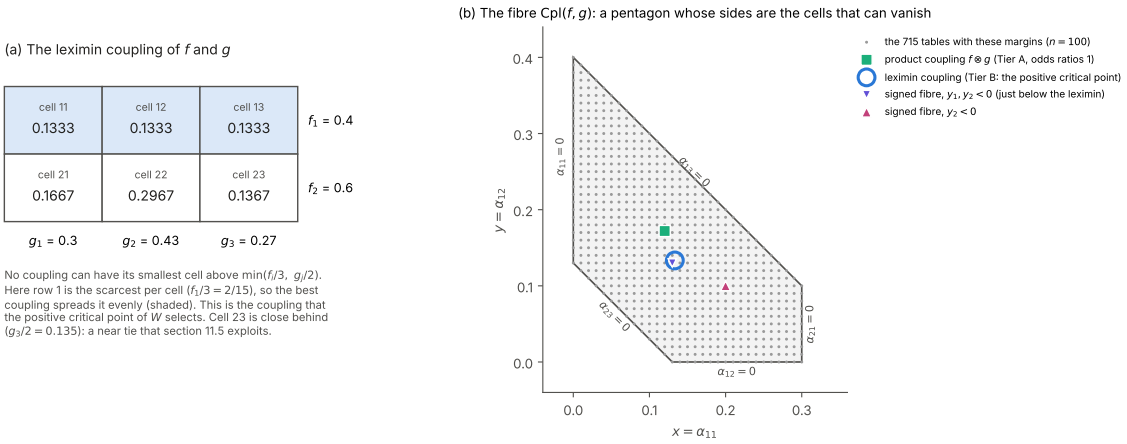


Figure 3: The example at a glance. (a) The leximin coupling: row 1 is the scarcest per cell, so it is spread evenly, and every other cell is larger. (b) The fibre, with the 715 tables as dots. Each side is a cell that can vanish. Marked: the product coupling, selected by Tier A; the leximin coupling, selected by the positive critical point of W; and the two real critical points with a negative coordinate (signed fibres).

12.2 Tier A on the fibre is Fisher's noncentral hypergeometric family

Proposition 22 (the conditional law of a table)**.** *Let S be the integer tables with margins Nf, Ng, written by (t_{11}, t_{12}), with weights $c_t = 1/\prod_{ij} t_{ij}!$.*

(1) For any cell parameters $\theta_{ij} > 0$, the law of a table under independent Poisson (or multinomial) sampling with means $\propto \theta_{ij}$, conditioned on its margins, is the monomial law $q(u) \propto c_t\, e^{\langle u,(t_{11},t_{12})\rangle}$. Here $u_1 = \log \frac{\theta_{11}\theta_{23}}{\theta_{13}\theta_{21}}$ and $u_2 = \log \frac{\theta_{12}\theta_{23}}{\theta_{13}\theta_{22}}$ are the two log odds ratios. This is Fisher's noncentral hypergeometric family. At $u = 0$ it is the central hypergeometric law, the null law of Fisher's exact test [45].

(2) By proposition 1, $\varphi(u) = \log \sum_t c_t e^{\langle u,t\rangle}$ is a Kähler potential. The moment map is the expected table, and the Kähler metric is the Fisher information of the odds ratios. Since S is the set of lattice points of the Delzant lattice polygon $N \cdot \mathrm{Cpl}(f,g)$, the form extends to all of $X = \mathrm{Bl}_2\mathbb{CP}^2$.

(3) At $u = 0$ the mean is exactly $N\, f \otimes g$: Tier A at odds ratios 1 selects the product coupling.

(4) Every interior coupling is the mean for exactly one u (Birch). In the example, the leximin coupling needs odds ratios $(e^{u_1}, e^{u_2}) \approx (0.82, 0.46)$.

Proof. (1) $\prod_{ij} \theta_{ij}^{t_{ij}}/t_{ij}!$ restricted to fixed margins depends on t only through c_t and $t_{11}u_1 + t_{12}u_2$. Substituting $t_{13}, t_{21}, t_{22}, t_{23}$ from the margins gives the two log odds ratios. (2) is proposition 1 and its compact case. (3) The central hypergeometric mean is $t_{i+}t_{+j}/N$ [30]. (4) is proposition 1(3). CF3 checks (1) and (3), the latter in exact rationals, and computes (4). □

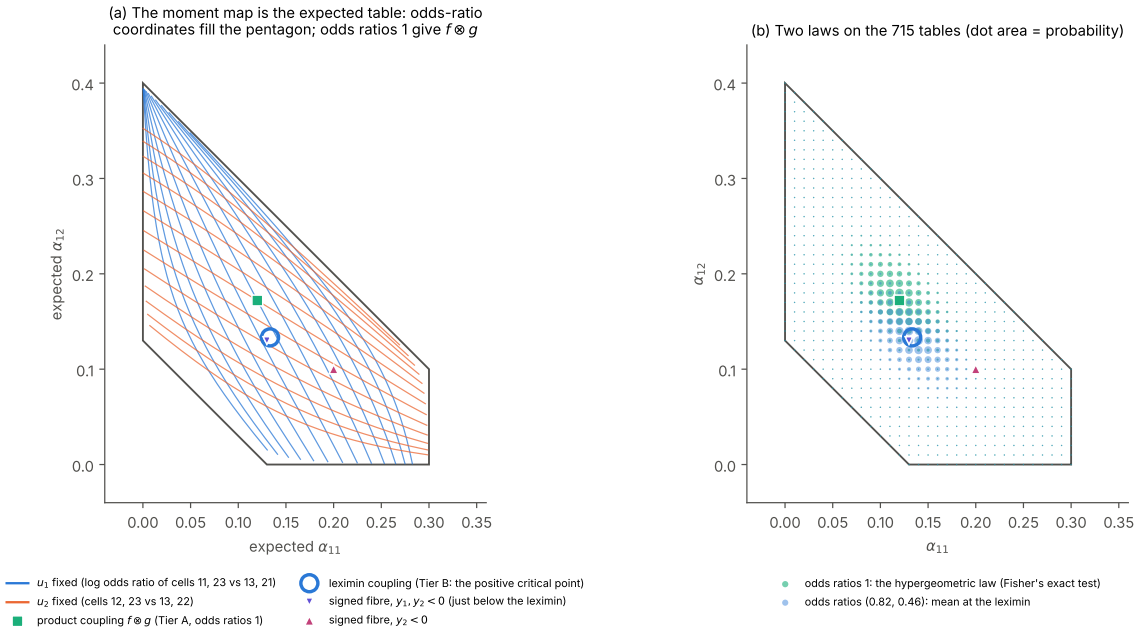


Figure 4: Tier A. (a) The moment map sends the plane of log odds ratios onto the open pentagon. Blue lines hold u_1 fixed and orange lines hold u_2 fixed. (b) Two laws on the 715 tables: the hypergeometric law (odds ratios 1, centred on $f \otimes g$), and the law with odds ratios $(0.82, 0.46)$, centred on the leximin coupling.

12.3 Tier B: the selected coupling is the leximin one

Theorem 23 (the max–min coupling)**.** *Let (f, g) be nondegenerate $n \times m$ margins, and let W be the potential of $\mathrm{Cpl}(f, g)$, a sum over its facet cells.*

(1) $h^* = \max_\alpha \min_{c\ facet} \alpha_c = \min\bigl(\min_i f_i/m,\ \min_j g_j/n\bigr)$.

(2) Every cell of a binding *row or column, one attaining the minimum, is a facet. If exactly one line binds, the leading set* K_0 *is that line's cells.*

(3) The candidate set has dimension $(n-2)(m-1)$ *for a binding row and* $(n-1)(m-2)$ *for a binding column. For* 2×3*, a binding row gives a single candidate, so leading order selects. A binding column gives a segment, and the next level selects.*

(4) The positive critical point of W *sits over the facet-leximin coupling, the lexicographic max–min of the facet cells. This is corollary 29, which follows from the selection theorem (theorem 27).*

Proof. (1) A row of m cells sums to f_i, so its smallest cell is at most f_i/m; columns are the same. Conversely, put h (the claimed value) in every cell and add the product of the residual margins $f_i - mh \geq 0$ and $g_j - nh \geq 0$, whose totals agree. This gives a coupling with every cell at least h. (2) A cell with $\alpha_c > 0$ on all of Cpl has $f_i + g_j > 1$. For a binding row i that would give $f_i > \sum_{k\neq j} g_k \geq (m-1)\min_k g_k$ and $f_i \leq (m/n)\min_k g_k$, so $n(m-1) < m$, which is impossible. If row i alone binds, the residual margins of the other rows and of all columns are positive, so the coupling of (1) has every other cell strictly above h^*. The row's cells, which average h^*, equal it at every maximiser. (3) The m cell functionals of a row sum to the constant f_i, so their normals span $m-1$ dimensions and contain 0 in the relative interior of their hull. Apply proposition 11(2) in dimension $(n-1)(m-1)$. (4) See corollary 29. CF6 checks it on 20 random rational margins at $T = 10^{-60}$. CF4 checks (1)–(3) on about 1000 random margins of shapes 2×3, 3×3, 2×4 and 3×4, against the facet-only programme. □

In words: *the selected coupling spreads the scarcest row or column evenly over its cells.* In the example, row 1 binds with $f_1/3 = 2/15$. So $h^* = 2/15$, $K_0 = \{11, 12, 13\}$, and the positive critical point sits over the leximin coupling $(2/15, 2/15)$. The product coupling $f \otimes g = (0.12, 0.172)$ has the single leading cell 13, at $0.108 < h^*$, so it is not a candidate. That is the Tier-B reason why "the product coupling carries no brane" [11].

Example 24 (the five critical points)**.** $W = y_1 + y_2 + T^{2/5}/(y_1y_2) + T^{3/10}/y_1 + T^{-13/100}y_1y_2$ has 5 critical points, twice the area 5/2 of its Newton polygon (Kushnirenko). Exact resultants at $T = 10^{-100}$ and 10^{-200} place them as follows.

- Three lie over the leximin coupling, one of them positive.
- One is real with $y_1, y_2 < 0$, over $(13/100, 13/100)$. Its leading cells are $\{11, 12, 23\}$, whose normals have the signed balance $(1,0) + (0,1) - (1,1) = 0$ but no positive one.
- One is real with $y_2 < 0$, over $(1/5, 1/10)$. Its leading cells are $\{12, 13, 21\}$, also with a signed balance only.

This is theorem 3(4) on the example. By the cited theorem [32], three torus fibres have non-vanishing Floer cohomology. This agrees with [11, Thm. 8.2]. CF5 checks the count, the positions, the signs and the leading sets.

12.4 The positive thimble on couplings

By theorem 8, the positive thimble with $\hbar = T^h$, read in $u = \log y/\log T$, tends to the uniform law on $Q_h = \{\alpha : \text{every facet cell} \geq h\}$ for $h < h^*$, and concentrates on argmax at h^*. On the fibre this has a plain reading. At $h = 0$ it is the uniform law on all couplings, which is Duistermaat–Heckman, and the limit of the uniform law on tables. As h grows it is uniform on the couplings

whose every cell is at least h. At $h^* = 2/15$ it is the leximin coupling (fig. 5; CF7). The thimble is thus a one-parameter family of probability laws on couplings, from "all equally likely" to "the fairest one".

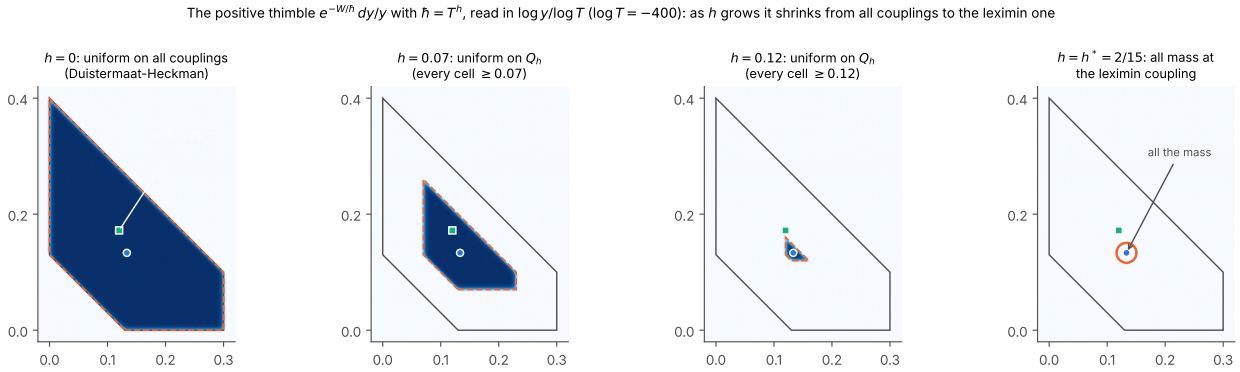


Figure 5: The positive thimble on the fibre, at four values of h, with $\log T = -400$. Dashed: the shrunk polygon Q_h, every cell at least h. The law is uniform on Q_h, and at $h^* = 2/15$ all its mass is at the leximin coupling (circle). The product coupling (square) is passed over.

12.5 Two kinds of wall, and one count of data

In the base, the margins (f, g) move the Kähler class of the fibre. Two families of walls cut the base.

- *Kähler-chamber walls* $f(I) = g(J)$. Across them the polygon gains or loses a side, which is a toric blow-up or blow-down [12]. These are the viability walls of the ledger.
- *Leximin walls*, where two of the ratios f_i/m, g_j/n tie. Across them the binding line, and with it the leading cells and the candidate dimension, jumps (theorem 23). These are the outcome walls of corollary 12, here explicit.

They are different walls (fig. 6(a)). The monotone point, with uniform margins, lies on all of them.

Example 25 (one count)**.** In the example, $f_1/3 = 2/15$ and $g_3/2 = 27/200$ differ by only $1/600$. Move one observation of 100 from column 3 to column 2, so that $g = (30, 44, 26)/100$. Then:

- column 3 binds, $h^* = 13/100$, and the leading cells are $\{13, 23\}$;
- the candidates become a segment, and the leximin coupling, selected at the next level, is $(27/200, 27/200)$;
- the real critical point with $y_1, y_2 < 0$ moves onto the leximin fibre, so the distinct valuations drop from three to two. By the cited theorem, so does the number of torus fibres with non-vanishing Floer cohomology.

The surface does not change: the nearest blow-up wall, $f_1 = g_2$, is three counts away. So in this example the least data that changes the perspective is one count, and it changes the leading cells and the Floer fibres, not the manifold (fig. 6(b); CF8, with exact resultants at $T = 10^{-100}$ and 10^{-300}).

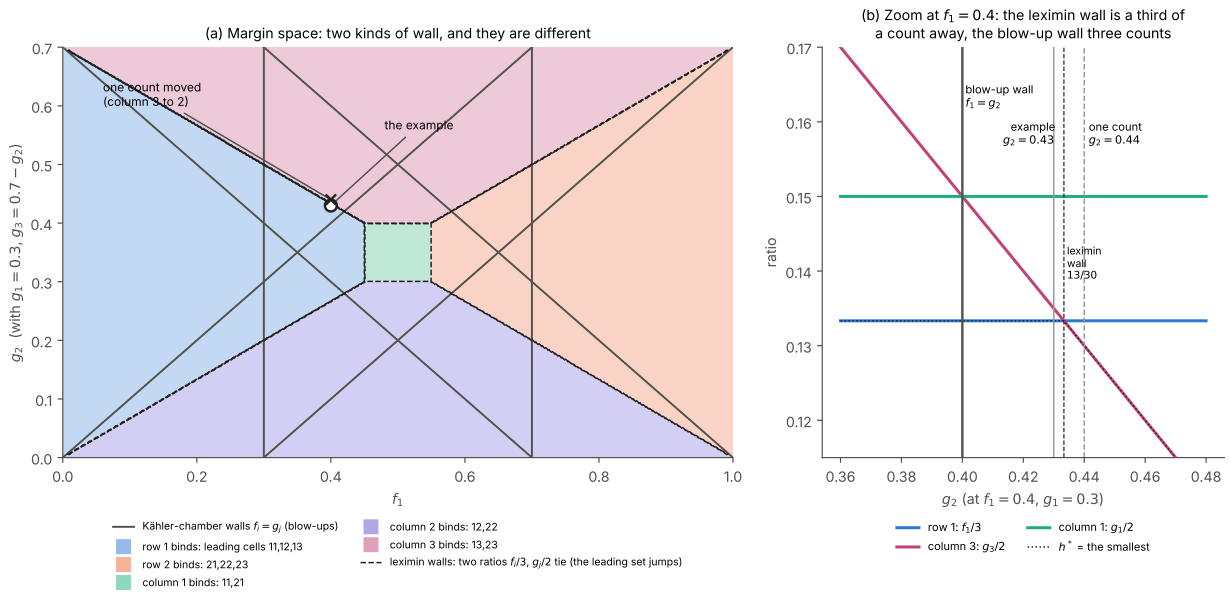


Figure 6: Margin space, in the slice $g_1 = 0.3$. (a) Colours show which row or column binds. Dashed lines are the leximin walls and solid lines the blow-up walls $f_i = g_j$. The example (circle) sits almost on a leximin wall, and one count (cross) crosses it. (b) Along $f_1 = 0.4$: the three ratios $f_1/3$, $g_3/2$, $g_1/2$, whose minimum is h^*. The leximin wall is a third of a count from the example, while the blow-up wall $f_1 = g_2$ is three counts away.

12.6 The selection theorem, and a column-binding example

Theorem 3 and proposition 11 put the valuation of the positive critical point in the candidate set. When that set is a point, this is the whole answer. When it is not, as for a binding column, the next level selects. This subsection proves that the levels continue all the way down, for any configuration.

Definition 26 (lexicographic max–min)**.** Let $\ell_k(u) = \langle v_k, u\rangle - \lambda_k$ with $v_1, \dots, v_m$ positively spanning $\mathbb{R}^n$. Put $A_0 = \mathbb{R}^n$ and $S_0 = \emptyset$. For $j \geq 1$, let h_j be the maximum over A_{j-1} of the least ℓ_k with k not yet fixed. Let A_j be the set where it is attained, and let S_j be the not-yet-fixed k with $\ell_k \equiv h_j$ on A_j. Stop when every k is fixed. Then A_J is a single point, the *lexicographic max–min point* u^{lex}.

Each S_j is nonempty (proposition 11(2), applied on A_{j-1}), and each A_j is bounded, so the recursion stops. The fixed normals span $\mathbb{R}^n$, so A_J is a point. On $\mathbb{CP}^2$ and on F_1 with $\alpha < 1/3$ there is one level. On F_1 with $\alpha > 1/3$ there are two, as in example 9.

Theorem 27 (selection)**.** *Let $a_k(L) \in [c, C] \subset (0, \infty)$, and let u_L minimise $F_L(u) = \sum_k a_k(L) e^{-L\ell_k(u)}$. Then $|u_L - u^{\mathrm{lex}}| = O(1/L)$ as $L \to \infty$. In particular, the valuation of the positive critical point of W_T, with $L = \log(1/T)$ and $u_L = \log y_T / \log T$, is the lexicographic max–min point of the facet functions, and the error at finite T is $O(1/\log(1/T))$.*

Proof. F_L is strictly convex and coercive, because the v_k positively span, so u_L is unique. We argue by induction on n; the case $n = 0$ is empty.

Level one is pinned to $O(1/L)$. For $\bar{u} \in A_1$, $F_L(u_L) \leq F_L(\bar{u}) \leq mCe^{-Lh_1}$. So each term satisfies $ce^{-L\ell_k(u_L)} \leq mCe^{-Lh_1}$, that is, $\ell_k(u_L) \geq h_1 - \kappa/L$ with $\kappa = \log(mC/c)$. By proposition 11(2)

there is a balanced law q supported exactly on S_1, and then $\sum_{k\in S_1} q_k\ell_k \equiv h_1$. Hence $\ell_k(u_L) \le h_1 + \kappa/(Lq_{\min})$ for $k \in S_1$.

The first level drops out. Let $D_1 = \{d : \langle v_k, d\rangle = 0,\ k \in S_1\}$ be the direction space of $E_1 = \operatorname{aff} A_1$, and P the orthogonal projection onto D_1. Then $Pv_k = 0$ for $k \in S_1$. So projecting $\nabla F_L(u_L) = 0$ gives, exactly,

$$\sum_{k\notin S_1} a_k(L)\, e^{-L\ell_k(u_L)}\, Pv_k = 0.$$

Write $u_L = p_L + w_L$, with $p_L \in E_1$ and $w_L \perp D_1$. The v_k, $k \in S_1$, span $D_1^{\perp}$, and $\langle v_k, w_L\rangle = \ell_k(u_L) - h_1 = O(1/L)$, so $|w_L| = O(1/L)$. Hence p_L is a critical point, and so the minimiser, of $F_L^{(1)}(p) = \sum_{k\notin S_1} a'_k(L)e^{-L\ell_k(p)}$ on E_1. Here $a'_k = a_k e^{-L\langle v_k, w_L\rangle}$ again lies in a fixed compact subset of $(0,\infty)$.

Induction. On E_1 the remaining ℓ_k have projected normals Pv_k that positively span D_1, because $A_1 = \{p \in E_1 : \ell_k(p) \ge h_1,\ k \notin S_1\}$ is bounded. Their lexicographic max–min point is u^{lex}. Indeed, off A_1 some remaining ℓ_k is below h_1, while on the relative interior of A_1 all exceed it, so the reduced recursion is levels $2, 3, \dots$ of the original. Since $\dim D_1 < n$, the induction hypothesis gives $|p_L - u^{\mathrm{lex}}| = O(1/L)$, and so $|u_L - u^{\mathrm{lex}}| = O(1/L)$. LX1–LX3 corroborate the statement and the mechanism, on 30 random configurations in multiprecision arithmetic. □

Remark 28 (provenance)**.** The procedure is the one Fukaya, Oh, Ohta and Ono use to locate a non-displaceable fibre [46]. Judd and Rietsch construct the tropical critical point recursively, and prove that it maximises the tropicalisation [57]. A related hierarchical selection governs the exponential-penalty trajectory in linear programming [37]. The statement is therefore adapted; the proof above is direct, and it gives the rate. The constant in $O(1/L)$ depends on the gaps between levels. That is why the near tie of example 25 converged slowly.

Corollary 29 (the selected coupling)**.** *For nondegenerate $n \times m$ margins, the positive critical point of the potential of* $\mathrm{Cpl}(f, g)$ *sits over the facet-leximin coupling. In* 292 *random margins of shapes* 2×3*,* 3×3 *and* 2×4*, this equals the leximin coupling over all cells (LX4). We have not proved that the two always agree.*

Example 30 (a column-binding fibre)**.** Repurpose Example 3.4(b) of [12]: $f = (2/5, 3/5)$, $g = (1/2, 3/10, 1/5)$. The fibre is the pentagon with vertices $(1/5, 0)$, $(2/5, 0)$, $(1/10, 3/10)$, $(0, 3/10)$, $(0, 1/5)$ and area $11/200$. Its facets are the cells $11, 12, 13, 22, 23$. Cell 21 never vanishes, since $f_2 + g_1 = 11/10$. The ratios are $f_i/3 = (2/15, 1/5)$ and $g_j/2 = (1/4, 3/20, 1/10)$, so column 3 binds. The levels are:

level	value	cells fixed	what is decided
1	$h_1 = 1/10$	$13, 23$	column 3 is spread evenly; the candidates form the open segment $x + y = 3/10$, $1/10 < x < 1/5$
2	$h_2 = 3/20$	$11, 12, 22$	along the segment, the least remaining cell is largest at the midpoint

So $u^* = (3/20, 3/20)$ (fig. 7(a),(b)). The endpoints are maximisers but not candidates: their leading sets, $\{11, 13, 22, 23\}$ and $\{12, 13, 23\}$, have no positive balance. At finite T,

$$x_L = \tfrac{3}{20} + \frac{\ln 2}{2L} + o(1/L),$$

because cells 11 and 22 coincide on the segment (both equal x) and balance cell 12 alone: $2e^{-Lx} = e^{-L(3/10-x)}$ (fig. 7(c)). Of the five critical points, the positive one and a real one with $y_1, y_2 < 0$ sit

over $(3/20, 3/20)$, both with this correction. The other three (one real, two complex) lie over the fibre $(2/15, 1/6)$ on the candidate segment, off the leximin point. Section 13.7 derives this exactly. The slow approach seen at finite T is the $1/\log(1/T)$ effect of their leading moduli $2^{\pm 1/3}$. The positive thimble shows the two levels as h grows: it is uniform on the segment at $h = 1/10$, uniform on $[h, 3/10 - h]$ between the levels, and concentrated at $3/20$ above (fig. 7(d)). The one-count example of example 25 is column-binding too, with levels $(13/100;\ 13, 23)$ and $(27/200;\ 11, 12)$. LX5–LX9 check every number here, the first exactly.

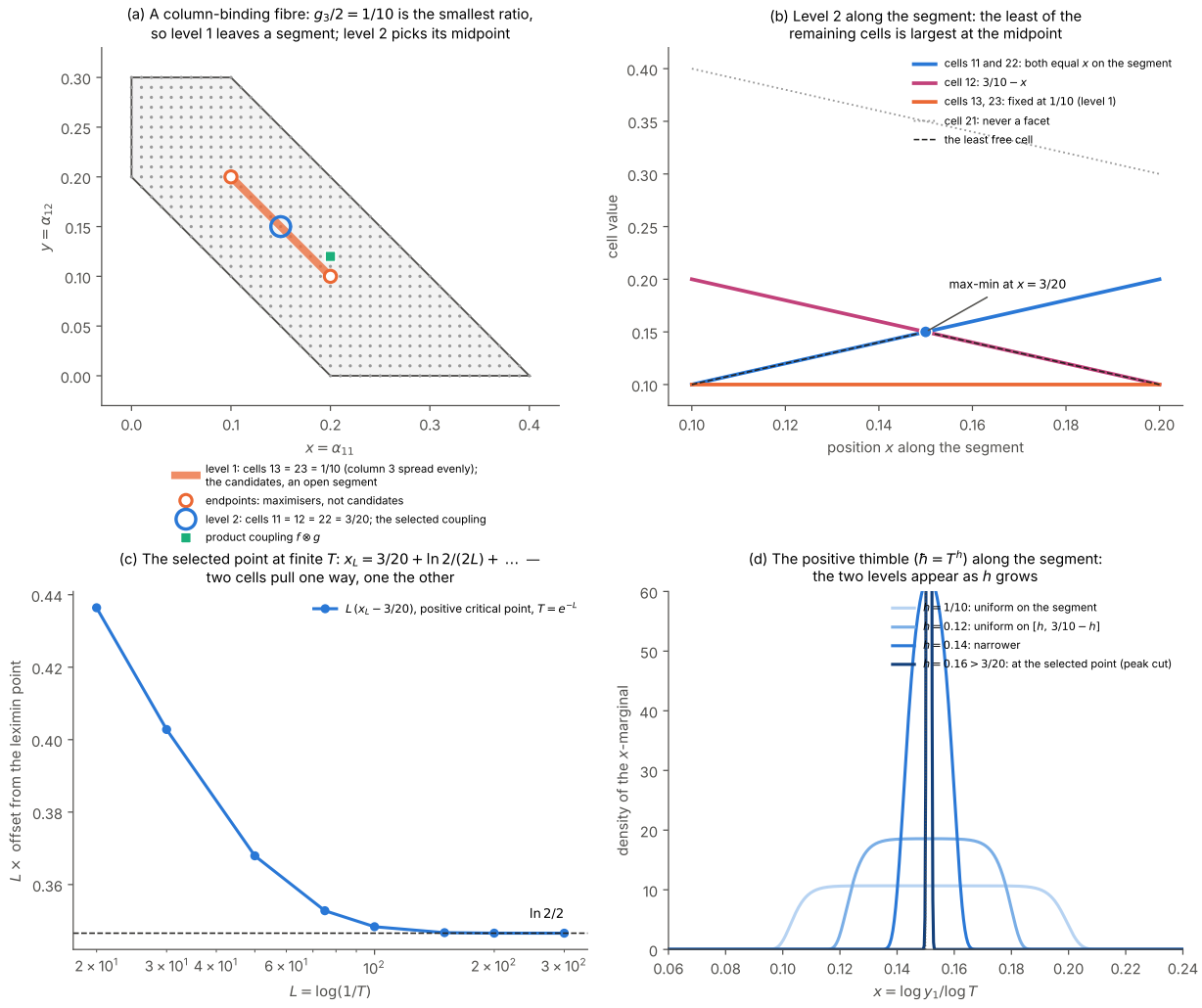


Figure 7: The selection theorem on a column-binding fibre. (a) Level 1 fixes cells 13 and 23 at $1/10$ and leaves a segment of candidates, whose endpoints are not candidates. Level 2 picks its midpoint. (b) Along the segment, the least remaining cell is largest at $x = 3/20$. (c) At finite T the selected point is offset by $\ln 2/(2L)$, because two cells pull against one. (d) The positive thimble along the segment shows both levels as h grows.

12.7 The example through every layer

layer	in the example	where
fibre	pentagon, area 1331/20000, facets $11, 12, 13, 21, 23$; 715 tables (Pick)	example 21; CF1–2
Tier A	Fisher's noncentral hypergeometric family; odds ratios 1 give $f \otimes g = (0.12, 0.172)$; the leximin needs $(0.82, 0.46)$	proposition 22; CF3
Tier B	$h^* = f_1/3 = 2/15$; row 1 binds; one candidate, the leximin coupling $(2/15, 2/15)$	theorem 23; CF4, CF6
selection	the valuation is the lexicographic max–min point, error $O(1/\log(1/T))$; a binding column needs a second level	theorem 27 and example 30; LX1–LX9
sign split	five critical points; signed fibres over $(13/100, 13/100)$ and $(1/5, 1/10)$	example 24; CF5
Floer (cited)	three fibres with non-vanishing Floer cohomology; none over $f \otimes g$	[32, 11]
thimble	uniform on couplings at $h = 0$; uniform on Q_h; the leximin coupling at h^*	theorem 8; CF7
walls	leximin wall 1/600 away in ratio; blow-up wall $f_1 = g_2$ three counts away	fig. 6
one count	column 3 binds; candidates a segment; three Floer fibres become two; same surface	example 25; CF8
not claimed	other thimbles, Stokes data, stability conditions, the category	section 15

Part III
Towards morphisms

13 Objects, morphisms and a categorical comparison

What this section shows, in plain words. So far we have found *which* torus fibres matter. A category also needs the arrows between them.

- Every critical point of W is an *object*: a torus fibre with a local system. Distinct objects have no arrows between them, and each object's arrows to itself form a Clifford algebra. For the probabilistic object, its quadratic form is the Fisher information of the conditioned law (fig. 8).
- Up to equivalence, an outcome wall does *not* change the category. It moves objects between fibres.
- It does leave two traces. The Clifford form of the probabilistic object, reduced modulo T, drops rank exactly when the candidate set stops being a point (fig. 9). And the Stokes walls, where the basis of thimbles mutates, crowd into a window of width $O(1/\log(1/T))$ around the outcome wall (fig. 10).

The computations are all on the critical points, values and Hessians of W. The Floer reading of those data is cited, and its status is marked in each statement.

13.1 Objects

By [32], a critical point c of W with valuation u_c gives the torus fibre $L(u_c)$ with a rank-one local system. Its holonomy has leading term y_c/T^{u_c}, and $L(u_c)$ has non-vanishing Floer cohomology. We call the pair an *object*. In our examples W has $\chi(X)$ critical points, twice the area of its Newton polygon: 4 for F_1 and 5 for $\mathrm{Bl}_2\mathbb{CP}^2$.

example	objects: fibre, type, leading holonomy
F_1, $\alpha = 1/5$	over the barycentre: positive $(1,1)$, complex (ζ,ζ), (ζ^2,ζ^2), with $\zeta^3=1$; over $(1/5,3/5)$: real-signed $(-1,1)$
F_1, $\alpha = 3/5$	all four over $(2/5,1/5)$: positive $(1,1)$, real-signed $(-1,1)$, complex $(i,-1)$, $(-i,-1)$
2×3 example	over the leximin coupling $(2/15,2/15)$: positive and two complex; over $(13/100,13/100)$: real $(-1,-1)$; over $(1/5,1/10)$: real $(1,-1)$
one count later	over $(27/200,27/200)$: positive, real $(-1,-1)$, two complex; over $(1/5,1/10)$: real $(1,-1)$

MO1 and MO2 check this table, the first part from the quartic and the second from exact resultants at $T = 10^{-200}$.

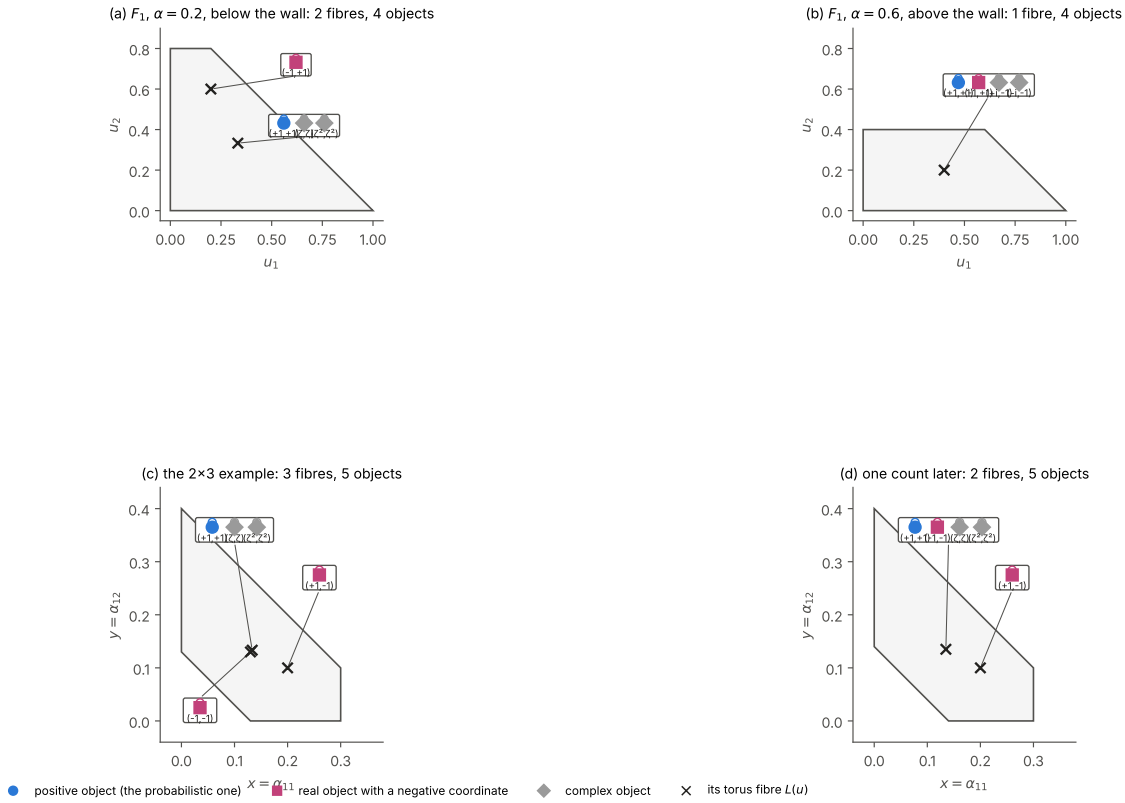


Figure 8: Objects over their torus fibres (crosses). Each box is a fibre; each marker is an object, labelled by its leading holonomy; each loop is its endomorphism algebra Cl_2. There are no arrows between distinct objects. (a)→(b): crossing the F_1 wall merges two fibres into one. (c)→(d): one count of data merges the leximin fibre with the nearby real fibre. In both cases the objects persist; only their fibres change.

13.2 Morphisms

Proposition 31 (morphisms between the objects)**.** *(1) Objects over distinct fibres have no morphisms: distinct moment fibres are disjoint, so their Floer complex is zero (exact).*

(2) Distinct objects over one fibre have distinct critical values $W(c) = m_0$. Morphisms are defined

only between objects with equal m_0, so the category splits into summands $\mathcal{F}_\lambda$ [10, 46], and they have no morphisms. This uses distinct m_0 only. Their holonomies also differ, but we do not use that to separate objects: an eigenvalue decomposition cannot by itself separate two objects with the same m_0. That these objects generate their summands is proved in the monotone case and conditionally otherwise [77]; our examples are not monotone.

(3) The Floer cohomology ring of an object is the Clifford algebra of the Hessian $Q = \sum_k c_k v_k v_k^\top$ of W at c, where c_k are the values of its terms [33, Thm. 5.6]. So it has dimension 2^n.

(4) For the positive object, $Q = W \cdot \mathrm{Cov}_w(v)$, where w is the law of the terms of W at y_T (mean exponent 0). Its quadratic form is the Fisher information of the conditioned law, scaled by W. It is real and positive definite.

Proof. (1) $L(u) = \mu^{-1}(u)$. (2) and (3) are the cited theorems; MO3 checks that the critical values (and, for the record, the holonomies) are distinct in every example. (4) $Q = \sum_k c_k v_k v_k^\top$ with $c_k = W w_k$, and $\sum_k w_k v_k = 0$. MO4 and MO5 check the identity, and the Clifford relations with explicit generators that span M_2, for all 18 objects. □

13.3 The A_∞ products

Proposition 32 (no higher products survive)**.** *Over the algebraically closed Novikov field Λ, let Q be nondegenerate. The $\mathbb{Z}/2$-graded algebra $\mathrm{Cl}(Q)$ is separable. For $n = 2$ it is the super matrix algebra $\mathrm{End}(\Lambda^{1|1})$. Its Hochschild cohomology vanishes in positive degrees. So every A_∞ structure on it, with $m_1 = 0$ and this m_2, is A_∞-isomorphic to the formal one: it is intrinsically formal [59, 73]. On the subcategory generated by these objects, the higher products m_k for $k \geq 3$ can be removed by an A_∞ isomorphism.*

Proof. A separable algebra has vanishing Hochschild cohomology in positive degrees. The obstructions to formality lie in $HH^{k,2-k}$ for $k \geq 3$, so they vanish. MO5 exhibits $\mathrm{Cl}(Q) \cong M_2$ for every object. The statement rests on the cited m_2 of proposition 31(3); we do not compute the A-side higher products themselves. □

So, within what the cited theorems give, the category generated by the objects is $\bigoplus_c \mathrm{Cl}(Q_c)$. It is semisimple, with one summand per critical point, matching $\dim QH^*(X) = \chi(X)$. On the mirror side, matrix factorisations at a nondegenerate critical point have the same Clifford description [42].

13.4 The Clifford shadow of selection

Over Λ, all Clifford algebras of nondegenerate forms of rank n are isomorphic, so (2)–(3) cannot see an outcome wall. The integral form can.

Theorem 33 (the Clifford shadow)**.** *Write the Clifford form of the positive object in the integral basis of $H^1(L;\mathbb{Z})$ and divide it by W. Then $Q/W = \mathrm{Cov}_w(v)$ has entries of non-negative valuation. Its reduction modulo T is $\mathrm{Cov}_{q^*}(v)$, where q^* is the limit law of theorem 3(2). Its rank is $\dim \sigma_0 = n - \dim(\textit{candidate set})$. So the reduced Clifford algebra is semisimple exactly when the candidate set is a single point, that is, exactly when leading order selects.*

Proof. w_k has valuation $\ell_k(u^*) - h^* \geq 0$, and $w \to q^*$ by theorem 3(2). $\mathrm{Cov}_{q^*}(v) = \sum_k q_k^* v_k v_k^\top$, because q^* has mean 0. Its rank is the dimension of the span of $\{v_k : k \in K_0\}$, which is $\dim \sigma_0$ because $0 \in \mathrm{relint}\, \sigma_0$. Then apply proposition 11(2). MO6 checks the convergence and the ranks. □

In the examples, the other eigenvalues have valuation equal to the gap between selection levels. On F_1 above the wall the small eigenvalue is $\sim T^{(3\alpha-1)/4}$, where $(3\alpha-1)/4 = h_2 - h_1$; it closes at the wall (fig. 9(a)). On the 2×3 fibre, one count turns rank 2 into rank 1, with valuation $1/200 = h_2 - h_1$ (fig. 9(b)). In the column-binding example the valuation is $1/20$. So the hierarchy of selection reappears as the hierarchy of scales of the probabilistic object's Clifford form.

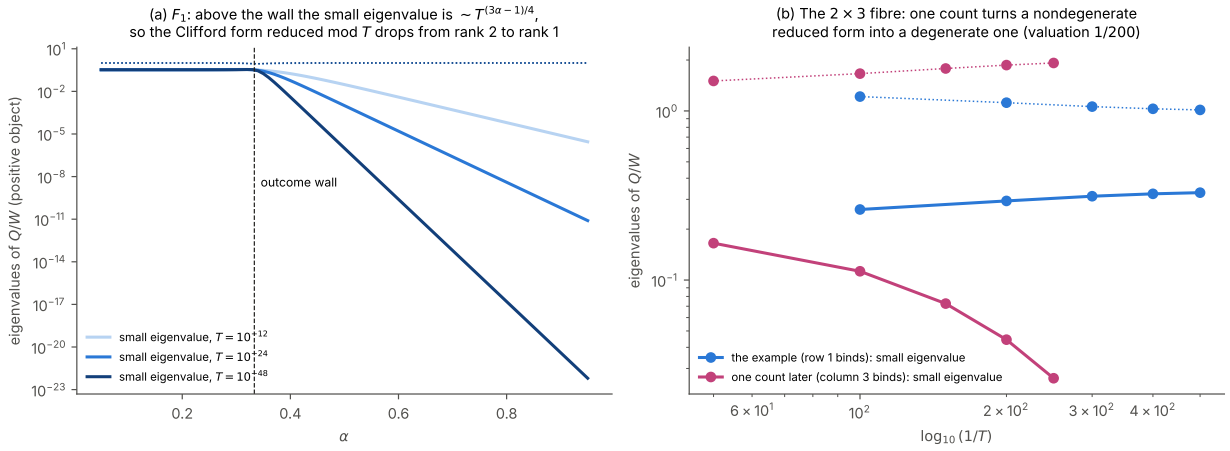


Figure 9: The Clifford shadow. (a) F_1: eigenvalues of Q/W for the positive object (solid: the small one; dotted: the large one) at three values of T. Above the wall the small eigenvalue tends to 0 like $T^{(3\alpha-1)/4}$, so the form reduced mod T has rank 1. (b) The 2×3 fibre, before and after one count. After the move the small eigenvalue decays like $T^{1/200}$.

13.5 Comparing the two sides of a wall

Three mechanisms could relate the two sides of a wall: an A_∞ functor, a continuation map, or a mutation. Here is what each does.

Theorem 34 (an A_∞ equivalence across the F_1 wall)**.** *On F_1, for every $T \in (0,1)$ and every $\alpha \in (0,1)$, the four critical points are distinct and nondegenerate. So they form four continuous branches in α, and the semisimple categories $\bigoplus_c \mathrm{Cl}(Q_c(\alpha))$ over Λ are related by algebra isomorphisms that depend continuously on α. For the positive object these are real: $A^\top Q(\alpha')A = Q(\alpha)$. In particular they are A_∞ equivalences across $\alpha = 1/3$. What changes at the wall is the* support*: which fibre carries which object. That is the merging of fibres in fig. 8.*

Proof. The discriminant $-27T^2 - 256T^{3-3\alpha}$ never vanishes (example 14(d)), so the critical points never collide, and nondegeneracy follows. A nondegenerate complex symmetric form has a symmetric square root, which gives the isomorphisms. MO7 checks the path $\alpha \in [1/5, 3/5]$ at $T = 10^{-24}$. The A-side reading is that of propositions 31 and 32, so the statement inherits their status. □

Continuation maps compare Floer theory of Hamiltonian-isotopic Lagrangians in one symplectic manifold. Here the evidence moves the Kähler class itself, and distinct fibres are not Hamiltonian isotopic. So there is no continuation map in the strict sense: the family comparison of theorem 34 takes its place.

Mutations belong to the category of thimbles, the Fukaya–Seidel category of the Landau–Ginzburg model. For a phase θ of $\hbar$, a distinguished basis of thimbles changes by a mutation when two critical values align along θ [74].

Proposition 35 (Stokes walls crowd the outcome wall)**.** *On F_1, the Stokes walls for the phases $\theta = 0.3$, 1 and $\pi/2$ lie within $4/\log(1/T)$ of $\alpha = 1/3$, for $T = 10^{-12}, 10^{-24}, 10^{-48}$, and $\log(1/T)\,(\alpha_\theta - 1/3)$ does not depend on T. Away from the window there are none. The mechanism is that $|W|$ is $\sim T^{\alpha}$ for the signed object and $\sim T^{1/3}$ for the barycentre objects. Outside a window of width $O(1/\log(1/T))$ one of them dominates, and the phases of the differences are frozen; inside it their ratio sweeps $(0, \infty)$, and every phase is crossed. For real $\hbar$ ($\theta = 0$) the positive thimble never mutates: it is the positive orthant for every λ (proposition 7).*

This is a computation, MO8 and MO9, with the mechanism sketched; we have not proved the window for general configurations.

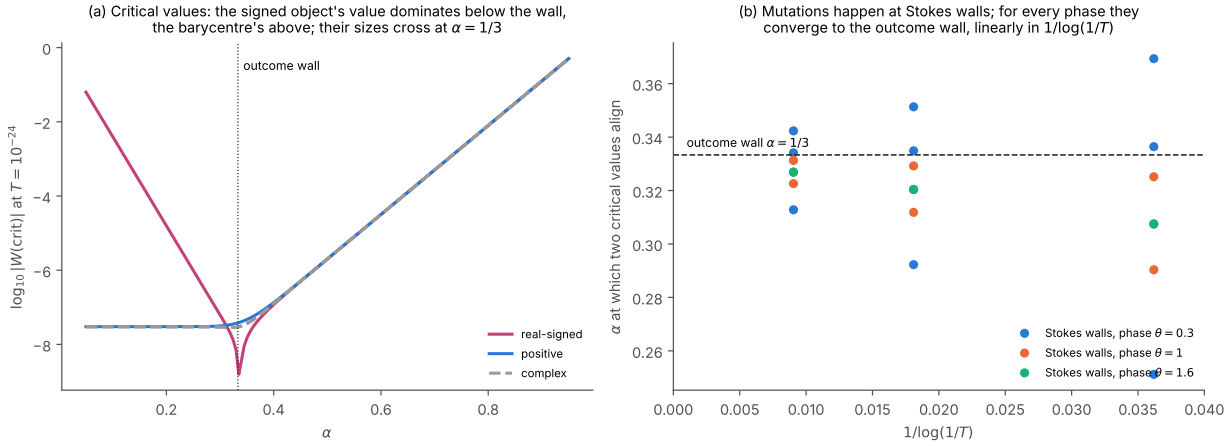


Figure 10: Mutations and the outcome wall on F_1. (a) The sizes of the four critical values at $T = 10^{-24}$: the signed object's value dominates below the wall, and the barycentre objects' values above it. (b) The Stokes walls for three phases, at three values of T, lie on lines through $(0, 1/3)$ in the variable $1/\log(1/T)$.

13.6 What sees the wall

invariant	across the outcome wall	status
which fibres carry objects	changes ($2 \to 1$; $3 \to 2$)	cited [32]; MO1–2
the category over Λ, $\bigoplus \mathrm{Cl}(Q_c)$	unchanged, up to A_∞ equivalence	theorem 34, on cited m_2
higher products	none survive	proposition 32, on cited m_2
the positive object's Clifford form mod T	rank drops ($2 \to 1$)	theorem 33, new
the thimble basis at phase $\theta \neq 0$	mutates within $O(1/\log(1/T))$ of the wall	proposition 35, computed
the positive thimble, real $\hbar$	never changes	proposition 7, proved

So a data-induced outcome wall is not a wall of the Fukaya category over Λ. It is a wall of its *integral* structure for the probabilistic object, and the place where, at finite T, the thimble bases mutate. The full A_∞ category over Λ_0, and split generation in the non-monotone case, are left open.

13.7 The column-binding example, classified

The column-binding fibre of example 30 is the first example whose objects are not all over the selected fibre, and whose holonomies are not all of modulus one. Stage IV (section 13.8) shows why: the example lies on a level-two leximin wall, and both features belong to the wall. We classify its five critical points exactly, then compute the morphisms between the resulting objects. Wall-crossing from this example is deferred.

The Floer convention. We use the convention of [46]. An object is a pair $(L(u), b)$ with $b \in H^1(L(u); \Lambda_0)$, whose holonomy coordinates are $y_i = T^{u_i} e^{b_i}$. Their leading coefficients $e^{b_{i,0}}$ are arbitrary nonzero complex numbers, not only those of modulus one, so local systems with non-unit leading modulus are included. On a compact toric Fano manifold the potential is W [32]. The pair has non-vanishing Floer cohomology iff y is a critical point of W, and $m_0(L(u), b) = W(y) \cdot 1$ [46]. For the Clifford description of the endomorphism ring we use the formula $Q = \operatorname{Hess} W$ at y [33, Thm. 5.6]. Cho states it in the setting of [32], and its use with non-unit leading coefficients is the one place where our citation goes beyond the original setting.

With $\tau = T^{1/10}$, the potential of this fibre is $W = y_1 + y_2 + \tau^4/(y_1 y_2) + \tau^3/y_2 + y_1 y_2/\tau^2$ (cells $11, 12, 13, 22, 23$).

Proposition 36 (the five critical points)**.** *(1)* Reduction. $y_1 \partial_1 W - y_2 \partial_2 W = y_1 - y_2 + \tau^3/y_2$. *So* $y_1 = s - \tau^3/s$ *with* $s = y_2$*, and the remaining equation is exactly*

$$q(s) = s^5 + \tau^2 s^4 - 2\tau^3 s^3 - 2\tau^5 s^2 + \tau^8 = 0.$$

Since $q(0) \neq 0$ *and* q *has no common root with* $s^2 - \tau^3$*, the critical points correspond one to one to the roots of* q*.*

(2) Newton polygon. *The lower hull of* q *has vertices* $(0, 8)$*,* $(3, 3)$*,* $(5, 0)$*. The edge of slope* $-5/3$*, with edge polynomial* $1 - 2c^3$*, gives three roots* $s \approx 2^{-1/3} \omega \, \tau^{5/3}$ *with* $\omega^3 = 1$*. The edge of slope* $-3/2$*, with edge polynomial* $c^2 - 2$*, gives two roots* $s \approx \pm\sqrt{2}\, \tau^{3/2}$*.*

(3) Distinct and nondegenerate for every $T > 0$. $\operatorname{disc}_s q = -\tau^{31}(256\tau^2 - 837\tau + 3456)$*, and the resultant of* q *with the Hessian determinant is* $\tau^{54}(256\tau^2 - 837\tau + 3456)$*. The quadratic has discriminant* $-2838375 < 0$*, so neither vanishes for* $\tau > 0$*.*

(4) Distinct critical values for every $T \in (0, 1)$. $W = 2s + \tau^4/(s^2 - \tau^3) + (s^2 - \tau^3)/\tau^2$ *at a critical point. The polynomial whose roots are the five critical values has discriminant* $-\tau^{225}(\tau - 1)^2(256\tau^2 - 837\tau + 3456)^3$*, which vanishes on* $(0, \infty)$ *only at* $T = 1$*.*

Proof. Each statement is an exact symbolic computation (CB1–CB4). In (2) the edge polynomials are read off the coefficients on each edge: $\tau^8 - 2\tau^3 s^3$ and $-2\tau^3 s^3 + s^5$. □

So there are five critical points, two fibres and five objects:

object	fibre	leading $(y_1, y_2)/T^u$	leading W	leading $\det(Q/W)$
P positive	$(\frac{3}{20}, \frac{3}{20})$	$(\frac{\sqrt{2}}{2},\ \sqrt{2})$	$2\tau + 2\sqrt{2}\,\tau^{3/2}$	$\sqrt{2}\,\tau^{1/2}$
N real $(-,-)$	$(\frac{3}{20}, \frac{3}{20})$	$(-\frac{\sqrt{2}}{2},\ -\sqrt{2})$	$2\tau - 2\sqrt{2}\,\tau^{3/2}$	$-\sqrt{2}\,\tau^{1/2}$
R real $(-,+)$	$(\frac{2}{15}, \frac{1}{6})$	$(-2^{1/3},\ 2^{-1/3})$	$-2\tau + 2^{2/3}\tau^{5/3}$	$-c\,\tau^{2/3}$
C_ω, $C_{\bar\omega}$ complex	$(\frac{2}{15}, \frac{1}{6})$	$(-2^{1/3}\bar\omega,\ 2^{-1/3}\omega)$	$-2\tau + 2^{2/3}\omega\,\tau^{5/3}$	$-c\,\bar\omega^2\tau^{2/3}$

Here $c = 3 \cdot 2^{-4/3}$, $\omega = e^{\pm 2\pi i/3}$, and $C_{\bar{\omega}}$ is the conjugate of C_ω. CB5 and CB6 check every entry against the roots at $\tau = 10^{-10}, 10^{-20}, 10^{-40}$, with relative errors $1.5 \cdot 10^{-4}$, $6.8 \cdot 10^{-8}$ and $1.5 \cdot 10^{-14}$. The two groups have opposite leading critical value, $\pm 2\tau$. Within a group the values separate at order $\tau^{3/2}$, or at order $\tau^{5/3}$ through the cube roots of unity (fig. 11).

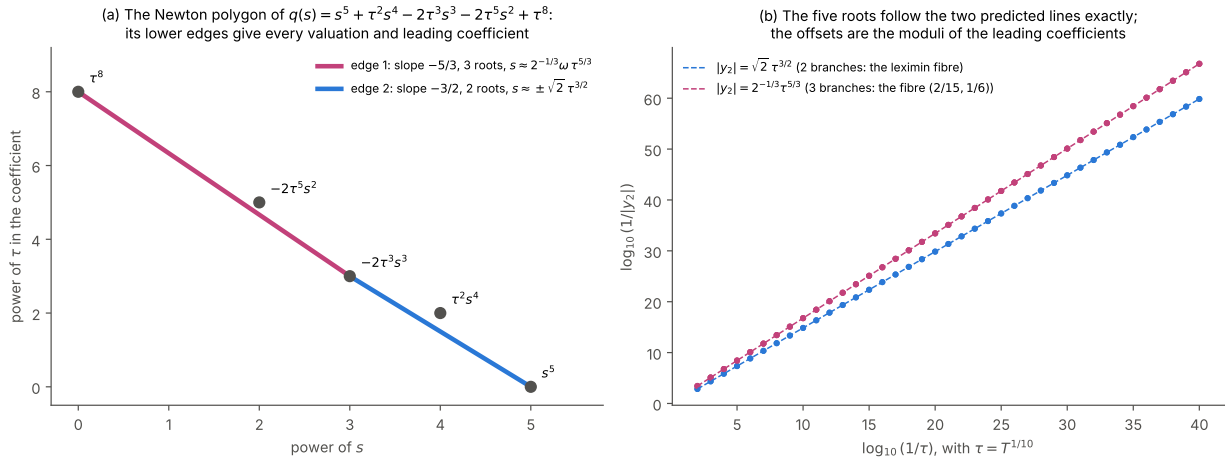


Figure 11: The column-binding example classified. (a) The Newton polygon of the quintic: one edge gives the three branches over $(2/15, 1/6)$, the other the two over the leximin fibre. (b) The five roots against τ follow the two predicted lines; the vertical offsets are the moduli of the leading coefficients.

Remark 37 (the drift explained)**.** At finite T a branch with leading coefficient ρ has apparent valuation $u - \log|\rho|/\log(1/T)$. For P, $\log(\sqrt{2}/2)$ gives the offset $\ln 2/(2L)$ of example 30. For the other three, $\log 2^{\pm 1/3}$ gives $\pm \ln 2/(3L)$. This was the slow "drift" seen there: the limits are exactly $(2/15, 1/6)$. The three sit there because their level-one terms have $y_1 y_2 \approx -\tau^3$. This sign makes the level-two terms (cells 11 and 22) cancel exactly, so the position is set by the next balance, of cell 12 against the constant term: $-2\tau^3 s^3 + \tau^8 \approx 0$. In the row-binding examples every leading coefficient has modulus one. Non-unit moduli appear here because a level of selection cancels.

Theorem 38 (reduction of the Clifford form)**.** *Let c be a nondegenerate critical point with valuation u_c. Let K be its leading set, the k with $\ell_k(u_c)$ minimal, equal to h_c. Write the term values as $c_k = \rho_k T^{h_c}(1 + o(1))$ for $k \in K$, and assume*

(H1) $\sum_{k \in K} \rho_k \neq 0$.

Then Q/W has entries of non-negative valuation and reduces modulo T to $M_c = \sum_K \rho_k v_k v_k^\top / \sum_K \rho_k$. Always $\operatorname{rank} M_c \leq \dim \operatorname{span}\{v_k : k \in K\}$. *Equality holds under*

(H2) the ρ_k, $k \in K$, are real and of one sign,

in particular for the positive object (theorem 33). Without (H2), cancellation can lower the rank.

Proof. By (H1), $W = (\sum_K \rho_k) T^{h_c}(1 + o(1))$, and every c_k has valuation at least h_c. Under (H2), $\sum_K \rho_k v_k v_k^\top$ is a semidefinite Gram matrix, whose rank is that of $\{v_k\}$. CB8 tests the theorem on 23 objects: F_1 at $\alpha = 1/5$ and $3/5$, the 2×3 example, its one-count neighbour, and this example. (H1) holds for all 23, and (H2) for 13. The rank equals the span dimension in all 23, including the 10 without (H2). An example is the signed object of F_1 at $\alpha = 1/5$, where $\rho = (-1, -1, 1)$ and $M_c = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. □

In the column-binding example $K = \{13, 23\}$ for all five objects, with $\rho = (1, 1)$ on the leximin fibre and $(-1, -1)$ on the other. (H2) holds, and $M_c = J = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, of rank 1, for every object. The reduced Clifford algebras are therefore all non-semisimple. The objects are told apart one order further:

$$P, N: \quad \frac{Q}{W} = J \pm \frac{\tau^{1/2}}{2} \operatorname{diag}\Big(\tfrac{\sqrt{2}}{2}, \tfrac{3\sqrt{2}}{2}\Big) + \dots,$$
$$R, C: \quad \frac{Q}{W} = J + 2^{-2/3}\bar{\omega}\,\tau^{1/3} \operatorname{diag}(1, -1) + \dots$$

For P and N the correction is first order in the small direction, so the small eigenvalue is $\pm(\sqrt{2}/2)\tau^{1/2}$, of valuation $1/20$ in T. The form is positive definite for P and indefinite for N. For R and C the first-order correction vanishes in the small direction, which is the cancellation the reduction theorem warns about. The small eigenvalue appears only at second order, $-\frac{3}{2}\,2^{-4/3}\bar{\omega}^2\tau^{2/3}$, of valuation $1/15$.

Proposition 39 (the Hom table)**.** *In the convention above, the five objects have the following morphisms.*

(1) Between the two fibres: none. The fibres are disjoint, so the unperturbed Floer complex is zero; this is a legitimate model, since the pair is transverse.

(2) Within a fibre: none. The critical values are distinct (proposition 36(4)), and morphisms are defined only between objects with equal m_0 [10, 46]. (Their holonomies are also distinct, but the argument does not use this.)

(3) On the diagonal: $\mathrm{Cl}(Q_c) \cong M_2$, with Q_c as above (cited, in the convention stated). No higher products survive (proposition 32).

So the category these objects generate is $\bigoplus_{c=1}^{5} \mathrm{Cl}(Q_c)$ over Λ. Over Λ_0, all five reduce to the rank-one form J. Generation of the Fukaya summands remains conditional, because the example is not monotone [77].

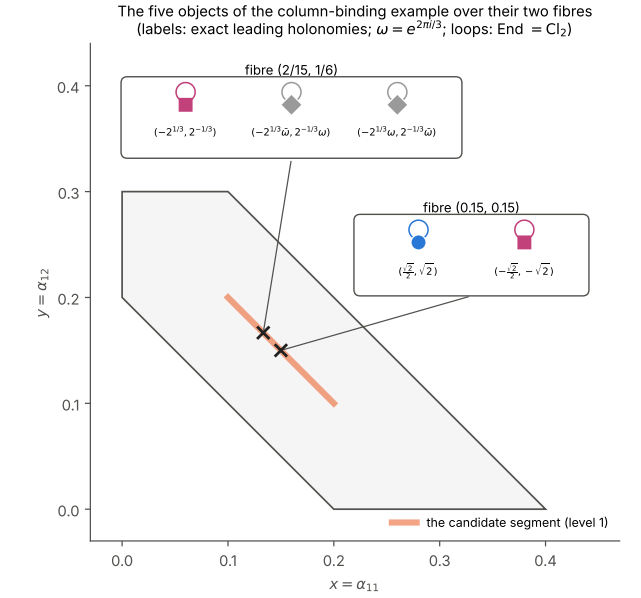
Proof. (1) $L(u) = \mu^{-1}(u)$. (2) and (3) are the cited statements together with proposition 36. CB7 checks the counts, the distinct values and holonomies within each fibre, and that explicit Clifford generators span M_2. □

13.8 Stage IV: a path through the column-binding example

Data so far moved margins by one count (a point) or along one direction of evidence (a line). Here we take a sheet of margins and one path across it, and we keep four questions separate along the path: do critical points collide, do critical values coincide, which fibres carry the objects, and what do the Clifford forms and the Stokes alignments see?

The sheet and the path. Fix $f = (2/5, 3/5)$ and let the column margins vary: $g = (1 - g_2 - g_3,\ g_2,\ g_3)$. Figure 13 shows this sheet with its blow-up walls $f_1 = g(J)$, which bound the Kähler chambers, and its leximin walls. The path is

$$g(\gamma) = \Big(\tfrac{7}{10} - \gamma,\ \tfrac{3}{10},\ \gamma\Big), \qquad \gamma \in \big[\tfrac{3}{20}, \tfrac{11}{40}\big].$$



Hom between the five objects ($c = 3 \cdot 2^{-4/3}$): only the diagonal survives.
First two: the leximin fibre; last three: the fibre (2/15, 1/6)

	P positive	*N* real (−, −)	*R* real (−, +)	C_ω complex	$C_{\bar{\omega}}$ complex
P positive	$\mathrm{Cl}(Q) \cong M_2$, $\det \sim \sqrt{2}\,\tau^{1/2}$, reduced: rank 1	0, same fibre, distinct m_0	0, disjoint fibres	0, disjoint fibres	0, disjoint fibres
N real (−, −)	0, same fibre, distinct m_0	$\mathrm{Cl}(Q) \cong M_2$, $\det \sim -\sqrt{2}\,\tau^{1/2}$, reduced: rank 1	0, disjoint fibres	0, disjoint fibres	0, disjoint fibres
R real (−, +)	0, disjoint fibres	0, disjoint fibres	$\mathrm{Cl}(Q) \cong M_2$, $\det \sim -c\,\tau^{2/3}$, reduced: rank 1	0, same fibre, distinct m_0	0, same fibre, distinct m_0
C_ω complex	0, disjoint fibres	0, disjoint fibres	0, same fibre, distinct m_0	$\mathrm{Cl}(Q) \cong M_2$, $\det \sim -c\,\bar{\omega}^2\tau^{2/3}$, reduced: rank 1	0, same fibre, distinct m_0
$C_{\bar{\omega}}$ complex	0, disjoint fibres	0, disjoint fibres	0, same fibre, distinct m_0	0, same fibre, distinct m_0	$\mathrm{Cl}(Q) \cong M_2$, $\det \sim -c\,\omega^2\tau^{2/3}$, reduced: rank 1

Figure 12: Top: the five objects over their two fibres, with exact leading holonomies; the leximin fibre carries P and N, the fibre $(2/15, 1/6)$ carries R, C_ω and $C_{\bar{\omega}}$. Bottom: the Hom table. Grey cells vanish because the fibres are disjoint, orange cells because the values of m_0 differ. The diagonal is $\mathrm{Cl}(Q_c) \cong M_2$, and every one reduces to the rank-one form J modulo T.

It stays in the pentagon chamber (facets $11, 12, 13, 22, 23$; no blow-up wall meets $\gamma \in (1/10, 3/10)$), and only the exponent of cell 23 moves. With $A = T^{2/5}$, $B = T^{3/10}$ and $C = T^{\gamma - 2/5}$,

$$W_\gamma = y_1 + y_2 + \frac{A}{y_1 y_2} + \frac{B}{y_2} + C\, y_1 y_2.$$

The column-binding example of section 13.7 is $\gamma = 1/5$.

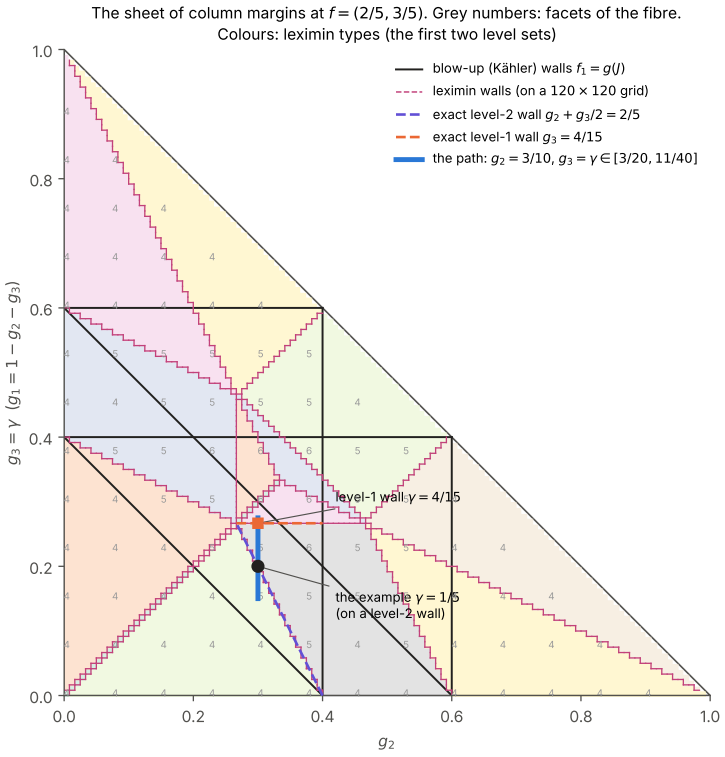


Figure 13: The sheet of column margins at $f = (2/5, 3/5)$. Black lines are blow-up walls. The stepped rose lines are leximin walls computed on a 120×120 grid (colours: leximin types), and the dashed lines are the two walls the path crosses, drawn exactly: $g_2 + g_3/2 = 2/5$ (level two) and $g_3 = 4/15$ (level one). Grey numbers count the facets of the fibre. The path (blue) runs inside one Kähler chamber, the pentagon, and crosses two leximin walls: the column-binding example (black dot) sits on a level-two wall, and the level-one wall (orange square) is at $\gamma = 4/15$.

Proposition 40 (the path, exactly)**.** *(1)* Two leximin walls. *For* $\gamma < 1/5$ *the levels are* $\gamma/2$ *(column* 3*),* $3/20$ *(column* 2*) and* $1/4 - \gamma/2$ *(cell* 11*). For* $1/5 < \gamma < 4/15$ *they are* $\gamma/2$ *and* $(4/5 - \gamma)/4$*, where the residual of row* 1 *binds. For* $\gamma > 4/15$ *row* 1 *binds at once, at* $2/15$*. At* $\gamma = 1/5$ *the cells* $11, 12, 22$ *tie at the second level, and at* $\gamma = 4/15$ *the cells* $11, 12, 13, 23$ *tie at the first. So* the column-binding example lies on a level-two leximin wall.

(2) Reduction. *The critical points are* $y_1 = (s^2 - B)/s$*,* $y_2 = s$*, where*

$$(s^2 - B)^2(1 + Cs) = As. \tag{1}$$

At $\gamma = 1/5$*, where* $C = A/B^2$*, this is the quintic of proposition 36.*

(3) Where the objects sit. *The Newton polygon of* (1) *changes exactly at the two leximin walls, and gives*

γ	*over the leximin fibre*	*elsewhere*
$(3/20, 1/5)$	4, *at* $(1/4 - \gamma/2, 3/20)$*:* $s = \pm T^{3/20}$, $y_1 = \pm T^{1/4-\gamma/2}$	R, *at* $(\gamma - 1/10, 2/5 - \gamma)$, *signs* $(+,-)$
$1/5$	2, *at* $(3/20, 3/20)$	3, *at* $(2/15, 1/6)$ *(proposition 36)*
$(1/5, 4/15)$	4, *at* $\frac{4/5-\gamma}{4}(1,1)$*:* $y_1 \approx y_2 = i^k T^{(4/5-\gamma)/4}$	R, *at* $(1/10, 1/5)$, *signs* $(-,+)$
$(4/15, 11/40]$	3, *at* $(2/15, 2/15)$*:* $y_1 \approx y_2 = \omega^k T^{2/15}$	N, *at* $(2/5 - \gamma)(1,1)$, *signs* $(-,-)$*; and* R

In every regime the leximin fibre carries the multiple root, as theorem 27 requires of the positive object.

Proof. (1) By theorem 23 and the recursion of definition 26. After column 3 binds, the segment has $a_{11} + a_{12} = r = 2/5 - \gamma/2$, and the facet cells on it are $11, 12, 22$. The residual row binds if $r < g_2 = 3/10$, that is, if $\gamma > 1/5$, and column 2 binds otherwise; row 1 binds at the first level once $\gamma/2 > 2/15$. (2) As in proposition 36: $y_1\partial_1 W - y_2\partial_2 W = y_1 - y_2 + B/y_2$. (3) Three dominant balances in (1) decide everything. Whether Cs or 1 dominates on the leximin roots switches at $\gamma = 4/15$. Whether C or A/B^2 dominates on the isolated root switches at $\gamma = 1/5$. And below $1/5$ the leximin roots have $s^2 = B(1 + o(1))$: writing $s^2 - B = \delta$ gives $\delta^2 \approx A/C$, so $y_1 = \delta/s$ has valuation $1/4 - \gamma/2$. PP2–PP5 check the levels, the Newton polygons at 10 rational γ, and the valuations and phases at $L = 200, 400$. □

So the peculiarities found in section 13.7 belong to the wall, not to column binding. The grouping $2+3$, the fibre $(2/15, 1/6)$ and the leading moduli $2^{\pm 1/3}$ occur only at $\gamma = 1/5$. On either side of it, four objects sit over the leximin fibre with leading moduli one. This is what the remark on the drift meant by a level of selection that cancels: there the linear coefficient $B^2C - A$ of (1) vanishes identically.

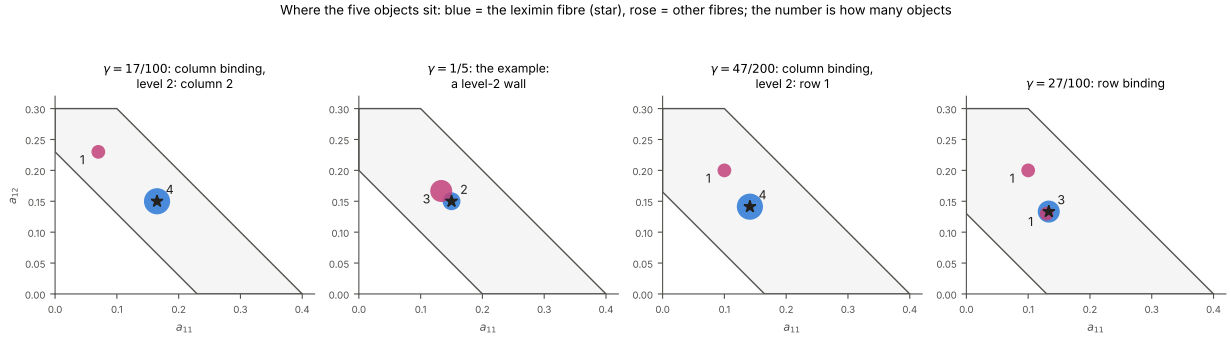


Figure 14: The five objects over the pentagon at four points of the path; the marker size and the number give how many objects sit over each fibre. Away from the wall $\gamma = 1/5$ the leximin fibre (blue, star) carries four objects, then three after the level-one wall.

Proposition 41 (what crosses each wall)**.** *Along the path, at fixed small* T*:*

(1) The level-two wall forces a collision. *At* $\gamma = 0.15$ *and* 0.18 *all five critical points are real; at* $\gamma \geq 0.22$*, three are. The coefficients of* (1) *are real and its leading coefficient* C *never*

vanishes, so the number of real roots can change only through a double root. Numerically the discriminant has exactly one zero on the path, at $\gamma_c(T)$, for $T = e^{-50}, e^{-100}, e^{-200}, e^{-400}$, with $\gamma_c = 0.19235,\ 0.19918,\ 0.199985,\ 0.19999999$: it approaches the wall from below, faster than $1/\log(1/T)$. There the two objects of signs $(+,-)$ meet and leave as the complex pair $I_\pm$. At that one parameter a critical point is degenerate.

(2) The level-one wall does not. *The discriminant has no zero on* $(0.21, 0.275]$. *The pair* $I_\pm$ *continues into* $C_\omega, C_{\bar\omega}$, *and* N *leaves the leximin fibre along the diagonal.*

(3) Critical values *never coincide on the path away from the collision, and the order of the real values is constant on each side (PP7). So outside* $\gamma_c(T)$ *the hypotheses of proposition 31 hold at every point: every off-diagonal Hom vanishes, and each endomorphism algebra is* $\mathrm{Cl}(Q_c) \cong M_2(\Lambda)$.

(4) The Clifford shadow sees both walls *(fig. 15(b)). The positive object's scale is* $h_2 - h_1 = 3/20 - \gamma/2$ *below* $1/5$ *and* $1/5 - 3\gamma/4$ *above it, so the scale has a kink at the level-two wall and closes at the level-one wall, where the reduced rank jumps from* 1 *to* 2 *(theorem 33). The four objects over the leximin fibre share this scale;* R *has rank* 2 *throughout.*

Proof. (1) Real-root counting, together with PP5 for the counts and PP6 for the zero. (2) PP6. (3) PP7, and proposition 31(2)–(3). (4) PP8 compares the slopes at $L = 400 \to 800$ with $h_2 - h_1$ from proposition 40(1). □

What (3) does and does not give needs saying plainly. At every parameter away from $\gamma_c(T)$ the five objects generate $\bigoplus_{c=1}^{5} \mathrm{Cl}(Q_c)$ over Λ, on both sides of the level-one wall. All such categories are equivalent over Λ, so this pointwise statement says little. It is not an A_∞ functor along the path, and non-collision alone would not give one. Over Λ_0 the positive object changes across the level-one wall: its reduced Clifford form goes from rank 1 to rank 2. Across the level-two wall the path meets a degenerate critical point at $\gamma_c(T)$, where the cited theorems do not apply.

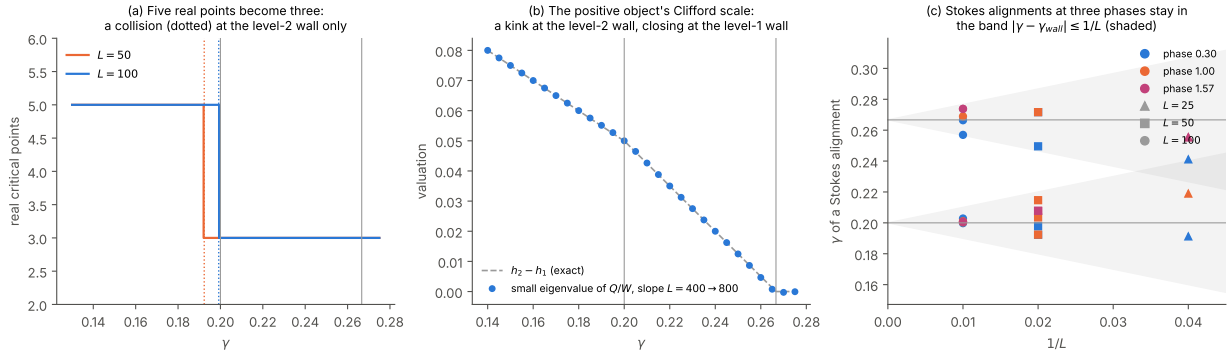


Figure 15: Along the path. (a) The number of real critical points at $L = 50$ and 100; the dotted lines are the zeros of the discriminant, one per L, at the level-two wall. (b) The small eigenvalue of Q/W at the positive object, against $h_2 - h_1$. (c) The positions of Stokes alignments at three phases and three values of L; the shaded bands are $|\gamma - \text{wall}| \le 1/L$.

Stokes alignments, separately. For the phases $\theta = 0.3$, 1.0 and $\pi/2$, and $L = 25, 50, 100$, every parameter where two critical values align, $\operatorname{Im} e^{-i\theta}(W_j - W_i) = 0$, lies within $1.5/L$ of one of the

two leximin walls, and at $L = 100$ within $1/L$; alignments occur near both walls (PP9; fig. 15(c)). This is the behaviour of proposition 35 on F_1, now near two walls at once. We do not claim it for every phase, nor that every leximin wall has such alignments nearby; near the level-two wall some alignments involve the pair born at the collision.

13.9 Composing couplings: Markov gluing and selection

The objects and morphisms of section 13 are cited from Floer theory. This subsection asks what composition the probabilistic side has of its own, and whether it has higher structure. It does not produce an A_∞ category. It separates a strict composition (the control) from one that is genuinely non-associative, and computes, exactly, the configuration that the associahedron K_4 then carries.

Definition 42 (couplings and two compositions)**.** An object is a finite set X with a full-support law p; a morphism $(X, p) \to (Y, q)$ is a coupling $\gamma \in \Pi(p, q)$. For a chain $\gamma_1, \dots, \gamma_k$ of consecutive couplings, its *fibre* is the polytope of multi-couplings on $X_0 \times \cdots \times X_k$ with the consecutive pair marginals $\gamma_1, \dots, \gamma_k$.

(1) *Markov gluing* composes by conditional independence over the shared marginal: $\gamma_2 \circ \gamma_1 = \gamma_1 D_q^{-1} \gamma_2$, with $D_q = \mathrm{diag}(q)$.

(2) *Selection* composes a block at once: $\sigma(\gamma_1, \dots, \gamma_k)$ is the marginal on $X_0 \times X_k$ of the facet-leximin point of the fibre. By theorem 27, applied to the fibre's potential, this is the $T \to 0$ limit of the positive critical point.

Proposition 43 (the control: Markov gluing)**.** *(1) Markov gluing is associative and unital, with the diagonal coupling D_p as identity.*

(2) The Markov multi-coupling of a chain is the unique point of largest entropy in its fibre, and its (X_0, X_k) marginal is the iterated composite.

(3) Every fibre is a nonempty convex polytope. So any assignment of points of $\Pi(p_0, p_n)$ to the faces of K_n extends to a continuous map $K_n \to \Pi(p_0, p_n)$ (piecewise linear on the barycentric subdivision) whose restriction to each face is determined by the points of that face. For Markov gluing every face receives the same point, so such fillers force no operation beyond the strict composition.

Proof. (1) Matrix multiplication, $(\gamma_1 D^{-1} \gamma_2) D^{-1} \gamma_3 = \gamma_1 D^{-1} (\gamma_2 D^{-1} \gamma_3)$, and $D_p \, D_p^{-1} \gamma = \gamma$. (2) By the chain rule, $H(X_0, \dots, X_k) \le H(X_0) + \sum_i H(X_i \mid X_{i-1})$, and the right side is fixed by the pair marginals; equality holds exactly when the chain is Markov, and entropy is strictly concave. (3) The Markov point lies in the fibre, and the target is convex. AS1 checks (1) and (2) in exact rationals, and that Markov gluing gives one composite on every face of K_4. □

This is the categorical-probability picture of Markov kernels [47], restricted to couplings. Part (2) places it in this note: the Markov composite is the Birch point of the chain model's fibre. Selection picks a different point of the same fibre, and that changes the composition law.

Faces as dissections. A face of K_n is a partial bracketing of n morphisms, equivalently a dissection of the $(n+1)$-gon with vertices $0, \dots, n$, whose sides $(i, i+1)$ are the γ_{i+1} and whose root side $(0, n)$ is the output [78]; combinatorially K_n is the secondary polytope of the $(n+1)$-gon [52]. The composite along a dissection is computed cell by cell: a cell with vertices $i_0 < \cdots < i_k$ is one selection in its own $(k+1)$-way table, whose consecutive margins are its sides, each an input

or the composite of a smaller cell. Vertices are full bracketings (binary selections), and the top cell is one selection in the $(n+1)$-way table. This is what makes face compatibility meaningful: the composite of a face is computed from the composites of the blocks it contains, with their own margins. AS2 checks the face counts $3, 11, 45$ for $n = 3, 4, 5$.

Proposition 44 (selection: what holds and what fails)**.** *Selection is unital (the diagonal coupling on either side returns the other) and commutes with transposition, $\sigma(\gamma_2^\top, \gamma_1^\top) = \sigma(\gamma_1, \gamma_2)^\top$. It is not associative: on* 30 *random rational chains of three couplings the two bracketings differ in* 26*.*

Proof. With a diagonal coupling the fibre is a single point. Transposition reverses the chain and preserves the fibre and its facets. The rest is AS5–AS6, in exact rationals. □

Theorem 45 (the K_4 configuration, exactly)**.** *Take binary laws $p_0, \dots, p_4$ with first entries $2/5, 1/3, 3/7, 1/4, 5/11$, and the couplings $\gamma_1, \dots, \gamma_4$ with $a_{00} = 13/60, 7/30, 1/80, 3/80$. The eleven faces of K_4 receive the endpoint couplings with these entries a_{00}:*

vertex (bracketing)	a_{00}	*edge to the next vertex (block)*	a_{00}
((12)3)4	173/960	[0..3]	997/5760
(1(23))4	61/384	[1..3]	197/1152
1((23)4)	61/320	[1..4]	14941/73920
1(2(34))	38/165	[2..4]	1249/5280
(12)(34)	93/385	[0..2]	15403/73920
top: one selection in the 5*-way table*	28573/147840	*Markov, every face*	2371469/12672000

A block $[i..j]$ is one selection in the table over $X_i, \dots, X_j$. The edge in each row joins that row's vertex to the next one, cyclically. All eleven are distinct, each edge composite lies strictly between its two vertices, and the top lies strictly inside their range. Markov gluing gives the one value in the last cell on every face.

Proof. Each composite is computed level by level by linear programming and the leximin point is solved exactly from the fixed cells and the margins (AS3). Every selection used here then carries an exact certificate (SX2): feasibility; the forced zeros and the dimension of the fibre; a witness face of dimension one less for each facet cell, and for every other cell an exact lower bound or a certified extra zero; for each level a balanced law $q > 0$ on the level set with $\sum_c q_c e_c$ in the row space of the margins and the earlier fixed cells, which bounds the level and forces the whole level set; and strictly increasing levels. So the composites are certified, not only computed. □

The same holds with more states, and then the configuration can be seen in the plane. For $p_0 = (2/5, 3/5)$ and $p_4 = (1/4, 1/3, 5/12)$, with ternary p_1, p_3 (fig. 16(b)), the eleven composites are distinct points of the two-dimensional polytope $\Pi(p_0, p_4)$, the top lies strictly inside the convex hull of the five bracketings, and the Markov composite lies outside it (AS4; exact orientation tests and certified selections, SX2–SX3). The image of the pentagon crosses itself: the edge through block [0..3] meets the one through [2..4], and so does the edge through [1..3]. So the map $K_4 \to \Pi(p_0, p_4)$ of proposition 43(3), built from these composites, is a configuration and not an embedding, and since the target is convex it carries no homotopy type of its own. Its content is where the eleven points sit.

Finite T. Replace the facet-leximin point by the positive critical point at $T = e^{-L}$ on each fibre. On the first three couplings of theorem 45, the two bracketings differ by $-41/960$ at $L = 25, 50, 100, 200$, to 10^{-9}: the binary composites already sit at their limits, and the top approaches its limit faster than $1/L$ (AS7). So the bracketings have no common leading term. The failure of associativity is visible at the tropical level, not only at higher order in T.

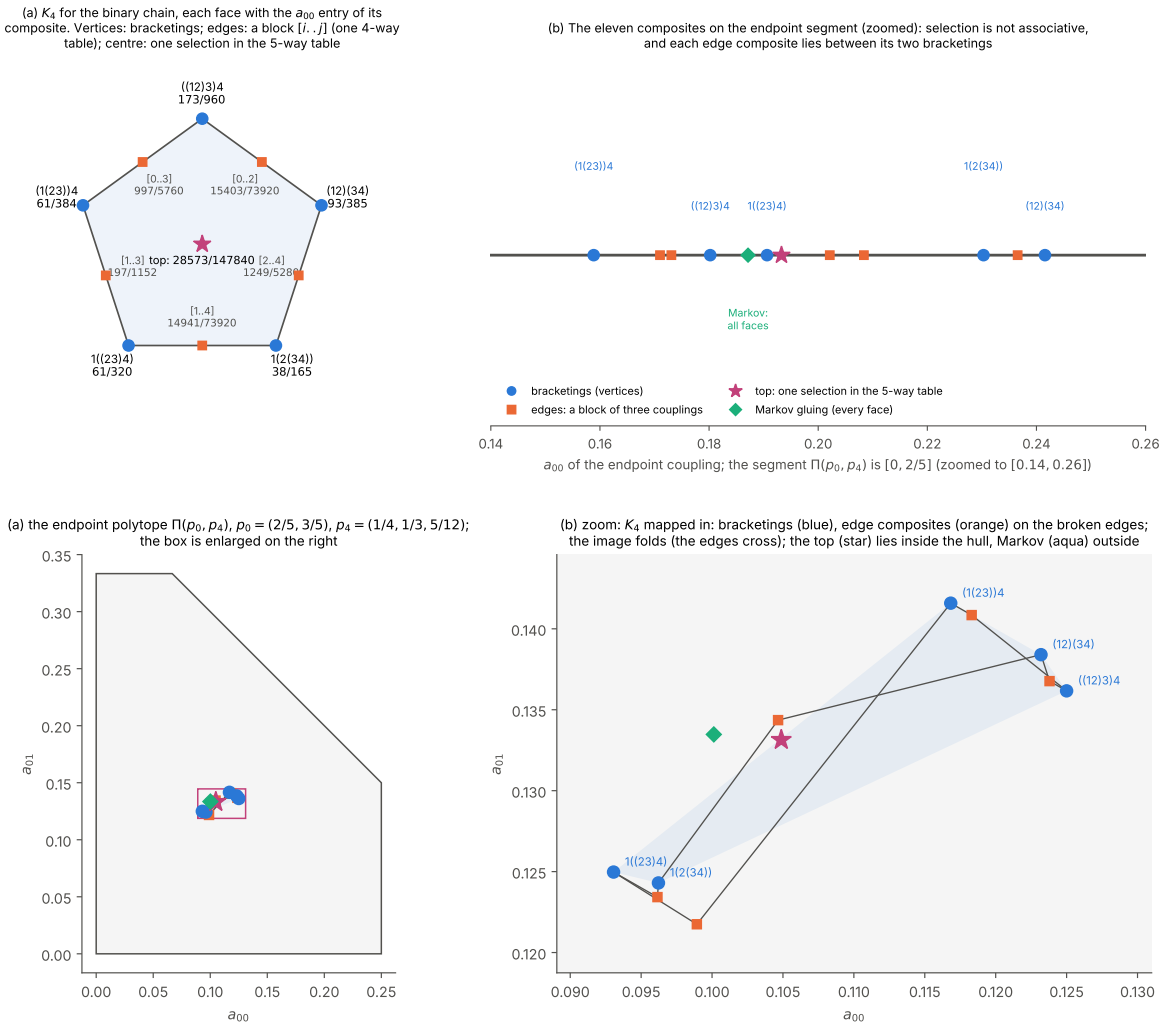


Figure 16: Composition by selection. Top: (a) K_4 for the binary chain of theorem 45, each face labelled with its composite; (b) the eleven composites on the endpoint segment, with the single Markov composite. Bottom: the chain with ternary intermediates; (a) the endpoint polytope; (b) K_4 mapped into it. The five bracketings (blue) and the five edge composites (orange) trace a folded pentagon; the top (star) is inside the hull, the Markov composite (aqua) outside.

Remark 46 (what this does and does not give)**.** An earlier draft conjectured that the associator might have a positive valuation, equal to a gap $h_j - h_1$ between selection levels, which would make it a candidate for a filtered m_3 that cannot be removed. That form is refuted here: the associator is of order one. Two questions remain open. On the locus where the tropical bracketings agree, does the finite-T associator have a T-adic valuation tied to the level gaps? And, before any of this can be called an A_∞ operation, one needs chain-level morphism complexes and a differential, with m_3 a specified homotopy between the two composites that satisfies the next coherence relation. Nothing here constructs those. What is established is the vertex and face data that any such structure on selection would have to interpolate.

13.10 Selection: a no-go result, continuity, and binary chains

Can the associator of section 13.9 be the m_3 of an A_∞ category whose morphism complexes are chains on the coupling polytopes? It can be, formally, and the first result says why that would carry no invariant. The rest of the subsection studies selection itself, where the information is.

Proposition 47 (no-go for contractible morphism spaces)**.** *Let $\mathcal{C}$ be an uncurved A_∞ category over $\mathbb{Q}$, $\mathbb{Z}$- or $\mathbb{Z}/2$-graded, whose objects are finite probability spaces and whose morphism complexes have $H^*(\hom(P,Q)) \cong H_*(\Pi(p,q);\mathbb{Q})$, for instance singular chains. Suppose m_2 induces on H^0 the composition of point classes, $[x]\cdot[y] = [\text{any composite}]$, with the class of the diagonal coupling as unit. Then:*

(1) the homology category has $\hom = \mathbb{Q}$ *in degree* 0 *for every pair of objects, composition* $1\cdot 1 = 1$*, and every two objects are isomorphic;*

(2) $\mathcal{C}$ is quasi-equivalent to the full subcategory on one object, and that is quasi-isomorphic to $\mathbb{Q}$ with $m_n = 0$ for $n \geq 3$.

So no choice of $m_3, m_4, \dots$ on such complexes is an invariant up to quasi-equivalence. In particular the associator of selection is always a boundary there: the two bracketed composites are points of one convex polytope, and the segment between them is a 1*-chain whose boundary is their difference. The proposition says nothing about categories built from coupling data in other ways.*

Proof. (1) $\Pi(p,q)$ is nonempty and convex, so $H_* = \mathbb{Q}$ in degree 0, spanned by the class of any point, and any composite of points is a point. Every class $1 \in \hom(P,Q)$ has the inverse $1 \in \hom(Q,P)$. (2) A full subcategory meeting every isomorphism class of the homology category is quasi-equivalent to the whole. Its one object has cohomological endomorphisms $\mathbb{Q}\cdot 1$; a cohomologically unital A_∞ algebra is quasi-isomorphic to a strictly unital minimal one [74], whose underlying space is $\mathbb{Q}\cdot e$, and strict unitality gives $m_n(e,\dots,e) = 0$ for $n \geq 3$. This holds for either grading. □

The exact data of theorem 45 are therefore data of the selection operation, not invariants of such a category. That redirects the work: to understand selection, and to look for higher structure where the morphism spaces are not contractible (section 15).

Theorem 48 (binary selection is Fréchet-midpoint gluing)**.** *Let X_0 and X_2 be binary and X_1 any finite set, with couplings $\gamma_1 \in \Pi(p_0,p_1)$ and $\gamma_2 \in \Pi(p_1,p_2)$. For each $b \in X_1$ put $r_b = \gamma_1(0,b)$, $s_b = \gamma_2(b,0)$ and $q_b = p_1(b)$. Then*

$$\sigma(\gamma_1,\gamma_2)_{00} = \sum_b \frac{L_b + U_b}{2}, \qquad L_b = \max(0,\, r_b + s_b - q_b), \quad U_b = \min(r_b, s_b),$$

the sum over b of the midpoints of the Fréchet–Hoeffding intervals of the slices. Markov gluing gives $\sum_b r_b s_b / q_b$ instead.

Proof. The pair-margin constraints involve one value of the middle variable at a time, so the fibre is the product over b of the transportation polytopes of the slices $X_0 \times \{b\} \times X_2$, with margins $(r_b, q_b - r_b)$ and $(s_b, q_b - s_b)$. With X_0, X_2 binary each slice is the segment $x_{0b0} \in [L_b, U_b]$: the other three cells are $r_b - x_{0b0}$, $s_b - x_{0b0}$ and $q_b - r_b - s_b + x_{0b0}$, and the Fréchet–Hoeffding bounds are where one of the four vanishes. A face of a product is a product of faces, so the facets of the fibre are the facets of single slices, and the facet functions of slice b are $x_{0b0} - L_b$ and $U_b - x_{0b0}$, the cells that vanish at its ends (several cells when the slice is tied, all with the same function). A degenerate slice, $L_b = U_b$, is a point and contributes nothing.

Now run the recursion of definition 26. Each facet function depends on one slice only, and the feasible set is a product. So at level j the maximum of the least free function is the least, over the slices with free functions left, of each slice's own next level; the slices attaining it fix their functions, and the others are untouched. By induction the recursion fixes, in each slice, exactly the levels of that slice's own recursion, and the selected point is the product of the slices' selected points. On a segment with facet functions $x - L$ and $U - x$ that point is the midpoint, at level $(U - L)/2$. SB1 checks the formula against the leximin computation on 60 chains, including ternary middles and tied slices. □

So selection glues each slice at the midpoint between its comonotone and countermonotone couplings, where Markov gluing takes the product $r_b s_b / q_b$, which also lies in $[L_b, U_b]$. The two agree exactly when the product sits at the midpoint of every slice. Two consequences follow at once. Since L_b and U_b are a maximum and a minimum of affine functions of the inputs, binary selection is continuous and piecewise linear on the whole product of the coupling polytopes, with no exceptional walls. And on uniform margins it has a closed form.

Corollary 49 (uniform margins: correlations compose by a signed minimum)**.** *If $p_0 = p_1 = p_2 = (1/2, 1/2)$, write a coupling by its correlation $\rho = 4\gamma_{00} - 1$. Then*

$$\rho_{\text{Markov}} = \rho_1 \rho_2, \qquad \rho_{\text{sel}} = \operatorname{sign}(\rho_1 \rho_2)\, \min(|\rho_1|, |\rho_2|).$$

In $v = -\log|\rho|$, Markov gluing adds and selection takes the maximum. On uniform binary chains selection is therefore associative, with unit $\rho = 1$.

Proof. With $q = 1/2$, theorem 48 gives $\gamma_{00} - \frac{1}{4} = \frac{1}{2}\big(|t_1 + t_2 - \frac{1}{2}| - |t_1 - t_2|\big)$ for $t_i = \gamma_{i,00}$, which is $\operatorname{sign}(d_1 d_2)\min(|d_1|, |d_2|)$ with $d_i = t_i - \frac{1}{4}$. The signed minimum is associative. SB2 checks both laws and associativity on a 21^3 grid, exactly. □

Proposition 50 (where selection is associative)**.** *For binary chains $\gamma_1, \gamma_2, \gamma_3$ with margins $p_0, \dots, p_3$, the associator $A = \sigma(\sigma(\gamma_1, \gamma_2), \gamma_3) - \sigma(\gamma_1, \sigma(\gamma_2, \gamma_3))$ is the explicit piecewise-linear function obtained by composing theorem 48 with itself. Moreover:*

(1) among the 9^4 margin tuples with entries in $\{1/10, \dots, 9/10\}$, only the uniform one makes A vanish on the whole test grid (a 5^3 screen leaves 17; a 13^3 grid leaves one);

(2) for non-uniform margins A can vanish on a set of positive volume: for the margins $2/5, 1/3, 3/7, 1/4$ of theorem 45, $A = 0$ on the open polyhedron of all (t_1, t_2, t_3) for which the 22 branch and feasibility inequalities of the formula keep the signs they have at $(13/60,\ 7/30,\ 29/240)$; likewise for the margins $3/20, 1/4, 3/4, 17/20$ around $(1/40,\ 1/48,\ 51/80)$;

(3) on exact 13^3 grids, $A = 0$ at between 15% and 55% of the points for six non-uniform tuples, and this set is not a union of classes of the dependence signs $\operatorname{sign}(t_i - p_i p_{i+1})$.

Proof. The formula is theorem 48. (2) At the given point every maximum and minimum in the formula is strict, so the same branches hold on the open polyhedron, where A is affine; its four coefficients are all zero, and the point lies inside, so the polyhedron has nonempty interior (SX1, exact). (1) and (3) are exact evaluations on rational grids, SB5–SB6; they are a census, not a classification of the locus. □

Example 51 (one parameter, exactly)**.** Take the margins $2/5, 1/3, 3/7, 1/4$ of theorem 45 with $t_1 = 13/60$, $t_2 = 7/30$, and let t_3 run over $[0, 1/4]$. Then $A(t_3)$ is piecewise linear with kinks at $1/30$, $13/120$, $2/15$, $3/20$, $143/840$, $41/210$, $7/30$ and slopes in $\{0, \pm\frac{1}{4}, \frac{1}{2}\}$. It vanishes exactly on the interval $[13/120, 2/15]$ and at the point $5/28$, changes sign there, and equals $-41/960$ at $t_3 = 1/80$, the example of section 13.9 (fig. 17(b); SB7). The interval is a slice of the full-dimensional cell of proposition 50(2). Independence, $t_3 = 3/28$, lies just outside it; we know no reason for this and record it only as an observation.

Continuity in general. On a region of inputs where the facet set and the level sets $S_1, \dots, S_J$ of the recursion are fixed, the selected point is the unique solution of a linear system whose matrix is fixed and whose right side is linear in the inputs, so selection is affine there (SB3 checks this exactly at 339 midpoints). Across changes of the facet set we have only evidence: along four paths of a $3 \times 2 \times 3$ chain the facet set changes at least ten times, and the composite never jumps (SB4; fig. 17(c)). A proof of continuity across these walls is open.

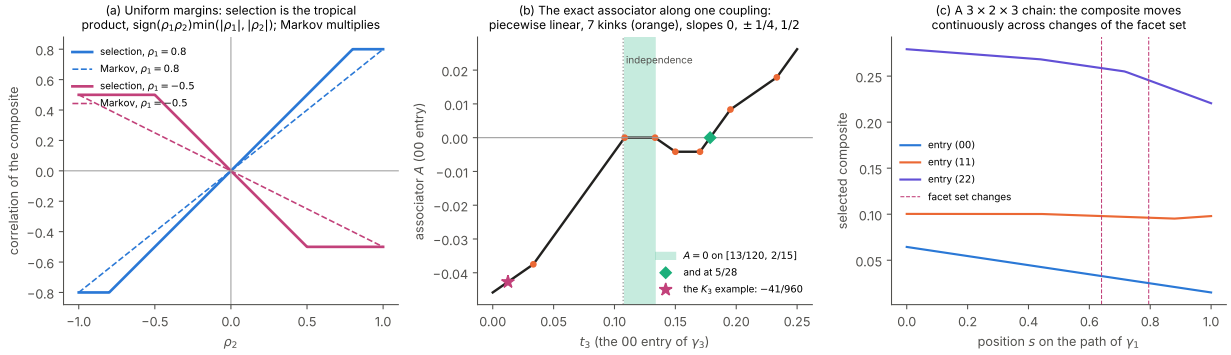


Figure 17: Selection on small chains. (a) Uniform binary margins: correlations compose by a signed minimum under selection and by a product under Markov gluing. (b) The exact associator of example 51, with its kinks, its interval and point of associativity, and the value $-41/960$ of section 13.9. (c) The selected composite along a path of a $3 \times 2 \times 3$ chain; the dashed lines are changes of the facet set, where nothing jumps.

13.11 A Floer reading of selection, at the level of objects

Composition of couplings is reduction in stages. The tables over $X_0 \times X_1 \times X_2$ with pair marginals γ_1, γ_2 form the fibre polytope $F(\gamma_1, \gamma_2)$; when it is Delzant it is the moment polytope of a toric manifold X_F, a symplectic reduction of the manifold of all tables. Both compositions pick a point of F and then take its (X_0, X_2) marginal. Markov gluing picks the point of largest entropy, the Birch point (proposition 43(2)). Selection picks the facet-leximin point, which by theorem 27 is the valuation of the positive critical point of the potential of X_F. The fibre over that point is the non-displaceable torus fibre that Fukaya, Oh, Ohta and Ono locate in any compact toric manifold

[46, Theorem 1.5 and §9], by the recursion of definition 26 (section 12.6, the remark on provenance). For binary chains this can be said exactly, and on both sides of Floer theory.

Proposition 52 (binary selection picks the Floer-nontrivial fibre)**.** *Let X_0 and X_2 be binary, and let k be the number of slices b with $L_b < U_b$ (theorem 48). Then $X_F \cong \prod_b \mathbb{CP}^1$, with areas $U_b - L_b$, which is Fano, and for a torus fibre L_c:*

(1) L_c carries a unitary local system with $HF \neq 0$ if and only if c is the product of the slice midpoints; there are 2^k such objects, $y_b = \pm T^{(U_b - L_b)/2}$ in slice coordinates [32];

(2) with local systems ρ valued in $\overline{\mathbb{F}}_2^*$, $HF(R, (L_c, \rho)) \neq 0$ for the real Lagrangian R if and only if c is the product of the midpoints; then $\rho = 1$ and $HF(R, L_c)$ is free of rank 2^k [8, Theorems 1.2 and 4.3].

So $\sigma(\gamma_1, \gamma_2)$ is the (X_0, X_2) marginal of the unique torus fibre of X_F that is Floer-nontrivial, on its own or against R. The Markov composite is the marginal of the Birch point of the same polytope, $c_b = r_b s_b / q_b$, which is a different point in 37 of 40 chains tested.

Proof. By the proof of theorem 48, $F = \prod_b [L_b, U_b]$ over the nondegenerate slices, with facet functions $x_b - L_b$ and $U_b - x_b$ of primitive normals ± 1; so $X_F = \prod_b \mathbb{CP}^1$, and SF1 confirms by exact traversal that F is Delzant with 2^k vertices and $-K \cdot C = 2$ on every invariant curve. In slice coordinates the potential is $W = \sum_b (y_b + T^{w_b}/y_b)$ with $w_b = U_b - L_b$. (1) Its critical points are $y_b = \pm T^{w_b/2}$, over the midpoints, and Cho–Oh identify the fibres with $HF \neq 0$ with the valuations of critical points (SF2). (2) Alston–Amorim show $HF(R, (L_c, \rho)) \neq 0$ if and only if $\nabla W_c(\rho) = 0$. In characteristic 2 the b-th component is $\rho_b T^{c_b - L_b} + \rho_b^{-1} T^{U_b - c_b}$, which vanishes only if the exponents agree and $\rho_b^2 = 1$, that is c_b is the midpoint and $\rho_b = 1$. There their differential $\sum_j \sqrt{\rho^{v_j}}\, T^{E_j/2} f_j$ is $\sum_b 2 T^{w_b/4} f_b = 0$, so the complex, free of rank 2^k, is its own cohomology (SF3). SF4 compares the Birch points. □

Remark 53 (what this explains, and how far it goes)**.** *Non-associativity.* Reduction in stages is associative; choosing a point is not compatible with it in general. Birch points are, which is why Markov gluing is associative. Floer fibres are not: $\sigma(\sigma(\gamma_1, \gamma_2), \gamma_3)$ takes the Floer fibre of one reduced space and reduces again, while the top cell takes the Floer fibre of the larger reduced space at once.

Beyond binary. For the binary K_4 example the fibres behind the eleven composites are, by exact traversal (SF5): the 14 distinct 3-way fibres are each $\mathbb{CP}^1 \times \mathbb{CP}^1$, so proposition 52 applies; of the 5 distinct 4-way fibres, three are Delzant and two are simple but not unimodular, so the toric variety is an orbifold, and none is Fano (a curve with $-K \cdot C = -4$); the 5-way fibre of the top cell is not Fano either: a complete exact traversal (34 166 vertices) finds it simple but not unimodular, so an orbifold, with a curve of $-K \cdot C = -32$. For the Delzant ones the selected point is the FOOO non-displaceable fibre. The equivalences (1) and (2) need the Fano hypothesis and are not claimed there, and the orbifold fibres would need orbifold Floer theory.

What is not claimed. This reads selection at the level of objects, one torus fibre per composite. It constructs no functor and says nothing about morphisms or products between these fibres; the categorical comparison of section 13.12 is untouched. The real Lagrangian enters only through [8]; that it corresponds to hard evidence is a hypothesis we do not test.

13.12 Three categorical directions, kept apart

Part III has touched three different constructions. They should not be confused.

construction	what is established here	what is missing
the toric-fibre category	objects are critical points of W over torus fibres; binary selection is read as the Floer-nontrivial fibre of the reduced space (proposition 52); Hom vanishes off the diagonal where the critical values differ; the endomorphisms are Clifford algebras; no higher products (propositions 31 and 32); all cited, in the convention of section 13.7	generation in the non-monotone case [77]; the category at a collision (proposition 41)
a coupling-polytope enhancement	Markov and selection compositions, the exact K_4 data (section 13.9); with point-class composition and diagonal units it is trivial up to quasi-equivalence (proposition 47)	nothing to add in that form; another enhancement would need morphism spaces that are not contractible
a Fukaya–Seidel category of W	only its combinatorial shadow: the critical points, their values, and where Stokes alignments occur (proposition 35, section 13.8)	gradings; Floer complexes between thimbles; the polygon counts that define μ^d; the directed A_∞ structure [74]; and a proof that the positive thimble is the probabilistic object in that category

13.13 A small dagger category for finite probability

The underlying category here is standard. What is new is a precise account of which of its symmetries selection keeps and which it breaks. Write $\gamma \otimes \gamma'$ for the Kronecker product of couplings, and $D_p = \mathrm{diag}(p)$.

Definition 54 (the category Cpl)**.** Objects are finite sets with full-support laws (X, p) (a state of mass zero is deleted). A morphism $(X, p) \to (Y, q)$ is a coupling $\gamma \in \Pi(p, q)$; composition is Markov gluing $\gamma_1 D_q^{-1} \gamma_2$, with identity D_p (definition 42). The dagger is the transpose, $\gamma^\dagger = \gamma^\top \in \Pi(q, p)$. The tensor product is $(X, p) \otimes (Y, q) = (X \times Y, p \otimes q)$ on objects and $\gamma \otimes \gamma'$ on morphisms, with the swap of factors as symmetry. A coarse-graining of (Z, r) along a map $f : Z \to Z'$ sends δ to δF, where F is the 0/1 matrix of f.

Writing $K = D_p^{-1}\gamma$ turns couplings into Markov kernels that carry p to q, and the transpose becomes Bayesian inversion [47, 34].

Proposition 55 (the structure Markov gluing carries)**.** Cpl *is a dagger symmetric monoidal category, and Markov gluing commutes with coarse-graining an end space:* $(\gamma D^{-1}\delta)F = \gamma D^{-1}(\delta F)$.

Proof. Associativity and units are proposition 43(1). $(\gamma_1 D^{-1}\gamma_2)^\top = \gamma_2^\top D^{-1}\gamma_1^\top$ and $D_p^\top = D_p$, so the transpose is an identity-on-objects involution that reverses composition. The mixed-product rule gives $(\gamma \otimes \gamma')(D \otimes D')^{-1}(\delta \otimes \delta') = (\gamma D^{-1}\delta) \otimes (\gamma' D'^{-1}\delta')$ and $(\gamma \otimes \gamma')^\top = \gamma^\top \otimes \gamma'^\top$. The last identity is associativity of matrix multiplication. NC1 checks all of these in exact rationals on random chains. □

Example 56 (forecast, umbrella, wet)**.** Let X say whether the forecast is rain, Y whether an umbrella is carried, Z whether one gets wet, with $P(\text{rain}) = 2/5$, $P(\text{carried}) = 1/3$, $P(\text{wet}) = 3/7$. Two separate data sets record the pairs:

$$\gamma_1 = \begin{pmatrix} 13/60 & 11/60 \\ 7/60 & 29/60 \end{pmatrix} \in \Pi(X, Y), \qquad \gamma_2 = \begin{pmatrix} 7/30 & 1/10 \\ 41/210 & 33/70 \end{pmatrix} \in \Pi(Y, Z),$$

rows and columns in the order (yes, no). No record contains all three. The two compositions give different laws for (forecast, wet):

$$\gamma_1 D_q^{-1} \gamma_2 = \begin{pmatrix} 23/112 & 109/560 \\ 25/112 & 211/560 \end{pmatrix}, \qquad \sigma(\gamma_1, \gamma_2) = \begin{pmatrix} 31/120 & 17/120 \\ 143/840 & 361/840 \end{pmatrix}.$$

These are the first two couplings of theorem 45. Figure 18 draws them, together with the two squares of proposition 58.

Proposition 57 (what selection keeps)**.** *For $\gamma \in \Pi(p, q)$ and $\delta \in \Pi(q, r)$:*

(1) closed: $\sigma(\gamma, \delta) \in \Pi(p, r)$*;*

(2) unital: $\sigma(\gamma, D_q) = \gamma$ *and* $\sigma(D_p, \gamma) = \gamma$*;*

(3) reversal: $\sigma(\gamma, \delta)^\top = \sigma(\delta^\top, \gamma^\top)$*;*

(4) relabelling: *for bijections α, β, κ of X, Y, Z, selection of the relabelled couplings is the relabelled selection;*

(5) a copied factor: *for every full-support law w, $\sigma(\gamma \otimes D_w,\ \delta \otimes D_w) = \sigma(\gamma, \delta) \otimes D_w$.*

Proof. (1) The end marginal of a table with pair marginals γ, δ has margins p and r. (2) is proposition 44. (3) and (4): the maps $x_{ijk} \mapsto x_{kji}$ and $x_{ijk} \mapsto x_{\alpha(i)\beta(j)\kappa(k)}$ carry the fibre bijectively onto the fibre of the transformed chain. They preserve every cell value, so they preserve the facets and each step of definition 26; the leximin point is unique, so it goes to the leximin point. (5) The pair marginal $\gamma \otimes D_w$ forces $x_{(i,a),(j,b),(k,c)} = 0$ unless $a = b$, and $\delta \otimes D_w$ forces $b = c$. So the fibre is the product over a of the scaled copies $w_a F(\gamma, \delta)$, through $x_{(i,a),(j,a),(k,a)} = w_a y^{(a)}_{ijk}$. Its facets are facets of single copies, and the facet functions of copy a are w_a times those of $F(\gamma, \delta)$. As in the proof of theorem 48, the recursion on a product whose facet functions each depend on one factor selects the product of the factors' own selected points. Scaling all of a factor's functions by $w_a > 0$ does not change that factor's selected point. NC2 and NC3 check (1)–(5) exactly. □

Proposition 58 (what selection breaks)**.** *With γ_1, γ_2 of example 56:*

(1) An independent factor. *With $\gamma' = \left(\begin{smallmatrix} 1/5 & 1/20 \\ 3/10 & 9/20 \end{smallmatrix}\right)$ and $\delta' = \left(\begin{smallmatrix} 1/10 & 2/5 \\ 1/2 & 0 \end{smallmatrix}\right)$, the 00 entry of $\sigma(\gamma_1 \otimes \gamma', \gamma_2 \otimes \delta')$ is 29/1400, while that of $\sigma(\gamma_1, \gamma_2) \otimes \sigma(\gamma', \delta')$ is 31/1200. So selection is not monoidal.*

(2) Coarse-graining an end. *With $\delta = \left(\begin{smallmatrix} 1/10 & 1/10 & 2/15 \\ 1/5 & 1/5 & 4/15 \end{smallmatrix}\right)$, merging the last two states of Z after selection gives 00 entry 11/90, and merging them before gives 17/120.*

(3) Coarse-graining the middle *changes both compositions. Markov gluing: for $\gamma = \left(\begin{smallmatrix} 1/10 & 1/10 & 1/5 \\ 1/5 & 3/20 & 1/4 \end{smallmatrix}\right)$, $\delta = \left(\begin{smallmatrix} 1/5 & 1/10 \\ 1/20 & 1/5 \\ 1/5 & 1/4 \end{smallmatrix}\right)$, merging the last two middle states moves the 00 entry from 79/450 to 73/420. Selection: for $\gamma = \left(\begin{smallmatrix} 1/8 & 9/32 & 3/32 \\ 3/16 & 1/4 & 1/16 \end{smallmatrix}\right)$, $\delta = \left(\begin{smallmatrix} 5/144 & 5/18 \\ 85/448 & 153/448 \\ 5/64 & 5/64 \end{smallmatrix}\right)$, it moves from 1283/8064 to 305/2016, and it moves on 10 of 15 random chains.*

In (1) and (2) Markov gluing passes both squares (proposition 55).

Proof. Exact computation, with every input above (NC4–NC6). □

Why Markov passes the tensor square and selection does not: the product chain's fibre contains $F(\gamma,\delta)\times F(\gamma',\delta')$ and also tables that are not products. For any table x in it, with marginals x_1,x_2 on the two systems, $H(x)\le H(x_1)+H(x_2)\le H(m)+H(m')$, where m,m' are the Markov points of the factors. Equality holds at $m\otimes m'$, so the maximum-entropy point of the product is the product of the maximum-entropy points. A least cell has no such inequality, and in (1) the leximin point of the product fibre is not a product.

property	Markov gluing	selection
associative	yes	no (proposition 44 and theorem 45)
unital; reversal (dagger); relabelling	yes	yes
tensoring by an identity D_w (a copied factor)	yes	yes
tensoring by an independent factor	yes	no
coarse-graining an end space	yes	no
coarse-graining the middle space	no	no
the point of the fibre it picks	largest entropy	lexicographically largest least slack

Remark 59 (status)**.** So selection is a geometric rule on fibres. It is unchanged by renaming states, by reversing direction and by copying information, but it changes when an independent system is added or an end is recorded at another resolution. Markov gluing is unchanged by all of these, and both change with the resolution of the middle. Together with the associator and the K_4 configuration, this is further evidence that selection does not give Cpl a second composition law; nothing here alters the three directions of section 13.12.

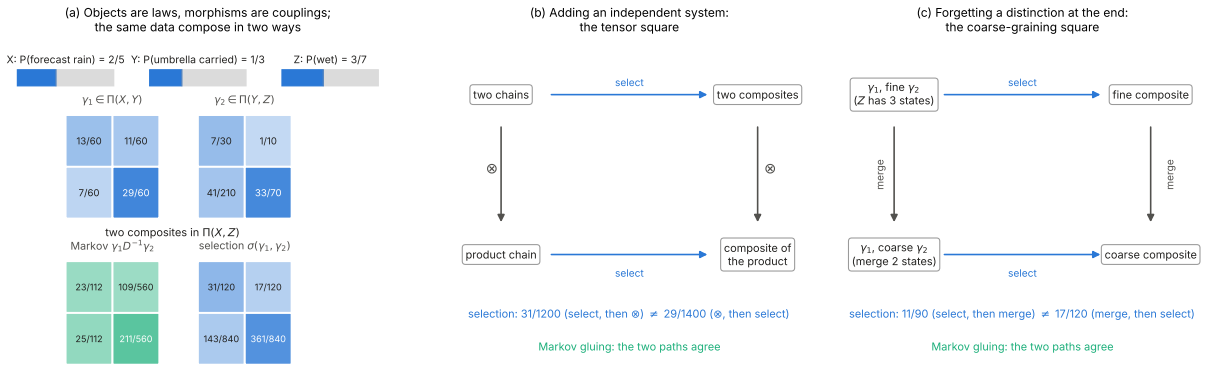


Figure 18: The small category on example 56. (a) Three binary laws, the two data sets, and the two composites they give. (b) The tensor square: selecting and then taking the product differs from selecting on the product chain. (c) The coarse-graining square: merging two end states before or after selection gives different answers. Markov gluing passes both squares.

13.14 How little it takes: poisoning, representation, and what the data determine

In example 56 take the decision "warn that rain is coming with you" when $P(\text{wet}\mid\text{forecast rain})\ge\tau=1/2$, computed from a composite. This subsection asks how little has to change for that decision to flip. Two kinds of change are compared: edits of the data themselves (poisoning), and changes in the representation underneath that leave the data as recorded untouched. For binary ends, what the data alone determine has a closed form.

Proposition 60 (what the data determine)**.** *Let X_0 and X_2 be binary, and use the notation of theorem 48.*

(1) The 00 *entries of the (X_0, X_2) marginals of all joint laws compatible with γ_1 and γ_2 fill the interval $[L, U] = [\sum_b L_b,\ \sum_b U_b]$.*

(2) Selection is its midpoint. So among all rules it has the least worst-case error over compatible joint laws, namely $(U - L)/2$. Markov gluing has worst-case error $\max(u_M - L, U - u_M)$, which is larger unless u_M is the midpoint.

(3) Splitting a middle state into two, consistently with the recorded data, can only shrink $[L, U]$.

(4) The data alone decide the threshold question exactly when $\tau p_0 \notin (L, U)$, with $p_0 = P(X_0 = 0)$.

Proof. (1) By the proof of theorem 48 the fibre is $\prod_b [L_b, U_b]$, and the 00 entry of the end marginal is $\sum_b x_{0b0}$. (2) The first claim is theorem 48, and the midpoint of an interval minimises the largest distance to its points. (3) Merging the two halves of the split state sends a compatible table of the refined chain to a compatible table of the original one, with the same end marginal. (4) follows from (1). NC7 checks (1) and (2) against linear programs over the fibre on 20 chains, and NC9 checks (3) on 195 refinements. □

In this language the interval is the *identified set* of the composite [62]. Neither composite is determined by the data: Markov gluing adds conditional independence given Y, and selection takes the centre.

Proposition 61 (edits inside the data)**.** *On binary chains, write p_0, p_1, p_2 for the first entries of the laws and t_1, t_2 for the* 00 *entries of γ_1, γ_2. Change γ_1 by a margin-preserving swap of size s (the* 00 *and* 11 *entries up by s, the others down). This moves $2|s|$ of the mass, so $2|s|$ is the fraction of records edited.*

(1) The Markov composite is affine in s with slope $(t_2 - p_1 p_2)/(p_1(1 - p_1))$, which is proportional to the covariance in γ_2.

(2) The selected composite is piecewise linear in s, with slopes in $\{0, \pm\frac{1}{2}, \pm 1\}$. Its slope is 0 *on the open region where $t_1 + t_2 < p_1$, $t_2 < t_1$, $(p_0 - t_1) + (p_2 - t_2) < 1 - p_1$ and $p_2 - t_2 < p_0 - t_1$: there, edits of γ_1 do not move it at all.*

Proof. (1) Differentiate $t_1 t_2/p_1 + (p_0 - t_1)(p_2 - t_2)/(1 - p_1)$. (2) Each slice midpoint $(L_b + U_b)/2$ is a half-sum of a maximum and a minimum of affine functions with slope 0 or ± 1 in t_1. Under the four inequalities, $L_0 = 0$, $U_0 = t_2$, $L_1 = 0$ and $U_1 = p_2 - t_2$, none of which depends on t_1. □

Example 62 (how little, both ways)**.** In example 56, in units of $P(\text{wet} \mid \text{forecast rain})$:

(1) *The data.* They determine only [7/24, 1], which contains τ. Markov gluing gives $115/224 \approx 0.513$, worst-case error $109/224 \approx 0.487$. Selection gives $31/48 \approx 0.646$, worst-case error $17/48 \approx 0.354$. Both warn.

(2) *Poisoning.* The smallest margin-preserving edits that flip each decision are, as fractions of records edited:

	edit γ_1	edit γ_2
Markov stops warning	$1/38 \approx 2.6\%$	$1/35 \approx 2.9\%$
selection stops warning	$9/70 \approx 12.9\%$	$2/15 \approx 13.3\%$
the data force "warn" ($L \geq \tau p_0$)	$1/6 \approx 16.7\%$	$1/6 \approx 16.7\%$
the data force "no warning" ($U \leq \tau p_0$)	$89/210 \approx 42.4\%$	$13/30 \approx 43.3\%$

(3) *Changes that cost no data.* Reveal a hidden distinction in the middle, say whether a carried umbrella is opened, with each half of mass 1/6 and consistent with both records (195 splits on a grid of step 1/120). Markov's answer then ranges over $[477/1120,\ 673/1120] \approx [0.426, 0.601]$, and it stops warning in 56 of them, with not one record changed. Selection's ranges over $[25/48,\ 17/24] \approx [0.521, 0.708]$ and warns in all of them. The identified set shrinks, its lower end rising to at most 5/12 and its upper end falling to 3/4, but it always contains τ. Record "wet" at a finer resolution (49 splits), or attach an independent system (14 random ones): Markov's answer does not move, by proposition 55, while selection's moves within [319/504, 37/56] and [0.604, 0.641]. None of these flips selection.

Figure 19 draws the three parts. NC8 and NC9 compute every number exactly. Each edit is an exact root, with the decision checked just before and just after it.

Both sides. A rule that composes probabilities without the joint data has to supply the missing structure, and each rule exposes a different weakness. Markov gluing supplies conditional independence given the middle. That makes it stable under every change of representation at the ends, but its answer hangs on a distinction in the middle that the data may not record: in the example, an unrecorded distinction flips it at no cost in data, and an edit of 2.6% of one data set does too. Selection supplies the centre of what the data allow. That gives it the least worst-case error, and it needs about five times the edit to flip here. In part of the input space it ignores edits altogether, which is robustness and also blindness to evidence. But it pays in representation: an independent system or a finer record of an end moves it, though Markov gluing is unaffected. What the data themselves compute is the interval. When the threshold lies inside it, every decision taken is a consequence of the rule, not of the data, and making the data decide takes 16.7% of the records in one direction and 42% in the other. These statements are exact for binary ends. For larger ends the identified set is a polytope containing both composites, and we do not claim that selection is its centre.

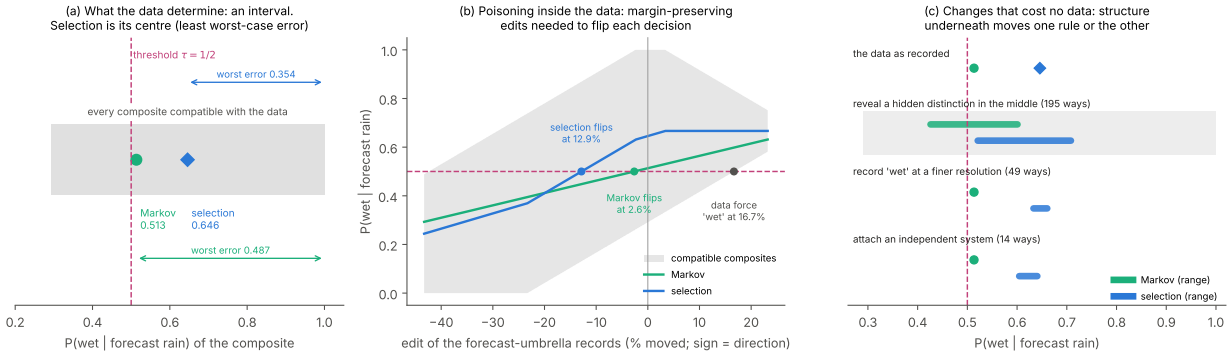


Figure 19: How little changes the decision in example 56. (a) What the data determine: the identified interval (grey), the two composites, and their worst-case errors; selection sits at the centre. (b) Margin-preserving edits of the forecast–umbrella records: Markov stops warning at 2.6%, selection at 12.9%, and the data themselves force a warning only at 16.7%. (c) Changes that cost no data. A hidden distinction in the middle moves Markov across the threshold and shrinks the interval; a finer record of "wet" and an independent system move selection only.

13.15 Selection as a section of the marginal projection

Sections 13.13 and 13.14 compared two rules on chains. Both have the same shape: from local data, choose one joint law that agrees with the data. This subsection makes that shape the object of study. A rule is a section of a projection. Which laws such a rule obeys is a separate question, and a category appears only after a further step.

The projection. Let V be a finite set of variables with finite state sets X_v, and let $\mathcal{C}$ be a cover of V by contexts $C \subseteq V$. Let $\mathcal{J} = \Delta(\prod_v X_v)$ be the simplex of joint laws, and

$$\pi : \mathcal{J} \longrightarrow \prod_{C \in \mathcal{C}} \Delta(X_C)$$

the map to the family of context marginals. Write $\mathcal{L}$ for the locally consistent families, those whose marginals agree on overlaps, and $\mathcal{D} = \pi(\mathcal{J}) \subseteq \mathcal{L}$ for the data that some joint law produces. The fibre over $d \in \mathcal{D}$ is $\mathcal{J}_d = \pi^{-1}(d)$, and the points of $\mathcal{L} \setminus \mathcal{D}$ are the contextual data [2]. A chain $X_0, \dots, X_n$ with contexts $\{X_{k-1}, X_k\}$ is the case of section 13.9, and there $\mathcal{L} = \mathcal{D}$ [82].

Definition 63 (sections and their laws)**.** A *section* is a map $s : \mathcal{D} \to \mathcal{J}$ with $\pi \circ s = \mathrm{id}_{\mathcal{D}}$. Let Γ be the group generated by relabelling states and by permutations of V that preserve $\mathcal{C}$; it acts on $\mathcal{J}$ and on $\mathcal{D}$, and π commutes with it. A section is *equivariant* if $s(\gamma d) = \gamma s(d)$ for $\gamma \in \Gamma$; *interior* if $s(d)$ lies in the relative interior of $\mathcal{J}_d$; *system independent* if it sends the product data of two independent systems to the product of their sections; and *natural* for a map $f : X_v \to X'_v$ if pushing the data forward along f and then taking the section gives the pushforward of the section.

The projection records which joint laws are compatible with the data. A section chooses one of them. Continuity, symmetry, system independence and naturality are further properties of that choice, and the projection implies none of them.

Theorem 64 (the projection)**.** *(1) $\mathcal{D}$ is the convex hull of the images of the deterministic laws δ_ω, $\omega \in \prod_v X_v$. For $d \in \mathcal{D}$ the fibre $\mathcal{J}_d$ is a non-empty polytope; for $d \in \mathcal{L} \setminus \mathcal{D}$ there is no compatible joint law.*

(2) For a face G of $\mathcal{D}$ let $\Omega(G) = \{\omega : \pi(\delta_\omega) \in G\}$. Then $\pi^{-1}(G)$ is the face of $\mathcal{J}$ spanned by $\{\delta_\omega : \omega \in \Omega(G)\}$, and for d in the relative interior of G

$$\dim \mathcal{J}_d = |\Omega(G)| - 1 - \dim G.$$

So the fibre is a single point over the whole of G exactly when $|\Omega(G)| = \dim G + 1$.

(3) $d \mapsto \mathcal{J}_d$ is Lipschitz in the Hausdorff metric on $\mathcal{D}$.

Proof. (1) π is linear and $\mathcal{J}$ is the simplex on the δ_ω. (2) Write $G = \mathcal{D} \cap \{\ell = 0\}$ with ℓ affine and $\ell \geq 0$ on $\mathcal{D}$. Then $\ell \circ \pi$ is affine and ≥ 0 on the simplex, so its zero set is the face spanned by the vertices where it vanishes, which are the δ_ω with $\omega \in \Omega(G)$. That face is a simplex of dimension $|\Omega(G)| - 1$, and π maps it affinely onto G. The fibre of an affine surjection of polytopes over a relative-interior point has dimension equal to the difference of the dimensions; fibres of linear projections of polytopes are the subject of [16]. (3) The fibres are the polyhedra $\{x \geq 0 : Ax = b(d)\}$ with fixed A, and these depend Lipschitz-continuously on b on the set where they are non-empty [83]. □

Theorem 65 (sections)**.** *(1) Every continuous, strictly concave ϕ on $\mathcal{J}$ gives a section $s_\phi(d) = \arg\max_{\mathcal{J}_d} \phi$, continuous on $\mathcal{D}$. For the entropy H the section s_H is continuous, interior, equivariant and system independent.*

(2) The leximin section s_{lex} (definition 26 applied to $\mathcal{J}_d$, with the cell values as slacks) is equivariant wherever it is defined. It is not system independent.

(3) If the stabiliser Γ_d of d fixes exactly one point of $\mathcal{J}_d$, every equivariant section takes that value at d.

Proof. (1) By theorem 64(3) and Berge's maximum theorem the set of maximisers depends upper-semicontinuously on d; it is a single point, so it is continuous. The entropy has infinite slope into the interior at the boundary of a face, so its maximiser is relatively interior. H is invariant under permutations of cells, and Γ acts by permuting cells and maps $\mathcal{J}_d$ to $\mathcal{J}_{\gamma d}$, which gives equivariance. System independence is subadditivity, as in section 13.13: $H(x) \le H(x_1) + H(x_2)$ for a joint law of two systems with marginals x_1, x_2, with equality at the product. (2) Equivariance is the argument of proposition 57(3)–(4): a cell permutation preserves the facets and every step of the recursion. Proposition 58(1) is a counterexample to system independence. (3) $s(d) = s(\gamma d) = \gamma s(d)$ for $\gamma \in \Gamma_d$, so $s(d)$ is the fixed point. □

Theorem 66 (from sections to a category)**.** *On chain covers, let $S = (S_n)_{n\ge 1}$ choose a section for every chain of every length, and set $\gamma_1 \circ_S \gamma_2 =$ the (X_0, X_2) marginal of $S_2(\gamma_1, \gamma_2)$.*

(1) $\circ_S$ is closed and unital, since the fibre over (γ, D_q) is one point.

(2) Call S inner-face consistent *if, for every $n \ge 3$ and $0 < i < n$, summing X_i out of $S_n(\gamma_1, \dots, \gamma_n)$ gives $S_{n-1}(\gamma_1, \dots, \gamma_i \circ_S \gamma_{i+1}, \dots, \gamma_n)$. Then $\circ_S$ is associative, and every bracketing of a chain equals the end marginal of S_n. So finite probability spaces, couplings and $\circ_S$ form a category.*

(3) If moreover S is equivariant (reversal included), system independent, or natural for end maps, the category is respectively a dagger category, monoidal, or compatible with coarse-graining the ends.

(4) Maximum entropy is inner-face consistent, and its category is Cpl *with Markov gluing. The leximin family is not inner-face consistent.*

Proof. (2) Both bracketings of $(\gamma_1, \gamma_2, \gamma_3)$ are end marginals of faces of S_3: summing out X_2 gives $S_2(\gamma_1, \gamma_2 \circ \gamma_3)$, and summing out X_1 gives $S_2(\gamma_1 \circ \gamma_2, \gamma_3)$. Both therefore equal the end marginal of S_3, and induction on n handles longer chains. (3) The three properties pass to end marginals. (4) The maximum-entropy law with given consecutive marginals is the Markov chain, and summing out an inner variable of a Markov chain gives the Markov chain of the composed kernel. For leximin, theorem 45 gives a top composite that differs from a bracketing; MS2 checks both statements. □

Inner-face consistency is sufficient for associativity; we do not claim it is necessary. In the language of the table in section 13.13, each law of a composition is a property of the section family: units are a property of the one-point fibres; dagger and relabelling are equivariance; the tensor square is system independence; coarse-graining is naturality; and associativity is coherence under marginalisation across different decompositions of a chain.

Open problem 67 (the associative-section problem)**.** Characterise the section families on chain covers that are unital, equivariant under relabelling and reversal, system independent, and inner-face consistent. Is maximum entropy the only one? Or do other associative families exist, with or without system independence?

Theorem 66 gives a sufficient mechanism. It does not say that every associative composition arises from an inner-face consistent family, nor that maximum entropy is the only such family. Shore and Johnson's theorem (remark 69) does not settle Problem 67 either, because its axioms quantify over a different class of rules. Leximin shows that equivariance and units do not imply coherence.

Remark 68 (associativity as a Beck–Chevalley condition; symmetry as a topos quotient)**.** Inner-face consistency asks that forgetting an inner variable commute with the chosen gluing. In the categorical treatment of Markov kernels, composition is exactly gluing followed by marginalisation of the middle variable [47, §§11–13]. That this composition is associative rests on the Markov property, which makes the relevant squares of pushforwards and pullbacks commute. We read that as a condition of Beck–Chevalley type. Theorem 66(4) is that statement for the maximum-entropy section. The selection associator of section 13.10 and the configuration of theorem 45 measure the failure of the same condition for the leximin section. The associativity failures are therefore failures of a Beck–Chevalley condition for a gluing that is not Markov; they do not come from the geometry of the spaces involved. Hyperconnected geometric morphisms do not account for them: geometric morphisms compose associatively, and a map of finite state spaces induces a localic morphism, whose hyperconnected part is trivial. Where topos theory does enter is symmetry. Equivariance of sections (theorem 65(3)) is a statement about Γ-sets, and hyperconnected quotients of a topos correspond to such actions [27]. We claim no more than this dictionary.

Theorem 66 also says where the selection associator of section 13.10 comes from: it measures the failure of the leximin family to be inner-face consistent.

Remark 69 (maximum entropy, with care)**.** Shore and Johnson [75] derive minimum relative entropy as the only updating rule satisfying four consistency axioms: uniqueness, invariance, system independence and subset independence. Their axioms quantify over all linear constraint sets on all finite spaces, and their reading has been debated [80]. Our sections are defined only on the fibres of one projection, so their theorem does not make maximum entropy the unique section with any particular list of the properties above. We therefore prove each law directly. The comparison table of section 13.13 shows which laws separate the two rules; system independence is one of them.

Remark 70 (three layers)**.** The sheaf-theoretic approach to contextuality [2, 3] asks whether compatible local data have a global probabilistic extension. Here a global section of the event sheaf is a *deterministic* assignment, and our fibre $\mathcal{J}_d$ is the polytope of probability distributions over those assignments: its vertices are exactly the deterministic global sections. In that language theorem 64(2) says that over a face G the data can use exactly the deterministic global sections in $\Omega(G)$, and that the mixture of them is pinned down exactly when they are affinely independent; corollary 74 and theorem 77 say this happens along the whole contextuality boundary of every cycle. We study the geometry of the space of extensions and the rules that choose among them. Information cohomology [14, 81] gives a complementary perspective on the identities that particular rules satisfy, maximum entropy above all. The three study different objects: the event sheaf, convex fibres of global distributions and their sections, and information structures with coefficient modules. The bridge between them is constructed here only as far as proposition 71.

The two sheaf axioms give a short dictionary. *Gluing*, the existence of a global section, is $\mathcal{J}_d \neq \varnothing$: no contextuality. *Locality*, the uniqueness of the glued section, is $\dim \mathcal{J}_d = 0$: the data determine the joint law. Positivity is the only obstruction to gluing. If signed measures are allowed, every locally consistent family has a global extension [2, 6], and these extensions form an affine space $\mathcal{A}_d = \{x : \pi(x) = d,\ \sum_\omega x_\omega = 1\}$ whose dimension is the same for every $d \in \mathcal{L}$ (here $15-8=7$

for the 4-cycle). The linear layer, as for a fine sheaf, has no obstruction; the fibre is its intersection with the simplex, $\mathcal{J}_d = \mathcal{A}_d \cap \mathcal{J}$. Contextuality is $\mathcal{A}_d \cap \mathcal{J} = \varnothing$. Corollary 74 and theorem 77 say that at the contextuality boundary the affine space, of dimension $2^n - 1 - 2n$ for the n-cycle, meets the simplex in a single point: where gluing is about to fail, the gluing is unique. How far $\mathcal{A}_d$ lies from the simplex is measured by the contextual fraction [1], the largest weight of a global subprobability that fits under the data.

Proposition 71 (what the cocycle gives)**.** *In the settings of [14, 81] the degree-one information cocycles are the multiples cH of Shannon entropy, and the cocycle condition is the chain rule $H(X,Y) = H(X) + H(Y \mid X)$. For a section defined by maximising cH on each fibre:*

(1) if $c > 0$ the maximiser is unique and the section is system independent. This uses the identity $H(P \otimes P') = H(P) + H(P')$, which is the cocycle condition, together with the inequality $H(P_{XY}) \le H(P_X) + H(P_Y)$, that is $I(X;Y) \ge 0$, which is concavity and not cohomology;

(2) if $c < 0$ the cocycle condition still holds, but the maximisers of cH are the minimisers of H, which are attained at vertices and need not be unique, so no section is defined. For the binary chain with all marginals $1/2$ and $\gamma_1 = \gamma_2 = \left(\begin{smallmatrix} 1/4 & 1/4 \\ 1/4 & 1/4 \end{smallmatrix}\right)$, the fibre is a square, and all four vertices have entropy $\log 4$, the minimum.

So the cocycle together with strict concavity gives system independence, and the cocycle alone does not.

Proof. (1) Strict concavity gives uniqueness; for system independence see the proof of theorem 65(1). (2) A concave function on a polytope attains its minimum at a vertex. In the example, the flip $x \mapsto 1/4 - x$ in either slice maps the fibre to itself and preserves entropy, and it permutes the four vertices; each vertex has four cells equal to $1/4$ and four equal to 0 (MS11). □

Concavity alone is not enough either. The Cressie–Read power family [18]

$$\phi_q(x) = \frac{x^q - x}{q(1-q)}, \qquad q \le 1,$$

gives a strictly concave cell functional for each q. It is Shannon's $-x \log x$ at $q = 1$ and Burg's $\log x$ at $q = 0$, up to terms constant on a fibre, and it moves toward the leximin point as $q \to -\infty$. MS11 maximises $\sum_c \phi_q(x_c)$ on the fibres of the forecast–umbrella chain, of a second binary chain and of their product, and measures how far the product chain's section is from the product of the two sections, in total variation (numerical, 40 digits):

q	1	0.99	0.9	0.5	0	-1	-4	-8
defect	$< 10^{-15}$	0.0008	0.0077	0.032	0.049	0.064	0.073	0.075

The facet-leximin selection has defect 0.0752. Within this family only the Shannon member passes the tensor square, and the defect grows steadily toward the leximin value. That is what Shore and Johnson's system-independence axiom predicts for separable functionals [75]. Here it is a computation on one pair of chains, not a proof of Problem 67; fig. 20 draws it.

13.16 The 4-cycle cover

The smallest cover with a cycle has four binary variables A_0, A_1, B_0, B_1 and contexts $\{A_i, B_j\}$: the cycle $A_0 - B_0 - A_1 - B_1 - A_0$ of the CHSH setting. Data are four 2×2 tables that agree on the

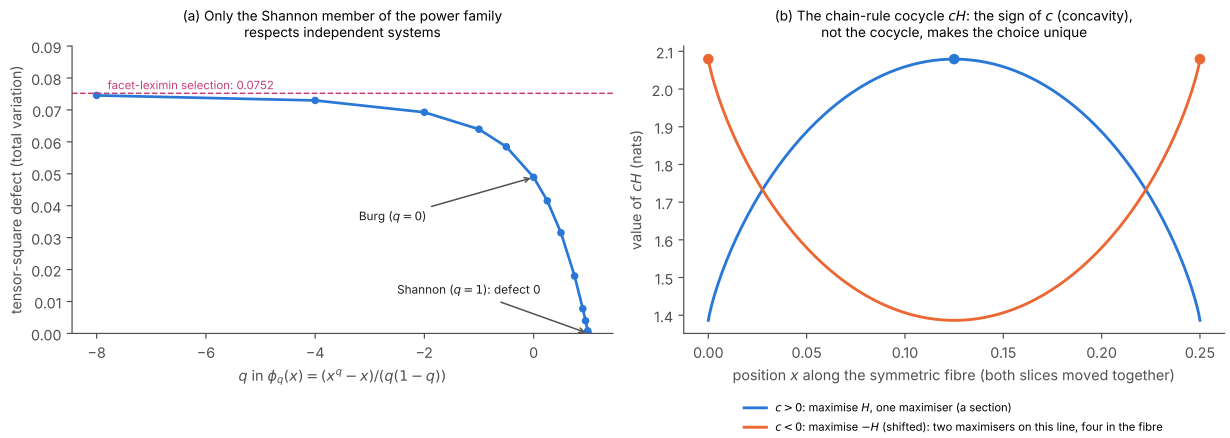


Figure 20: (a) The tensor-square defect of the section that maximises $\sum_c \phi_q(x_c)$: zero only at Shannon ($q = 1$), rising toward the facet-leximin value. (b) The chain-rule cocycle cH on a symmetric fibre: for $c > 0$ it has one maximiser, for $c < 0$ several (four in the fibre). Uniqueness comes from the sign of c, that is from concavity, not from the cocycle.

single marginals. They are coordinatised by eight numbers, the four entries $P(A_i = 0, B_j = 0)$ and the four probabilities $P(A_i = 0)$, $P(B_j = 0)$. Then $\mathcal{J}$ is the simplex on 16 outcomes, of dimension 15, $\mathcal{D}$ has dimension 8, and the generic fibre has dimension 7.

Proposition 72 (the marginal polytope of the 4-cycle)**.** *$\mathcal{D}$ has* 16 *vertices, the images of the deterministic laws, and exactly* 24 *facets. Sixteen of them are the local inequalities of $\mathcal{L}$, the nonnegativity of the sixteen table entries. The other eight are the Clauser–Horne inequalities [17], for example*

$$P(A_0B_0) + P(A_0B_1) + P(A_1B_0) - P(A_1B_1) \;\le\; P(A_0) + P(B_0),$$

where $P(A_iB_j)$ stands for $P(A_i = 0, B_j = 0)$ and $P(A_0)$ for $P(A_0 = 0)$; the others are obtained by choosing which context carries the minus sign and by the four with ≥ -1 in place of ≤ 0. So $\mathcal{D} = \mathcal{L} \cap \{\mathrm{CH}\}$, which is Fine's theorem [26]. On the symmetric path with uniform marginals and correlations $\lambda, \lambda, \lambda, -\lambda$, whose CHSH value is $S = 4\lambda$, the data lie in $\mathcal{L}$ for $\lambda \le 1$ and in $\mathcal{D}$ exactly for $\lambda \le 1/2$. Past $S = 2$ there is no joint law, up to the PR box at $\lambda = 1$.

Proof. Exact enumeration: every hyperplane through eight affinely independent vertices that has all sixteen vertices on one side (MS3); MS5 checks the path. □

Proposition 73 (fibres over the boundary)**.** *$\mathcal{D}$ has* 2721 *faces. By theorem 64(2) the fibre dimension over the interior of a face G is $|V(G)| - 1 - \dim G$, and the counts are:*

(1) the interior: fibre dimension 7*;*

(2) the 16 *positivity facets, each with* 12 *vertices: fibre dimension* 4*. A structural zero leaves a four-parameter family of joint laws;*

(3) the 8 *Clauser–Horne facets, each with* 8 *vertices: these are* 7*-simplices, so every point of a closed CH facet, including all* 1752 *faces lying in CH facets, has exactly one compatible joint law;*

(4) the remaining lower faces, where the fibre dimension is 0 *on* 400 *further faces and* 1, 2 *or* 3 *on the others (fig. 21(a)).*

Proof. Exact face enumeration (MS4), checked against the numerical fibre dimension at an interior point of 40 sampled faces. □

So the collapse of the fibre at the boundary of $\mathcal{D}$ is a fact to be proved, not a consequence of being on the boundary. It holds at the CH facets because each is a simplex, and it fails at the positivity facets. Those facets are facets of $\mathcal{L}$ as well: crossing them makes a table entry negative, not the data contextual. The contextuality boundary is the part of $\partial\mathcal{D}$ that lies in the relative interior of $\mathcal{L}$, and it consists of points of CH facets.

Corollary 74 (the contextuality boundary is rigid)**.** *For the 4-cycle, every point of $\partial\mathcal{D}$ that is not on a positivity facet has exactly one compatible joint law. Ambiguity at the boundary of $\mathcal{D}$ survives only at structural zeros.*

Proof. Such a point lies on a facet of $\mathcal{D}$ that is not a positivity facet, hence on a CH facet, and proposition 73(3) applies. □

That a CHSH facet carries eight deterministic points is a direct count (theorem 77(1) with $n = 4$). The consequence for identification, that a unique joint law exists along the whole contextuality boundary, follows at once; we have not found it stated in this form. Section 13.19 shows that it holds for every cycle.

13.17 At the contextuality boundary: sections and robustness

Fibre collapse and agreement of the sections are different questions, so we track them separately.

Proposition 75 (the symmetric path)**.** *On the symmetric path the stabiliser of the data, among the automorphisms of the 4-cycle combined with bit flips, contains a group of order* 16 *(all of it for $\lambda = 2/5$), and the fixed set of that group in the fibre is a single point for every $\lambda \in [0, 1/2]$. By theorem 65(3) every equivariant section therefore takes the same value; maximum entropy and leximin agree. The unobserved quantity $q = P(A_0 = 0, A_1 = 0)$, whose variables are never measured together, has identified interval $[\lambda/2, (1-\lambda)/2]$. Its width $(2-S)/4$ goes to* 0 *at the CH facet, where the single joint law gives $q = 1/4$.*

Proof. The fixed equations together with the fibre equations have rank 16 and do not depend on λ (MS6, exact). The interval is computed by linear programming at 21 values of λ, to 10^{-9} (numerical). □

Without symmetry the two sections disagree, and the question becomes whether the disagreement survives at the boundary. MS7 follows straight paths from one interior point to a generic point of three kinds of face (numerical):

toward	fibre width (largest range of a cell)		disagreement $\frac{1}{2}\|s_H - s_{\text{lex}}\|_1$	
	at 0.9	at the face	at 0.9	at the face
a CH facet	0.051	0	0.053	0
a positivity facet	0.377	0.364 (fibre 4)	0.048	0.051
a 6-face with 10 vertices	0.302	0.320 (fibre 3)	0.025	0.034

The disagreement of any two sections is at most 8 times the fibre width, since $\frac{1}{2}\|x-y\|_1 \le \frac{1}{2}\cdot 16\max_c |x_c - y_c|$. So the sections are forced together exactly as fast as the fibre closes, and only where it closes. As the data approach a CH facet every rule gives nearly the same joint law, and the identified set of every unobserved quantity shrinks with the fibre. A rule then cannot be steered without moving the data. At a structural zero the data still allow a four-parameter family, so rules disagree and decisions depend on the rule, as in section 13.14. One step past the CH facet no joint law is left, and every section fails at once. In this sense the contextuality boundary is where marginal data come closest to determining the joint law, immediately before they become inconsistent with every joint law.

13.18 Simpson's reversal in the same framework

A chain cover never records all three variables together, so the sign of an association between the ends given the middle is not in the data. It is chosen by the section.

Proposition 76 (Simpson's reversal)**.** *(1)* Selection sensitivity: made by the rule. *For* $\gamma_1 = \left(\begin{smallmatrix} 5/24 & 11/24 \\ 1/24 & 7/24 \end{smallmatrix}\right)$ *and* $\gamma_2 = \left(\begin{smallmatrix} 3/23 & 11/92 \\ 15/22 & 3/44 \end{smallmatrix}\right)$*, the selected joint law has conditional covariance of* (X_0, X_2) *equal to* $1/276$ *given* $X_1 = 0$ *and* $1/99$ *given* $X_1 = 1$*, but marginal covariance* $-31/4048$*: a Simpson reversal [76]. The Markov section has conditional covariance* 0 *in both slices and marginal* $-49/3036$*. The compatible marginal covariances fill* $[-95/1518,\ 287/6072]$*, and at the upper end the conditional covariances are* ≥ 0 *as well, so the same data admit a joint law with no reversal.*

(2) Information-loss sensitivity: made by coarse-graining the middle. *For* $\gamma_1 = \left(\begin{smallmatrix} 4/21 & 3/14 & 2/21 \\ 1/7 & 2/21 & 11/42 \end{smallmatrix}\right)$ *and* $\gamma_2 = \left(\begin{smallmatrix} 1/9 & 2/9 \\ 65/252 & 13/252 \\ 5/32 & 45/224 \end{smallmatrix}\right)$*, Markov gluing gives marginal covariance* $85/4032 > 0$ *with three middle states and* $-43/6272 < 0$ *after merging the last two. Selection goes from* $13/448$ *to* $-1/42$*.*

Proof. Exact computation (MS8, MS9). □

The two mechanisms differ. In (1) the data leave the joint law underdetermined, and the reversal comes from how a rule completes it; Markov gluing is not neutral either, since setting the conditional association to zero is also a completion. In (2) the reversal comes from forgetting a distinction in the conditioning variable, which is the classical aggregation form of Simpson's paradox. The first is a sensitivity to the choice of section, the second a sensitivity to information loss, and the framework keeps them apart.

What the sections support. Taking sections 13.15 to 13.18 together:

(1) The projection alone sorts the data: empty fibre (contextual), a point (determined), or a positive-dimensional polytope (identified up to a set).

(2) A section is a rule. On chains, maximum entropy is inner-face consistent and gives the dagger symmetric monoidal category CpI. Leximin keeps the symmetries but gives no category.

(3) On the 4-cycle, composition is not the question. What survives is the section and its symmetry, and contextual data have no section at all.

Simpson's reversal and the sign of a composed association are properties of the section or of the resolution of the middle, not of the recorded data. These are the precise forms of the statements that motivated this part. The underlying category is standard, and the new content is the section viewpoint with its exact compatibility and failure results.

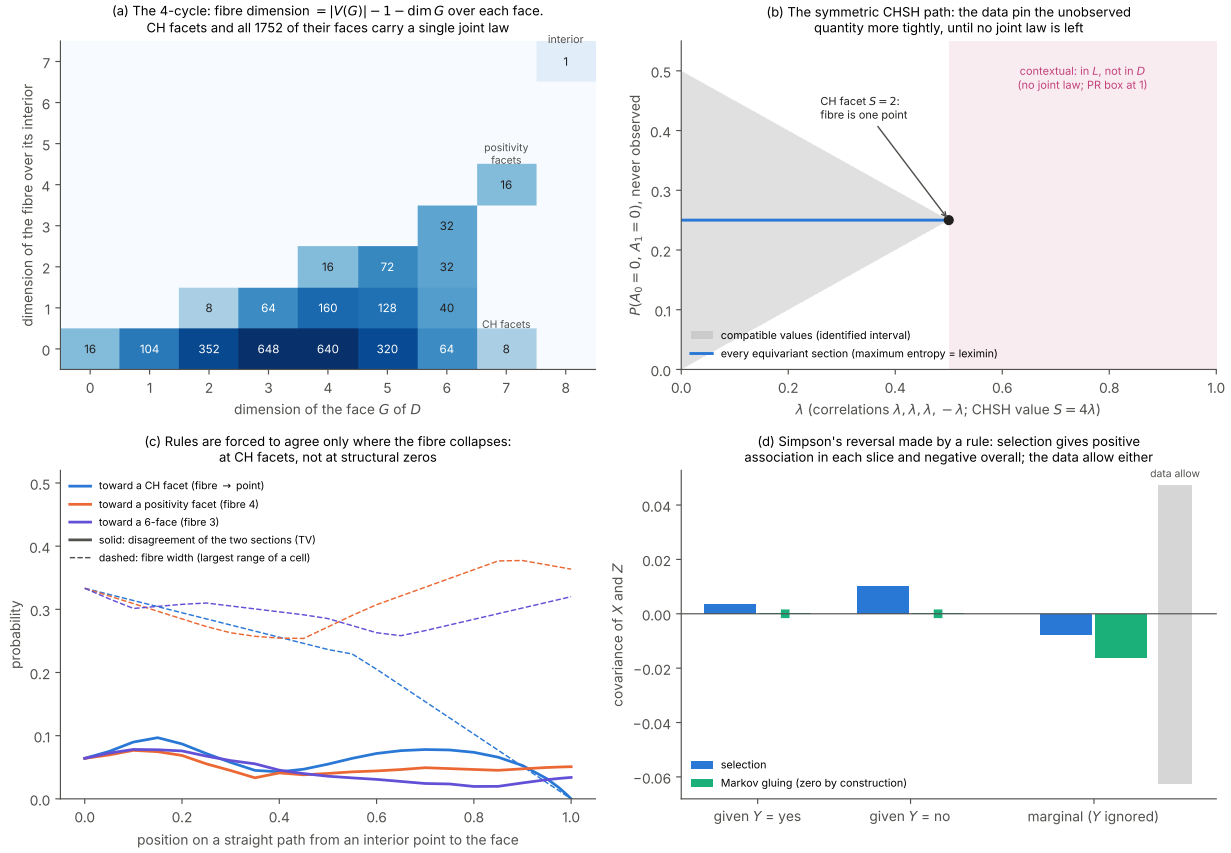


Figure 21: The 4-cycle and Simpson. (a) Faces of the marginal polytope $\mathcal{D}$ by dimension, against the dimension of the fibre over their interior. The Clauser–Horne facets are simplices, so there the data determine the joint law; positivity facets leave a four-dimensional fibre. (b) The symmetric CHSH path: the identified interval of the never-observed $P(A_0 = 0, A_1 = 0)$ closes at $S = 2$, where every equivariant section is forced to $1/4$, and past it the data are contextual. (c) Straight paths to three kinds of face: the two sections disagree until the fibre closes, and only a CH facet closes it. (d) A Simpson reversal made by selection: positive association within each slice of X_1 and negative overall. Markov gluing has zero association within the slices, and the data allow either sign overall.

13.19 Every cycle: the contextuality boundary is a union of simplices

The n-cycle cover has binary variables $X_0, \dots, X_{n-1}$ and contexts $\{X_i, X_{i+1}\}$, indices mod n; $n = 4$ is the CHSH case and $n = 5$ the KCBS case. Data are the n entries $P(X_i = 0, X_{i+1} = 0)$ and the n singles, so $\dim \mathcal{D} = 2n$, and the generic fibre has dimension $2^n - 1 - 2n$. For $n \geq 4$, Araújo et al. [9] show that the facets of $\mathcal{D}$ are the $4n$ local inequalities and the 2^{n-1} noncontextuality inequalities

$$\sum_i \gamma_i \langle X_i X_{i+1} \rangle \;\leq\; n - 2, \qquad \gamma_i \in \{\pm 1\}, \;\; \prod_i \gamma_i = -1,$$

in ±1 notation, where $\langle X_i X_{i+1}\rangle = 1 - 2P(X_i \neq X_{i+1})$ is linear in the data. For $n = 3$ our exact enumeration gives the same list.

Theorem 77 (rigidity of the contextuality boundary for every cycle)**.** *For every $n \geq 3$:*

(1) each noncontextuality facet of $\mathcal{D}$ contains exactly $2n$ deterministic points and is a $(2n-1)$-simplex. So every point of the contextuality boundary, the part of $\partial\mathcal{D}$ in the relative interior of $\mathcal{L}$, has exactly one compatible joint law;

(2) each local facet contains $3 \cdot 2^{n-2}$ deterministic points, so the fibre over its interior has dimension $3 \cdot 2^{n-2} - 2n$; this is 4 for $n = 4$, 14 for $n = 5$ and 36 for $n = 6$.

Proof. (1) A deterministic assignment $x \in \{\pm 1\}^n$ has edge values $e_i = x_i x_{i+1}$ with $\prod_i e_i = 1$, and $\sum_i \gamma_i e_i = n - 2k$, where k counts the edges with $e_i \neq \gamma_i$. Since $\prod \gamma_i = -1$, k is odd, so the inequality holds, with equality exactly when $k = 1$. Choosing the one edge where $e_i \neq \gamma_i$ (n ways) fixes every e_i, and x is then fixed up to a global flip (2 ways): $2n$ points. They span a facet, of dimension $2n - 1$, so they are affinely independent, and the facet is a simplex. Theorem 64(2) then gives a one-point fibre over every point of it, its faces included. (2) The local facet $P(X_i = a, X_{i+1} = b) \geq 0$ contains the assignments that avoid the pair (a, b) on that edge, which are three quarters of the 2^n. Apply theorem 64(2) with $\dim G = 2n - 1$. □

MS10 recomputes all facets of $\mathcal{D}$ exactly for $n = 3, 4, 5$, independently of [9]: 16, 24 and 36 facets, the counts $4n + 2^{n-1}$, with the vertex numbers above and the noncontextuality facets in exactly the γ form. For $n = 6, 7, 8$ it checks that every listed inequality is valid and a facet with the stated vertex count. Completeness of the list there is taken from [9]. So the rigidity found for CHSH is not special to the 4-cycle. On every cycle the contextuality boundary is a union of simplices, and the data there determine the joint law. The ambiguity left at a structural zero grows exponentially in n, while the contextuality boundary stays rigid. Whether the noncontextuality facets of other scenarios are simplices is a separate question that we have not tested. Where they are not, theorem 64(2) measures how much ambiguity survives at the boundary.

13.20 How little poisoning, in many dimensions

Section 13.14 measured edits of one binary chain. Here the cover is general, the decision concerns a quantity that is never observed, and three different things are counted: how many cells of the data an attacker must change, how much can change with no data edit at all, and how a change over time is explained afterwards.

Setting. Data $d \in \mathcal{D}$ give the context tables, whose entries are affine in d: $Ed + e_0$, with $4n$ entries for the n-cycle. An edit δ costs $c(\delta) = \frac{1}{2}\|E\delta\|_1$, the total variation summed over the context tables. When each context is a separate sample this is the fraction of records edited, summed over the samples, and on one table it is the $2|s|$ of proposition 61. An edit is *undetectable* when $d + \delta \in \mathcal{D}$. Then the tables still agree on shared margins and are noncontextual, so no margin check or contextuality test sees it. The undetectable directions lie in the *consistent-edit space* $\Delta = \operatorname{range} E$, of dimension $\dim \mathcal{D}$ ($2n$ for the n-cycle). The decision is $w \cdot s(d) \geq \tau$ for a section s and a quantity w on joint laws whose variables are never observed together, here $P(X_0 = X_2)$ on the n-cycle and $P(A_0 = A_1)$ on CHSH. Write $[L(d), U(d)]$ for its identified interval.

Definition 78 (flip costs)**.** For a decision with $w \cdot s(d) > \tau$: $c_{\text{some}}(d)$ is the least cost of an undetectable edit after which $L \leq \tau$ (some rule flips), $c_{\text{all}}(d)$ the least cost after which $U \leq \tau$ (every rule flips), and $c_s(d)$ the least cost after which $w \cdot s \leq \tau$.

Proposition 79 (two thresholds)**.** *(1)* $c_{\text{some}} \le c_s \le c_{\text{all}}$ *for every section* s*. The same order holds for the first flipping time along any fixed direction.*

(2) c_{some} *is the value of one linear program: minimise* $\frac{1}{2}\|E(\pi x - d)\|_1$ *over joint laws* x *with* $w \cdot x \le \tau$*. The edited data* πx *lie in* $\mathcal{D}$ *by construction. The set* $\{U \le \tau\}$ *is the union of the half-spaces* $y \cdot d' + y_0 \le \tau$ *over the vertices* (y, y_0) *of the dual polyhedron* $\{\pi^\top y + y_0 \mathbf{1} \ge w\}$*, so* c_{all} *is the least of finitely many linear programs.*

(3) $c_{\text{some}}(d) = 0$ *exactly when* $L(d) \le \tau$*. When the decision is not identified, a change of rule flips it with no change to the data.*

Proof. (1) $w \cdot s(d') \in [L(d'), U(d')]$ for every d'. (2) $U(d') = \max\{w \cdot x : \pi x = d',\, x \ge 0,\, \mathbf{1} \cdot x = 1\}$ is, by linear-programming duality, the minimum of $y \cdot d' + y_0$ over the dual vertices. (3) follows from the definition. PF3 compares the costs at 80 points, and the first flipping times for the maximum-entropy section along 18 directions (numerical). □

A *circuit* of Δ is a nonzero edit whose entry vector $E\delta$ has minimal support. Two vectors are *conformal* when no coordinate has opposite signs in them. Every vector of a subspace is a conformal sum of at most $\dim \Delta$ circuits [69, 67], and the cost is additive over a conformal sum.

Proposition 80 (sparsity of undetectable flips)**.** *(1) Some cheapest undetectable flip, for* c_{some} *or* c_{all}*, is a conformal sum of at most* $\min(a, \dim \Delta)$ *circuits. Here* a *is the number of constraints of the flip region (pieces of* $\{L \le \tau\}$ *or* $\{U \le \tau\}$*, and facets of* $\mathcal{D}$*) that are tight at it.*

(2) For the n*-cycle,* Δ *has* 188 *circuits for* $n = 4$ *(supports* $4, 6, 8, 9$*, in numbers* $20, 32, 72, 64$*) and* 481 *for* $n = 5$ *(supports* $4, 6, 8, 10, 11$*). Every support is at most* $2n + 1$*. For* $n = 4, \dots, 8$ *there are exactly* $5n$ *circuits of support* 4*: the* n *margin-preserving swaps of proposition 61, and* $4n$ *shifts of one single margin, each moving mass between two cells in each of the two tables that contain the variable.*

Proof. (1) Fix the sign pattern of a cheapest flip δ^*. The edits in Δ with that pattern form a cone generated by the circuits conformal to it, and on that cone the cost is linear, $c(\sum \lambda_k C_k) = \sum \lambda_k c(C_k)$. Minimising it subject to the region's linear constraints is a linear program in $\lambda \ge 0$, and it attains the value $c(\delta^*)$. A vertex solution has at most as many nonzero λ_k as tight constraints, and any cheapest flip is a conformal sum of at most $\dim \Delta$ circuits by the result just cited. (2) Exact enumeration (PF1): every circuit is the kernel of $\dim \Delta - 1$ independent rows of E, and each candidate is verified with integer arithmetic. □

How often a single circuit is enough is a numerical question (PF2). Take a gap of 0.01 below L, and 30 random interior data points for each n. One support-4 circuit is already a cheapest flip in 30/30, 30/30, 28/30, 28/30 and 24/29 of the cases for $n = 4, \dots, 8$; for $n = 8$ that is 4 cells out of 32. On the 4-cycle, vertex optima use one circuit in 37 of 40 cases at gap 0.01, and one, two or three circuits in 19, 16 and 5 cases at gap 0.05. So the smallest poisoned footprint does not grow with the number of variables. It is four cells in two tables, or in one, and larger gaps need a few circuits because the flip region has corners. This complements the results for samples [64, 70, 65]. Those count records or coordinates of the inputs. Here the count is of cells of a marginal table, and the undetectable edits form a structured subspace.

Placement and the hidden structure. Let the hidden structure be the support S of the true joint law, the set of readings that occur. The data then lie in the relative interior of the face $G(S)$ of $\mathcal{D}$, the smallest face containing $\pi(S)$. Let $\Pi(S)$ be the number of signed circuits along which a small step stays in $\mathcal{D}$. It is the same at every point of the relative interior, and it counts the places an attacker can start an undetectable edit.

Proposition 81 (placement)**.** *(1) $S \subseteq S'$ implies $\Pi(S) \leq \Pi(S')$. Simpler hidden structure leaves fewer undetectable placements.*

(2) $\dim G + \dim(\textit{fibre over } G) = |V(G)| - 1$. *The freedom of the hidden structure splits into room to move the data undetected (the face dimension) and room for the rule to move the answer at no data cost (the fibre dimension).*

(3) On the 4-cycle, over all 2721 *faces: a vertex allows* 13 *of the* 376 *signed circuits, a CH facet* 252*, a positivity facet* 287*, and the interior all* 376*. At fixed face dimension a larger fibre mostly gives more placements, for instance* 165.4*,* 170.0 *and* 186.7 *on average for fibres* 0*,* 1*,* 2 *at dimension* 5*. The one exception is dimension* 6*, where fibre* 1 *averages* 204.6 *against* 207 *for fibre* 0*.*

Proof. (1) $G(S) \subseteq G(S')$, so every facet containing $G(S')$ also contains $G(S)$, and the feasible directions at $G(S)$ are those at $G(S')$ cut by more inequalities. (2) is theorem 64. (3) is an exact count with rational facet normals and integer circuits (PF4), with 0 violations of (1) over all inclusions of faces. □

At matched sparsity the split matters. Among faces with 8 readings, the rigid CH facets allow more undetectable placements (252) than faces with fibre 1 (about 205) or fibre 2 (about 187). Rigidity does not make placements scarce. It removes the free route: every flip must move the data. With a budget the decisive question is whether the decision is identified at the point (PF5). Take a gap $g = 0.05$ below the true value, on the sparsity-matched faces of Study 18 and three interior points. Wherever the decision is identified (all CH and fibre-1 faces tested, and one fibre-2 face), the cheapest flip costs at least g, and at most 4 single circuits achieve one within $1.5g$. Wherever it is not identified (the other two fibre-2 faces and the interior points), $c_{\text{some}} = 0$ along all 376 placements, while c_{all} ranges from 0.14 to 0.45. A defender who reports $[L, U]$ instead of one rule's value forces the attacker to pay c_{all}.

Over time: removal as a contravariant functor. Decisions are recomputed as data arrive, so the hidden structure, the face and the interval all move. An *information state* is a pair (B, C): a set of batches and a set of observed contexts, ordered by inclusion. Its object $R(B, C)$ is the integer pooled tables of the contexts in C over the batches in B. For $(B', C') \leq (B, C)$ the *removal map* $R(B, C) \to R(B', C')$ subtracts the counts of the batches in $B \setminus B'$ and forgets the contexts in $C \setminus C'$.

Proposition 82 (removal)**.** *(1) Removal maps compose, and removing nothing is the identity, so R is a functor from the opposite of the poset of information states to sets of tables. Removing batches and forgetting contexts commute.*

(2) Forgetting contexts can only widen the identified interval: $C' \subseteq C$ implies $[L_C, U_C] \subseteq [L_{C'}, U_{C'}]$. So the value of a context, the width it removes, is nonnegative.

(3) On a stationary stream with pooled clean data d, add one batch of k undetectable units with data p. Then $d_t = (1 - q_t)d + q_t p$ with $q_t = k/(N_t + k)$. Every rule is flipped exactly while

$q_t \geq q^*$, *where* q^* *is the least crossing point* $(\tau - y \cdot d - y_0)/(y \cdot (p - d))$ *over the dual vertices with* $y \cdot (p - d) < 0$. *So the flip lasts while* $N_t \leq k(1 - q^*)/q^*$.

Proof. (1) Subtraction of counts and deletion of coordinates compose and commute. (2) Fewer constraints give a larger fibre. (3) U is a minimum of affine functions, so it is concave along the segment from d to p. Its sublevel set there is an interval ending at p, and it starts at the first crossing of a dual-vertex function. PF6 checks (1) on 200 random chains and squares with integer counts, PF7 checks (2) at every time and every sub-cover, and PF8 checks (3) at 30 times (numerical). □

The removal maps answer why a decision changed. Compare the object at the later state with the objects obtained by removing one batch, or forgetting one context, at a time. The changes that revert the decision are the explanations. The exact comparison is cheap here. When it is not, a first-order estimate, the gradient of the decision value times the removed batch's contribution, ranks the candidates, as in the influence approach of [64]. An explanation need not be unique: near a threshold, removing a clean batch can also revert a decision.

Example 83 (one poisoned batch)**.** Batches of 1200 units, each reported in all four contexts of CHSH, decision $P(A_0 = A_1) \geq 0.45$ by maximum entropy (PF6–PF9, fig. 23). Until $t = 5$ the world is the first CH-facet law of Study 18. Its interval has width 0, so the decision is identified and robust. From $t = 6$ the hidden structure changes to 0.8 times that law plus 0.2 uniform, the data move into the interior, and the interval opens. At $t = 10$ an attacker adds the cheapest batch that flips every rule: 2298 units, all of one reading, against 12000 clean units ($q^* = 0.161$). Every rule is flipped at $t = 10$ only, but the maximum-entropy decision stays flipped for $t = 10$ to 15. At each of those times, removing the poisoned batch is the only single removal that reverts the decision. The first-order estimate of its effect is within 6% of the exact change (0.075 against 0.072 at $t = 10$). At the last time, forgetting a context widens the interval by 0.295 or 0.38. The batch is consistent and noncontextual, so no marginal test detects it, but it consists of a single reading, and comparing the objects over time detects it at once.

Remark 84 (reading a change cell by cell)**.** The removal maps attribute a change to batches; conformal sums attribute it to cells. A change $\delta = d_t - d_{t-1}$ between consistent states lies in Δ, so it is a conformal sum $\sum_k \lambda_k C_k$ of at most $\dim \Delta$ unit-cost circuits. Nothing cancels, so $c(\delta) = \sum_k \lambda_k$, and the first-order effect on the decision value ϕ splits the same way, $\nabla\phi \cdot \delta = \sum_k \lambda_k \nabla\phi \cdot C_k$. The decomposition is not unique. But the least share of $c(\delta)$ that any decomposition must put on circuits of more than four cells is an invariant of δ, computed by one linear program; call it the *coordination* of δ. Figure 22 shows each kind of circuit on the four CHSH tables. In example 83 (PF10), the clean steps before the attack, including the change of hidden structure at $t = 6$, have coordination 0: they are sums of margin shifts and swaps. The poisoned step has coordination 0.659, and removing the poisoned batch later has 0.652. The clean steps after the attack have 0.393, because each partly undoes the poison. The poisoned step's three largest contributions to the decision are a margin shift (-0.025), a 6-cell circuit (-0.022) and a 9-cell circuit (-0.014), out of a first-order total of -0.067. One example does not make coordination a detector. It shows what the conformal-sum theorem buys: an undetectable change has a canonical vocabulary of elementary edits, and its cost and its first-order effect are additive over them.

What is proved here is elementary: the two thresholds, the circuit bound, monotonicity of placement, the removal laws and the additivity of cost over conformal sums. The circuit census and the face table are exact computations, and the frequencies of one-circuit optima, the budget comparison and the time example are numerical. We have not found these statements for marginal data in the literature on poisoning, which counts records or input coordinates.

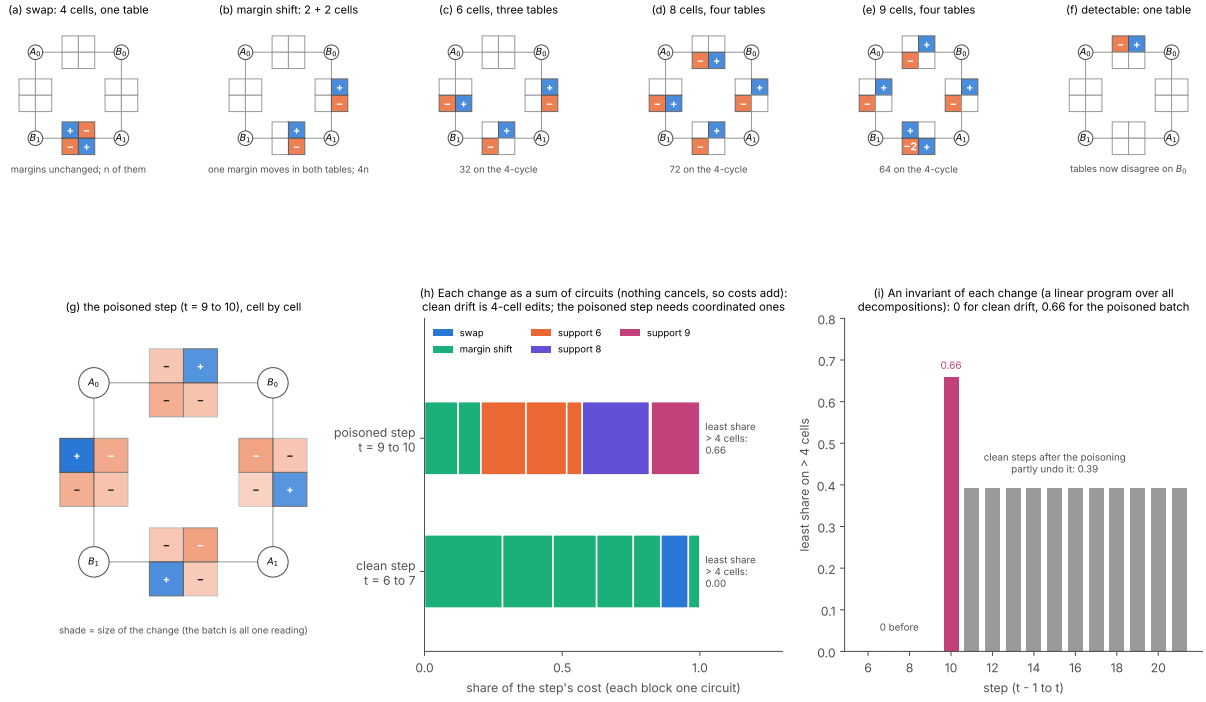


Figure 22: The kinds of undetectable edit, and one change read as a sum of them. Each diagram shows the four tables of the CHSH cover on the sides of the cycle A_0–B_0–A_1–B_1 (rows: the A variable; columns: the B variable). A + cell gains mass and a − cell loses it. (a) A swap: four cells in one table, all margins fixed. (b) A margin shift: one single margin moves, by two cells in each of the two tables that contain it. (c)–(e) Circuits of 6, 8 and 9 cells that coordinate three or four tables. (f) A detectable edit: one table alone, so the tables disagree on a shared margin. (g) The poisoned step of example 83, cell by cell. (h) Its conformal decomposition beside a clean step's: clean drift is a sum of 4-cell edits, the poisoned step needs coordinated ones. (i) The coordination (the least share any decomposition puts on circuits of more than 4 cells) of every step.

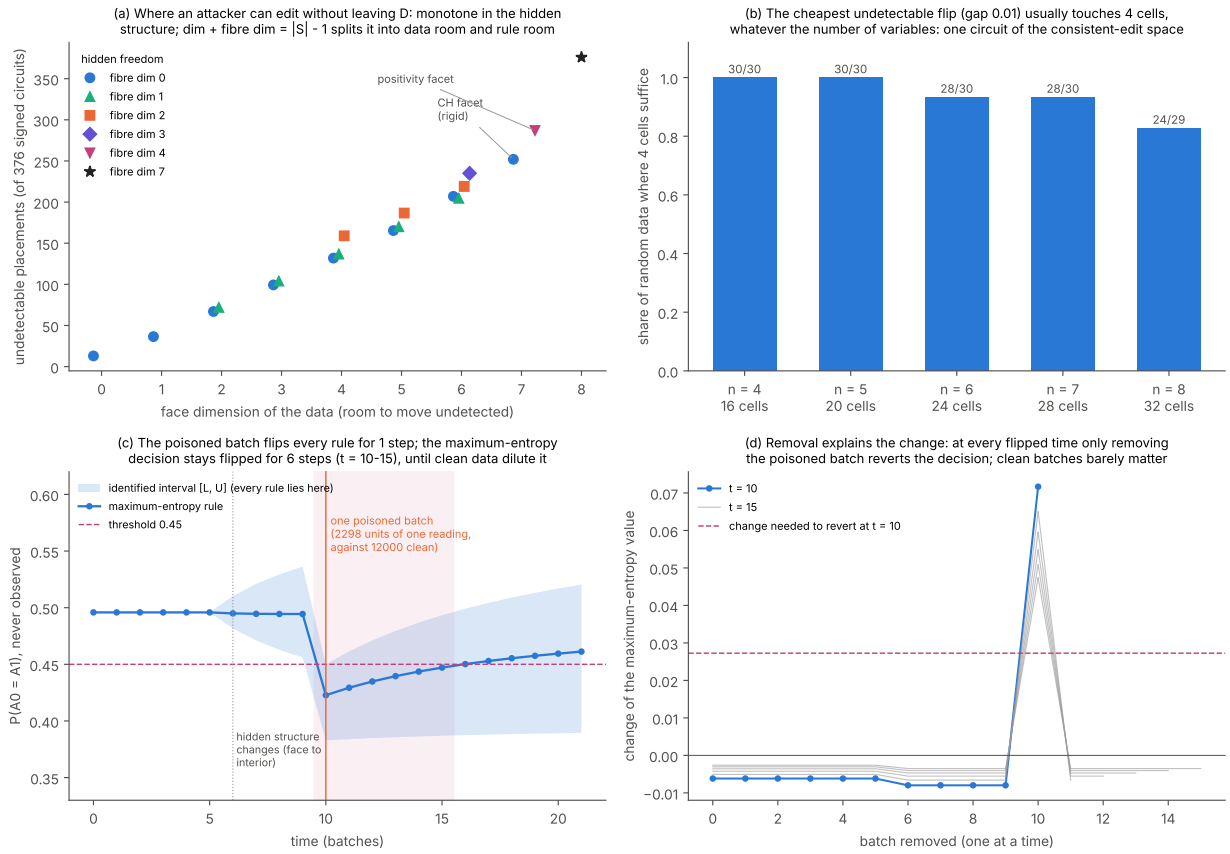


Figure 23: How little poisoning. (a) Undetectable placements (signed circuits that keep the data in $\mathcal{D}$) against the face dimension of the data, by fibre dimension, over all 2721 faces of the 4-cycle. The count is monotone in the hidden structure, and $\dim G + \dim(\text{fibre}) = |S| - 1$ splits that structure into room for the data and room for the rule. (b) The cheapest undetectable flip (gap 0.01) is usually one circuit of 4 cells, for every n up to 8. (c) One poisoned batch at $t = 10$, after the hidden structure changed at $t = 6$. The identified interval (shaded) flips below the threshold for one step; the maximum-entropy rule stays flipped for six. (d) Removing one batch at a time at the flipped times: only the poisoned batch reverts the decision.

13.21 The geometry of poisoning

A poisoning edit moves the data d, and with them the fibre J_d of compatible joint laws. Three effects can be separated: the fibre can *deform* (change dimension or size), the data can *cross* the contextuality boundary (the fibre empties), and the *selected* law can move. Section 13.20 gave the costs of flips. Here each effect gets a precise statement, and the decision gets a radius that the defender can compute from the data alone.

Proposition 85 (certified radii and the three effects)**.** *Let $\mathcal{V}_L$ and $\mathcal{V}_U$ be the dual vertices with $L(d) = \max_{(y,y_0)\in\mathcal{V}_L}(y\cdot d + y_0)$ and $U(d) = \min_{(y,y_0)\in\mathcal{V}_U}(y\cdot d + y_0)$, and let $\mathcal{C}_1$ be the circuits scaled to unit cost.*

(1) Certified radii. *L and U are Lipschitz in the edit cost, with constants $\Lambda_L = \max_{\mathcal{V}_L\times\mathcal{C}_1}|y\cdot C|$ and $\Lambda_U = \max_{\mathcal{V}_U\times\mathcal{C}_1}|y\cdot C|$. Hence $c_{\mathrm{some}}(d) \geq (L(d)-\tau)/\Lambda_L$ and $c_{\mathrm{all}}(d) \geq (U(d)-\tau)/\Lambda_U$.*

(2) Deformation. *The width $U - L$ changes by at most $(\Lambda_L + \Lambda_U)\, c(\delta)$, while the fibre dimension can jump. By theorem 64 it rises from $|V(G)| - 1 - \dim G$ to $|V(G')| - 1 - \dim G'$ when an edit moves the data from a face G into a larger face G'.*

(3) Crossing. *The cost distance from $d \in \mathcal{D}$ to the contextuality boundary is $\min_h \sigma_h(d)/\max_{C\in\mathcal{C}_1}(-h\cdot C)$ over the CH facets $h\cdot d + h_0 \geq 0$, where $\sigma_h(d) = h\cdot d + h_0$ is the slack. Along a line $d + tv$, the data first become contextual at $t = \min\{\sigma_h(d)/(-h\cdot v) : h\cdot v < 0\}$, and from then on the fibre is empty and the contextual fraction positive.*

(4) Selection. *L and U are continuous whatever the rule. A section given by a continuous objective with a unique optimum on every fibre (maximum entropy, or a fixed linear objective with unique optima) is continuous on $\mathcal{D}$. A selected law can jump only through ties or a rule that depends on the data discontinuously.*

Proof. (1) A maximum (or minimum) of affine functions is Lipschitz with constant equal to the largest dual norm of their gradients. The dual norm of y for the cost c is $\max y\cdot\delta$ over the unit cost ball, whose extreme points are the unit circuits [69]. A flip must move L (resp. U) by $L - \tau$ (resp. $U - \tau$). (2) Apply (1) to both ends. The dimension statement is theorem 64. (3) In a polyhedral norm, the distance to a hyperplane is the slack divided by the dual norm of its normal. For a point of a convex polytope, the distance to the boundary is the least distance to a facet hyperplane. The line formula is the first zero of the slacks. (4) The fibre map is continuous with compact values, being Lipschitz [83]. By Berge's maximum theorem the optimal set is upper hemicontinuous, hence continuous where it is a single point. L and U are optimal values, which (1) shows to be Lipschitz. □

For the decision quantity $P(A_0 = A_1)$ on CHSH, the constants are exact (PG1). There are six dual vertices on each side and 376 unit circuits, and $\Lambda_L = \Lambda_U = 1$; for $P(A_0 = 0, A_1 = 0)$ both constants are $1/2$. So no undetectable flip of the some-rule decision costs less than the gap $L - \tau$. Over 75 cases at 39 data points the bound is attained in 45 and is never off by more than a factor 2 (PG2). For every rule the median ratio of the actual cost to the bound is 1.10. The distance formula in (3) agrees with a direct linear program to machine precision (PG4). Along 20 straight data paths, both maximum entropy and a fixed-objective vertex rule move by at most 0.019 in total variation per step of 0.01 (PG6). So, unlike what one might expect, rules that pick vertices do not jump as the data move, as long as their optimum is unique. What poisoning can make discontinuous is the decision at its threshold, and the dimension of the fibre, but not its width and not a well-defined rule.

Example 86 (poisoning in stages)**.** Start from the first CH-facet law of Study 18, where $P(A_0 = A_1) = 0.496$ is identified (the fibre is a point), with 12000 clean units and threshold $\tau = 0.45$ (PG5, fig. 24). Compare two poisoning routes that add units of a single reading.

- *Route A*, the cheapest every-rule attack, injects the reading $(0, 1, 1, 0)$. That reading is a vertex of the same CH facet, so the data slide along the facet. At 600, 1200 and 1800 units the fibre is still a point, with $L = U = 0.472$, 0.451, 0.431. Some rule and every rule flip together, at a poisoned share of 0.0924 (about 1222 units).

- *Route B* injects $(1, 0, 1, 0)$, which lies off the facet. After the first unit the fibre dimension jumps from 0 to 7, but the width starts at 0 and grows at most twice as fast as the cost. The interval is $[0.472, 0.567]$ at 600 units, $[0.451, 0.633]$ at 1200, $[0.413, 0.747]$ at 2400, $[0.354, 0.640]$ at 4800 and $[0.275, 0.498]$ at 9600. Some rule flips at the same share 0.0924, but every rule flips only at 0.4977, about 11900 units: 5.4 times the share and nearly ten times the units. The maximum-entropy value first rises to 0.516 and falls below τ only after about 6060 units. The distance to the contextuality boundary is 0 throughout route A, and on route B it is 0.095, 0.182, 0.333, 0.571 and 0.500 at those stages.

- *Overreach.* Mix in the PR box that violates this CH facet. From the rigid start the data become contextual at once: the fibre is empty and the contextual fraction positive for every share above 0. From route B at 2400 units they become contextual at share 0.2501, where the facet slacks predict 0.2500. In all 31 cases the fibre is empty exactly when the contextual fraction is positive.

The cheapest attack on a rigid point keeps the point rigid. It slides along the facet, so the decision stays identified and simply moves. Deformation is what an attack off the facet produces, and it buys ambiguity rather than a decision: the interval straddles the threshold for a long time before every rule flips. A defender who reports $[L, U]$ sees route B as rule-dependent from share 0.0924 on, and route A as a clean flip. At a rigid point, complete determination lies next to inconsistency. One contextual push empties the fibre and is detected at once. So an attacker must stay inside $\mathcal{D}$, and a contextuality alarm is free for the defender. The literature's geometries of poisoning (bilevel, influence, metric) describe the learner. These statements describe the space of compatible explanations. The two meet in the first-order estimate of remark 84.

13.22 Bayesian inversion and sections

Bayesian inversion reverses a stochastic map relative to a prior. From p and $K(y \mid x)$ it gives $q = pK$ and $K^\dagger(x \mid y) = p(x)K(y \mid x)/q(y)$ wherever $q(y) > 0$. Categorically it is disintegration of the joint law in the other direction [34]. Parzygnat separates ordinary inverses, disintegrations and Bayesian inverses, and extends them to quantum Markov categories [68]. Bohinen and Perrone show that when an (a.s.-compatibly representable) Markov category has conditionals, the multiplication of its probability monad is weakly cartesian [63, Thm. 3.1]. This covers the Giry monad on standard Borel spaces [63, Cors. 3.2, 3.9]. They avoid calling the property a Beck–Chevalley condition, because that term has a different established meaning for fibrations and descent [63, Rem. 2.25]. Remark 68 uses the term only for our own reading. This subsection places inversion relative to sections.

Proposition 87 (inversion and sections)**.** *(1) In* Cpl *the dagger is Bayesian inversion: if* $\gamma(x, y) = p(x)K(y \mid x)$*, then* $\gamma^\top(y, x) = q(y)K^\dagger(x \mid y)$*. The dagger is total, although inversion of kernels is partial: at a null event every choice of* $K^\dagger(\cdot \mid y)$ *gives the same* $\gamma^\top$*. So*

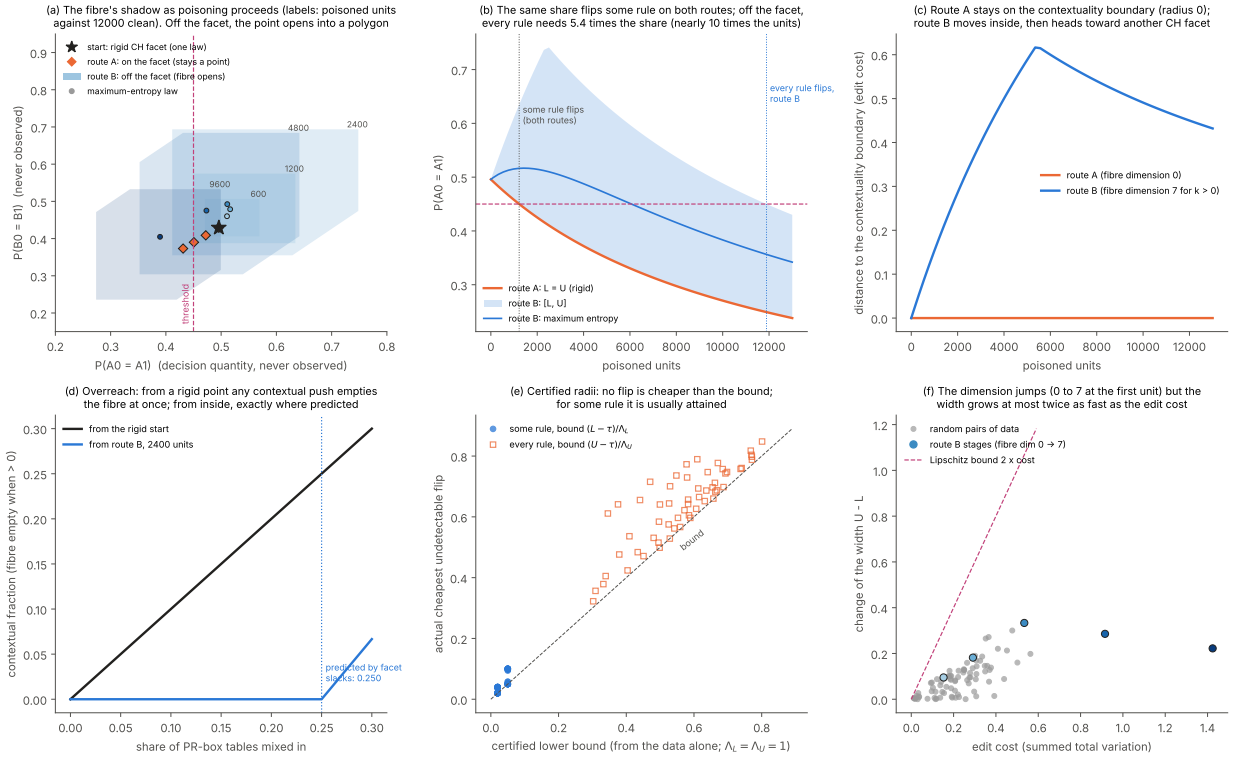


Figure 24: The geometry of poisoning, in stages (example 86). (a) The shadow of the fibre on two never-observed quantities. From a rigid CH-facet point (star), route A (diamonds) stays a point; route B opens into polygons as poisoned units are added (labels), with the maximum-entropy law inside. (b) The identified interval: the same share flips some rule on both routes, but flipping every rule off the facet takes nearly ten times as many units. (c) Distance to the contextuality boundary. (d) A PR-box overreach: from the rigid point any contextual push empties the fibre; from inside, exactly where the facet slacks predict. (e) Certified radii against actual cheapest flips. (f) Width change against edit cost: the dimension jumps, the width obeys the Lipschitz bound.

for one coupling, inversion is a symmetry and not a choice. Given the prior and the kernel, the fibre is a point.

(2) Let s be a section of the chain cover that is reversal-equivariant: $x_{ijk} \mapsto x_{kji}$ carries $s(\gamma_1, \gamma_2)$ to $s(\gamma_2^\top, \gamma_1^\top)$. Then composition by s commutes with inversion: $(\gamma_1 \circ_s \gamma_2)^\top = \gamma_2^\top \circ_s \gamma_1^\top$. Markov gluing (the dagger functor of Cpl*) and selection (proposition 57(3)) both qualify.*

(3) A rule that is not reversal-equivariant need not commute. The vertex of the chain fibre minimising $\sum G_i H_k x_{ijk}$, with $G = (0, 2, 1, 3)$ and $H = (0, 1, 2, 3)$, commutes on only 13 of 40 random chains with ends of up to 3 states. On chains with binary ends, every rule that minimises a fixed linear objective with a unique optimum commutes.

(4) Null events never reach a composite, while structural zeros do matter. A middle state with $q(y) = 0$ contributes nothing to any composite, whatever kernel is attached to it. A zero cell with full-support margins leaves every Bayesian inverse unique, yet the composite's identified interval keeps positive width, so the rule still decides.

Proof. (1) $\gamma^\top(y, x) = p(x)K(y \mid x) = q(y)K^\dagger(x \mid y)$, and the row of a null y is 0 whatever $K^\dagger(\cdot \mid y)$ is. (2) Reversal maps the fibre of (γ_1, γ_2) bijectively onto that of $(\gamma_2^\top, \gamma_1^\top)$. The end marginal of a reversed table is the transpose of the end marginal. (3) For binary ends the fibre is the box $\prod_b [L_b, U_b]$ in the coordinates x_{0b0}, which reversal fixes. A linear objective acts on it through the coefficients $c_{0b0} - c_{0b1} - c_{1b0} + c_{1b1}$, which reversal also fixes, so the optimum is the same point. The count is exact (IS3). (4) The row y of γ_2 is $q(y)K(y, \cdot) = 0$ (IS4). The interval is exact on six chains with a zero cell (IS5), where Markov gluing and selection disagree in five. In the sixth, Markov gluing happens to be the midpoint. IS1 and IS2 check (1) and (2) exactly on 60 and 40 random rational examples. □

So the question whether Bayesian inversion is a canonical section of the marginal projection has a precise answer. It is not a section but a symmetry of the projection. Every composition rule that respects the symmetry commutes with it, and a rule that does not respect it can fail to commute; we have not shown that every rule that commutes is reversal-equivariant. A prior and a kernel determine the joint law, so inverting them involves no selection. Selection enters only when the joint law is not determined, as for the composite of a chain, and there it is compatible with inversion when the rule is reversal-equivariant. Information destroyed by a map is not recovered by inverting it. In the merging example $f(a) = f(b) = 0$, $f(c) = 1$ with prior $(0.2, 0.5, 0.3)$, the inverse $P(a \mid 0) = 2/7$, $P(b \mid 0) = 5/7$ is fixed by the prior, not chosen.

Remark 88 (order drift as Bayesian inconsistency)**.** A model that generates variables one at a time defines, for each order σ, a joint law Q_σ, the product of its conditionals in that order. Bayes' rule between two orders says $Q_\sigma = Q_{\sigma'}$: the conditionals of one order are the Bayesian inverses of the other's. The order drift of Studies 17 and 18, the mean total variation between the Q_σ, therefore measures how far a learned model is from being a single probabilistic object that inverts consistently. Study 18's consistency penalty is the squared residual of Bayes' rule in log form at sampled assignments. On the sparsity-matched faces it lowered the drift from 0.315, 0.281 and 0.174 to 0.011, 0.007 and 0.010 (IS6, from the frozen output). A model with drift has no single joint law and so no single inverse: a question asked forward and backward can get answers that no joint law reconciles. By proposition 87(2), only a coherent model whose completion rule is reversal-equivariant answers both directions consistently.

13.23 The geometry of decisions and of the value of information

The fibre J_d is a credal set in the sense of imprecise probability: a polytope of compatible laws, finitely generated by its vertices. This subsection asks which part of its geometry matters for a decision, what information is worth, and how stable a decision is under a disturbance whose cause the decision-maker cannot see. Actions a carry utilities u_a on joint readings, so $\mathbb{E}_P u_a = u_a \cdot P$ is linear in P. Two kinds of information are kept apart:

- *structural* information is a new context, a pair never measured together; it shrinks the fibre;
- *instance* information is a signal Z about one unit's state; it leaves the population fibre unchanged.

Proposition 89 (decisions over a fibre)**.** *(1) The utility image $\{(u_a \cdot P)_a : P \in J_d\}$ is a polytope. A decision is* identified*, meaning one action is optimal for every compatible law, exactly when the image lies in one cell of the arrangement of indifference hyperplanes $u_a \cdot P = u_b \cdot P$. The fibre's dimension plays no role.*

(2) The value of resolving the fibre, under minimax regret, is $R(d) = \min_a \max_b U_{ab}(d)$, where U_{ab} is the upper end of the identified interval of $(u_b - u_a) \cdot P$. It needs $|A|^2$ linear programs, and $R(d) = 0$ exactly when the decision is identified. If a_0 is optimal for some law in J_d, then $R \le \max_b(U_{a_0 b} - L_{a_0 b})$, so resolving the fibre is worth at most its width along utility differences. R is Lipschitz in the edit cost, with constant the largest Lipschitz constant of the U_{ab} (proposition 85(1)). Observing more contexts never raises it. It is 0 on a rigid face for every utility.

(3) Mixing data mixes fibres: $\lambda J_d + (1-\lambda) J_{d'} \subseteq J_{\lambda d + (1-\lambda) d'}$. So along any segment of data L is convex and U and the width $U - L$ are concave. Poisoning by adding batches moves along such segments.

(4) Take act *and* abstain *with symmetric payoffs $\mathbb{1}[A_0 = A_1] - \tau$ and $\tau - \mathbb{1}[A_0 = A_1]$. Then $R = 2\min\{(\tau - L)_+, (U - \tau)_+\}$, which is concave along segments where it is positive. The value of randomising, the gain of the best mixed action over the best pure action under the worst compatible law, is exactly $R/2$.*

Proof. (1) A linear image of a polytope is a polytope, and an action is optimal throughout J_d exactly when every $U_{ab} \le 0$. (2) Minimax regret over a convex set of laws and finitely many actions: the inner maximum over P of $(u_b - u_a) \cdot P$ is a linear program. If a_0 is optimal at $P_0 \in J_d$, every $L_{a_0 b} \le 0$, so $U_{a_0 b} \le U_{a_0 b} - L_{a_0 b}$. Lipschitz continuity: minima and maxima preserve Lipschitz constants. Monotonicity: more contexts give a smaller fibre, hence smaller maxima. Rigid faces: every $U_{ab} = -U_{ba}$, so some action has all $U_{ab} \le 0$. (3) Convex combinations of compatible laws are compatible with the combined data, and L and U are optimal values over these sets. (4) Here $U_{\text{act,abstain}} = 2(\tau - L)$ and $U_{\text{abstain,act}} = 2(U - \tau)$. Each is concave along a segment, and so is their minimum. Acting with probability x earns $(2x - 1)(\phi - \tau)$. When $L < \tau < U$, the choice $x = 1/2$ guarantees 0, while the best pure action guarantees $-\min(\tau - L, U - \tau)$. Checks VG1–VG3 and VG5 test this numerically: the Lipschitz ratio is at most 0.53 over 40 pairs, and removing a context raises R in 118 of 120 cases. □

Proposition 90 (instance information)**.** *Let Z be a signal of accuracy α about $T = \mathbb{1}[A_0 = A_1]$, and let $\phi = P(T = 1)$. For the payoffs of proposition 89(4), write $a_z = P(T = 1, Z = z) - \tau P(Z = z)$.*

(1) Under a single law (any section), the value of the signal is $\sum_z |a_z| - |\sum_z a_z| \geq 0$. *This is Good's inequality as a triangle inequality [20]. It is* 0 *exactly when no posterior lies on the other side of* τ *from the prior, so the signal has decision value only when it can move the decision.*

(2) A robust decision-maker who commits to a policy $z \mapsto a$ *can always ignore the signal, so the signal's robust value is never negative [61].*

- *With mixed policies allowed, the minimax theorem makes the robust value* $\min_{\phi\in[L,U]} \sum_z |a_z| - \min_{\phi\in[L,U]} |\phi - \tau|$, *which is* 0 *for an uninformative signal.*
- *With pure policies only, the robust value equals the mixed-policy value plus the value of randomising without the signal, minus the value of randomising with it.*

(3) Information and value differ. Below a critical accuracy the signal has positive mutual information with T *but no decision value. For fixed* ϕ *the value is largest when* $\tau = \phi$, *whereas the entropy of* T *does not depend on* τ.

Proof. (1) The Bayes value with the signal is $\sum_z |a_z|$ and without it $|\sum_z a_z|$; the triangle inequality is an equality exactly when all a_z share a sign. (2) Constant policies are available, so the maximum over policies cannot fall. Payoffs are linear in ϕ and in the mixture, so the minimax theorem applies, and the robust value with mixing is the minimum over ϕ of the Bayes value. The decomposition for pure policies holds by definition. (3) Below the critical accuracy both posteriors stay on the prior's side of τ. Checks VG4 and VG7 test this. □

Theorem 91 (when local data determine the value of information)**.** *Let* $Z = z(\omega)$ *be a signal that is a function of the reading. Its classical value* $\mathrm{VOI}(P) = \sum_z \max_a (u_a \mathbb{1}_z)\cdot P - \max_a u_a \cdot P$ *is continuous and piecewise linear on* J_d. *On each* policy cell, *where the action* a_0 *without the signal and the rule* $z \mapsto a_z$ *with it are both optimal, it equals* $g \cdot P$ *for the* value functional $g = \sum_z \mathbb{1}_z (u_{a_z} - u_{a_0})$. *The following are equivalent:*

(i) the local data determine the value of information: VOI *is constant on* J_d;

(ii) the value functional of every policy cell that is full-dimensional in J_d *is orthogonal to the direction space of* J_d;

(iii) on the support of the fibre, each such value functional is cover-additive, *a sum* $\sum_k h_k(\omega_{C_k})$ *of functions of the observed contexts.*

In particular, a determined global law determines the value of information. A determined value does not determine the law, and determination of the value is independent of whether the decision itself is identified ($R = 0$*).*

Proof. On a policy cell each maximum is attained by the cell's actions, so VOI $= g \cdot P$ there. The cells are polyhedra, finitely many, and they cover J_d. (i)⇒(ii): on a full-dimensional cell, a direction v of J_d with $g \cdot v \neq 0$ changes VOI inside the cell. (ii)⇒(i): the full-dimensional cells cover J_d, because the others lie in finitely many proper affine subspaces. VOI is continuous and has zero derivative along J_d on each full-dimensional cell, so it is constant on the connected set J_d. (ii)⇔(iii): on the fibre's support S, the direction space is $\{v : \ v|_{S^c} = 0,\ \pi v = 0,\ \mathbf{1} \cdot v = 0\}$. Its orthogonal complement, restricted to S, is the span of the rows of the marginal map, that is, of the indicators of context events. For the last sentence: a point fibre has no directions. The cases below show the other two claims. □

The criterion is checkable: the cells are cut out by linear inequalities, and full-dimensionality is a slack linear program. The envelope $[\min, \max]$ of VOI over J_d is computed exactly by one linear program per cell and sense. On 80 random problems (cover-additive or general utilities; signals on one variable or a parity of two; interior data and the fibre-0, 1 and 2 faces of Study 18), the criterion holds exactly when the envelope is a point, in all 80 (VD1). The four cases of fig. 25(a), each with the signal B_0 (VD2):

- On a rigid face the value of information is determined (0.081), because the law is.
- In the interior, with a decision and a signal inside the context (A_0, B_0), the fibre has dimension 7 and the value is still determined (0.081).
- The running decision on the never-observed pair is not determined: the value ranges over $[0, 0.292]$.
- An identified decision ($R = 0$) can still have an undetermined value: $[0, 0.201]$.

The three determinations (of the global law, the decision and the value of information) therefore separate. When the value functional lives in one context, the value of information is computed from that table alone, so it stays defined for contextual data, where no global law exists. For a decision on $A_0 = 0$ with the signal B_0, a PR box (contextual fraction 1) has value 0.45 (VD3). This is the decision-value counterpart of the reconstruction of global choices from local problems in [79]: there, local optima glue exactly when the local utilities come from one global function. Here, the value of information glues exactly when the value functionals are sums over the contexts.

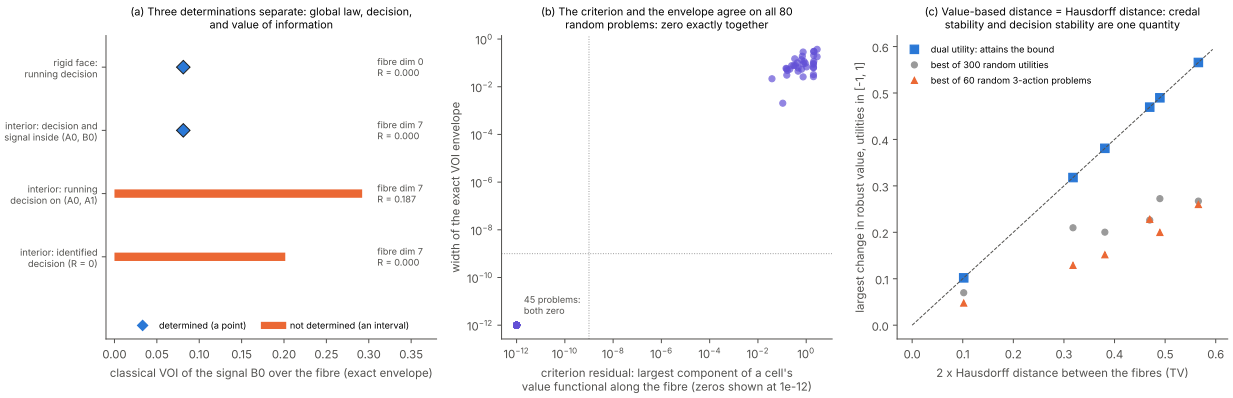


Figure 25: Determination of the value of information (theorem 91) and value-based distance (proposition 93). (a) The exact envelope of the classical value of the signal B_0 over the fibre in four cases: the global law, the decision (R) and the value of information are determined independently. (b) On 80 random problems, the criterion's residual and the envelope's width vanish together. (c) The largest change in robust value over utilities in $[-1, 1]$ equals twice the Hausdorff distance between the fibres, and is attained by the dual utility.

Example 92 (what information is worth on the staged routes)**.** Read example 86 through these quantities, with $\tau = 0.45$ (VG6–VG8, fig. 26).

- *Route A* keeps the fibre a point, so $R = 0$ at every stage, although the decision flips at about 1222 poisoned units. A robust decision-maker sees no reason to collect more data: the

cheapest attack is invisible to value-of-information monitoring. The signal about the unit is still worth up to 0.296, most near 1200 units, where the selected value 0.451 sits on the threshold.

- On *route B*, R is positive exactly between 1222 and 11890 units, where the interval straddles the threshold, and is concave there. It peaks at 0.258 near 6550 units, and randomising recovers half of it. The fibre's shadow at 2400 units is cut by the threshold into the laws that favour each action, and R is twice the smaller overhang, 2×0.037.

- *Envelope.* With pure policies, the robust value of a signal of accuracy 0.8 leaves the envelope of classical values over the fibre: at 2400 units it is 0.332 against $[0.018, 0.297]$, and at 6000 units 0.409 against $[0.159, 0.297]$. It does so by exactly the value of randomising, 0.037 and 0.119. With mixed policies the robust value is 0.295 and 0.290, inside the envelope.

- *Three gains.* At 2400 units the three gains from one signal differ. The identification gain is 0, since the fibre is unchanged. The information gain $I(T;Z)$ rises smoothly from accuracy 1/2. The Bayes decision gain is 0 until the critical accuracy 0.566 and linear after it. The robust gain with mixing has no kink there, because under ambiguity the robust baseline is already the randomised 0.

Proposition 93 (value-based distance is Hausdorff distance)**.** *For $d, d' \in \mathcal{D}$,*

$$\sup\Big\{\big|\max_a \min_{P\in J_d} u_a\cdot P - \max_a \min_{P\in J_{d'}} u_a\cdot P\big| :\ |u_a|\le 1\Big\} = 2\,d_H^{\mathrm{TV}}(J_d, J_{d'}),$$

and the supremum is attained by a single action. So the worst-case change in robust decision value, over all bounded decision problems, is the Hausdorff distance between the sets of compatible laws.

Proof. For one action the robust value is the support function $h_J(u) = \min_{P\in J} u\cdot P$. For convex compact sets, $\sup_{\|u\|_*\le 1} |h_J - h_{J'}|$ is the Hausdorff distance in the norm whose dual is $\|\cdot\|_*$. Here $\|\cdot\|_* = \|\cdot\|_\infty$, so the distance is in ℓ_1, which is twice total variation. A maximum over actions of 1-Lipschitz functions of the set is again 1-Lipschitz, so several actions cannot exceed the bound. VD4 finds the attaining utility from the linear-programming dual of the ℓ_1 distance from the farthest vertex. It reaches $2d_H^{\mathrm{TV}}$ to 10^{-7} on six random pairs, while 300 random utilities and 60 random three-action problems stay below. □

This is the counterpart, for credal fibres, of the value-based distance between information structures of Gensbittel, Pęski and Renault [53]. It answers how the two kinds of stability are connected: the credal-stability numbers above are, exactly, worst-case changes in robust decision value.

Stability, from the decision-maker's side. A model that keeps accepting data cannot tell genuine drift from fluctuation or poisoning. Histories that are equally likely under drift and under poisoning cannot be told apart by any procedure that sees only the history. So what matters to the decision-maker is how far a recommendation can be disturbed, whatever the cause. Five quantities answer this, and they are different.

- *Decision invariance.* The least edit cost that changes the recommendation is the certified radius of proposition 85: no undetectable disturbance smaller than $(L-\tau)/\Lambda_L$ can flip a decision that some rule would make, nor one smaller than $(U-\tau)/\Lambda_U$ for every rule.

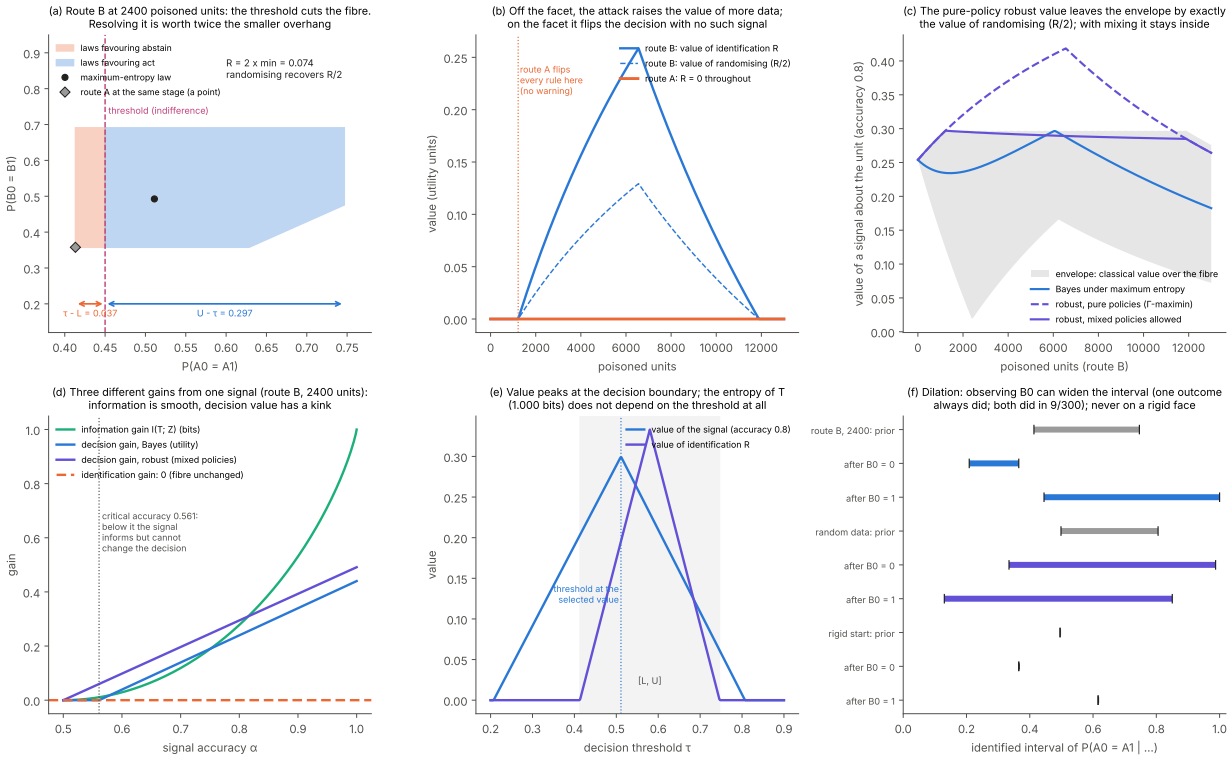


Figure 26: What information is worth (example 92). (a) The fibre's shadow at 2400 units on route B, cut by the threshold into laws favouring each action: the value of identification is twice the smaller overhang. (b) The value of identification along both routes: route A flips the decision with no value of information at all. (c) A signal about the unit along route B: Bayes under maximum entropy, the robust value with pure and with mixed policies, and the envelope of classical values over the fibre. (d) Information, identification and decision gains against the signal's accuracy. (e) The value of the signal and of identification against the threshold: largest at the decision boundary. (f) Dilation: observing B_0 can widen the interval of the never-observed quantity, though never on a rigid face.

- *Worst-case regret.* This is R, which is Lipschitz and so moves at most 2 per unit of cost in the example.

- *Credal stability.* The Hausdorff distance in total variation between fibres, computed exactly from their vertices (half the worst-case change in robust decision value, by proposition 93), stays within 1.11 times the edit cost along route B. It is 0.007, 0.029, 0.157 and 0.456 after 25, 100, 600 and 2400 units, while the fibre dimension jumps from 0 to 7 at the first unit.

- *Dilation.* Observing B_0 widened the interval of $P(A_0 = A_1)$ for some outcome at all 300 random data points tested, and for every outcome at 9. One example goes from $[0.499, 0.806]$ to $[0.334, 0.987]$ and $[0.130, 0.850]$. This is the dilation of imprecise probability, which Das identifies as the source of failures of the value of evidence [20]. It never occurs on a rigid face.

- *Diachronic stability* in Das's sense: commit to one law in the credal set and act by its conditionalisations [20]. A section is such a commitment, and the square "select, then condition" against "condition, then select" tests it. On marginal data the square fails structurally: after conditioning on $B_0 = z$ only the contexts containing B_0 stay observable, so re-selecting uses less information. The re-selected maximum-entropy law differs from the committed one by 0.084 to 0.274 in total variation at random points. On the rigid faces it differs by 0.220 to 0.459, and the decision value there by up to 0.189, because re-selection throws away a fully identified law (VG8). Commitment is therefore the stable policy here, and Good's inequality holds for it by proposition 90(1).

The order drift of learned models (remark 88) is a failure of exactly this commitment. Stability is not safety: a stable model can be poisoned along route A, and an unstable one may be tracking genuine change. Dynamically, pooling makes a single disturbance decay like $1/N_t$, with no exponential rate, and proposition 82(3) gives the exact duration of a one-batch flip. Under exponential forgetting at rate η, disturbances decay at the Lyapunov rate $\log(1 - \eta) < 0$. Whether topological summaries of a decision trajectory (persistent homology of sliding windows, braids formed by the ranked values of competing actions) warn before the decision margin closes is a question for a controlled study, not settled here.

Relation to the literature. The ingredients of this subsection are established. What is new is their joint quantitative treatment over marginal fibres.

- Abramsky and Brandenburger characterise when local tables glue to a global law [2], with cohomological witnesses of failure [3]. The contextual fraction measures an operational advantage in games and computation [1], which is not the decision value of acquiring information.

- Vigneaux's information cohomology characterises entropy as a cocycle [81]; an entropy is not a value of information.

- Tohmé, Caterina and Gangle treat the reconstruction of global choices from local problems sheaf-theoretically, and identify contextuality as an obstruction to it [79]. Theorem 91 is the analogous statement for the value of information.

- Iskandar separates a rule-specific value of information from the envelope of classical values over a credal set, and gives non-negativity under Γ-maximin with constant policies [61]. Proposition 90(2) agrees, and adds the exact decomposition of the excess over the envelope into the value of randomising.

- Das's practical stability [20] is the commitment that a section provides.
- Gensbittel, Pęski and Renault measure information structures by value [53]; proposition 93 does so for credal fibres.

Our contribution is the combination over marginal fibres: identified decisions, the regret from ambiguity that cannot be resolved, structural against instance information, when local data determine the value of information, and how all of these respond to poisoning.

Fisher geometry and decision relevance. Moskowitz, Russell and Lawless propose to value information information-geometrically [66]. Laws form a statistical manifold with the Fisher metric g, and they call information valuable at a point when it increases the "complexity" $\sqrt{\det g}$, the coefficient of the volume form. In the normal family a change of mean then has null value, and a decrease of the standard deviation is valuable. The toric Kähler structure of proposition 1 carries the same metric, with the phases of the torus as further coordinates. The next proposition places all of this against the decision geometry of this subsection.

Proposition 94 (Fisher geometry and decision relevance)**.** *(1)* Phases are decision-null. *Every expected utility, hence every robust value, regret and value of information, depends on an amplitude $z = \sqrt{P}\,e^{i\theta}$ only through $P = |z|^2$. On real amplitudes the Fubini–Study distance is half the Fisher–Rao distance $d_{\mathrm{FR}}(P,Q) = 2\arccos\sum_x \sqrt{P_x Q_x}$.*

(2) The fibre is the Fisher-null space of the data. *At an interior law, the Fisher information of one reading in each observed context, $I_{\mathrm{obs}} = \sum_c A_c^{\top}\operatorname{diag}(1/A_cP)A_c$ on $\{v : \mathbf{1}\cdot v = 0\}$, has null space exactly the direction space of J_d. The Fisher information of a complete reading is positive definite. So the criterion of theorem 91 says: the value of information is determined exactly when every full-dimensional cell's value functional is* Fisher-visible*, i.e. annihilates the null space of I_{obs}.*

(3) Decision value is Lipschitz in Fisher–Rao distance. *For utilities with $|u| \le 1$, $|u\cdot P - u\cdot Q| \le 2\,\mathrm{TV}(P,Q) \le 2\sqrt{1-\mathrm{BC}^2} = 2\sin(d_{\mathrm{FR}}/2)$, where $\mathrm{BC} = \sum\sqrt{P_xQ_x}$.*

(4) The converse fails both ways. *Points of one fibre can be far apart in Fisher–Rao distance and have the same value of information. Points at Fisher–Rao distance tending to 0 can lie on either side of a policy wall and take different actions; only the value is continuous.*

(5) Volume coefficients are coordinate artefacts. *In the coordinates (μ,σ^k) of the normal family, $\sqrt{\det g} = \sqrt{2}/(|k|\sigma^{k+1})$. This decreases in σ for $k = 1$, is constant for $k = -1$ and increases for $k = -2$. The maps $(\mu,\sigma) \mapsto (a\mu + b, a\sigma)$ act transitively by isometries, so every function of a point that is invariant under isometries is constant on the normal family.*

Proof. (1) $\mathbb{E}_P u = \sum_x u_x|z_x|^2$, and $|\langle\sqrt{P},\sqrt{Q}\rangle| = \mathrm{BC}$. (2) $v^{\top} I_{\mathrm{obs}} v = \sum_c\sum_k (A_cv)_k^2/(A_cP)_k$, which vanishes iff $A_cv = 0$ for every observed context, that is, iff v is a direction of J_d (at an interior law the fibre's support is everything). A complete reading gives $\sum_x v_x^2/P_x > 0$. The restatement is theorem 91, since annihilating the null space is orthogonality to the fibre's directions. (3) $2\,\mathrm{TV} = \sum|\sqrt{P}-\sqrt{Q}|\,|\sqrt{P}+\sqrt{Q}| \le \|\sqrt{P}-\sqrt{Q}\|\,\|\sqrt{P}+\sqrt{Q}\| = \sqrt{(2-2\mathrm{BC})(2+2\mathrm{BC})}$ by Cauchy–Schwarz, and $\cos(d_{\mathrm{FR}}/2) = \mathrm{BC}$. (4) FV4 gives both examples. (5) In (μ,σ), $g = \operatorname{diag}(1/\sigma^2, 2/\sigma^2)$; change coordinates with $d\sigma/dv$ for $v = \sigma^k$. The Fisher metric is invariant under a bijection of the sample space, and $x \mapsto ax + b$ maps $N(\mu,\sigma^2)$ to $N(a\mu+b, a^2\sigma^2)$. □

FV1–FV5 check each part. In (2) the null space has dimension 7 at 30 interior laws, and Fisher-visibility agrees with the exact VOI envelope in 40 of 40 decision problems. In (3) there is no excess in 2000 random pairs. In (4) two vertices of a fibre lie at Fisher–Rao distance 2.33 (TV 0.74) and have the same VOI 0.0812, while laws at Fisher–Rao distance 0.002 on either side of the wall $P(A_0 = A_1) = 0.45$ take different actions with a VOI gap below 10^{-16}. The certificate in (5) is exact (sympy). Part (1) is the probabilistic face of the objects in fig. 31. Over the leximin fibre of example C there are eight critical points with the same valuation and different leading holonomies (example 112). Floer theory separates them, and no decision does: they all carry the law of the moment point.

Three consequences for valuing information. First, statistical distinguishability is not decision relevance. The Fisher metric measures how well data separate laws, while the value of information measures whether the separation changes an optimal act. By (2)–(4), the two agree only through a one-sided Lipschitz bound and a common null space. Second, a valuation of information that is meant to be geometric has to be invariant, and by (5) the coefficient $\sqrt{\det g}$ is not. Taking σ^{-2} (the precision) as the coordinate reverses the ranking of [66], and the volume *form*, which is invariant, assigns no value to a point. A mean shift has null value under their measure, yet it can move $\mathbb{E}_P u_a$ across a threshold and so carry positive value of information in the sense of proposition 90. Third, what the Fisher geometry contributes here is the null space in (2): it locates the unresolved directions, and theorem 91 decides whether they matter.

13.24 Algebra of the operations, and selection that is not concave

Marginalisation is linear, composition of kernels is polynomial, and conditioning, Bayesian inversion and Markov gluing are rational. Maximum entropy on a cover is the hierarchical log-linear (toric) model, which meets each interior fibre in exactly one point (Birch's theorem) [25]. Four consequences are worth recording.

Proposition 95 (algebra of the operations)**.** *(1)* Removable singularities. *Each term $\gamma_1(i,j)\gamma_2(j,k)/q(j)$ of Markov gluing is at most $\min(\gamma_1(i,j), \gamma_2(j,k))$. So gluing extends continuously across $q(j) = 0$, and so does the dagger. The kernels do not: the row $K^\dagger(\cdot \mid j)$ can converge to different limits along sequences whose couplings converge to the same limit.*

(2) Exact and inexact rules. *Leximin and Markov gluing send rational data to rational laws: one is a sequence of linear programs, the other a product. Maximum entropy on a cover that is not decomposable need not. On the 4-cycle with uniform singles and every adjacent agreement probability $3/4$, the maximum-entropy value of $P(A_0 = A_1)$ is $(1+\sqrt{3})/4$, while the identified interval is $[1/2, 1]$ and leximin gives $3/4$.*

(3) Counting explanations. *Let the marginal counts at sample size N be integral. The number of integer joint tables of size N with those counts is an Ehrhart quasi-polynomial in N of degree equal to the fibre dimension [15]. In particular it is 1 on a rigid face whenever it is not 0.*

(4) Selection that is not concave. *A rule that minimises a concave objective over the fibre selects a vertex. As the data move, it jumps wherever two vertices tie, at a Maxwell point. The minimum-entropy rule, which picks the "simplest explanation", is one such rule. Maximum entropy and rules with unique optima are continuous (proposition 85(4)).*

Proof. (1) $\gamma_1(i,j) \le q(j)$ and $\gamma_2(j,k) \le q(j)$. For the kernels, take $\gamma_1(\cdot, j) = \varepsilon e_1$ along one sequence and εe_2 along another: both columns tend to 0, but the normalised rows are e_1 and e_2 (AL1). (2)

With zero fields and a common coupling, which symmetry forces, the maximum-entropy law is an Ising ring of four spins with $t = \tanh J$. Its adjacent correlation is $(t+t^3)/(1+t^4)$, and the correlation at distance two is $2t^2/(1+t^4)$. Setting the first to $1/2$ gives $t^4-2t^3-2t+1=0$, so $t+1/t = 1+\sqrt{3}$, and the second correlation is $2/((t+1/t)^2-2) = (\sqrt{3}-1)/2$. Hence $P(A_0 = A_1) = (1+\sqrt{3})/4$. AL2 checks this exactly with sympy and against iterative fitting to 10^{-10}. (3) The tables form the lattice points of N times the fibre, a rational polytope of the fibre's dimension. (4) A concave function attains its minimum over a polytope at a vertex. The vertices move continuously with the data within a basis, so the selected law can change discontinuously only where two of them give equal values. □

Rationality explains a pattern in this note's checks: the exact certificates use leximin and Markov gluing, while maximum entropy is always numerical. For the counts (AL3, fig. 27): on the three CH-facet laws of Study 18 the count is 1 at every $N = 240m$, $m = 1, \ldots, 5$. On the fibre-1 faces it is linear, for example $69, 137, 205, 273, 341$. On the fibre-2 faces it is quadratic, for example 1655, 6460, 14416, 25523, 39781. The fibre dimension is thus the growth rate of the number of datasets that explain the data. The integer moves between them are the Markov bases of Diaconis and Sturmfels [40]: the kernel counterparts of the edit circuits of proposition 80, which move the data rather than the explanation.

For selection that is not concave (AL4), take three straight data paths of 61 steps. The minimum-entropy law jumps by 0.82, 0.44 and 0.69 in total variation within one step, each time at a point where the two best vertices tie to within 10^{-13}. Maximum entropy on the same paths moves by at most 0.009 per step. On the path shown in fig. 27, the minimum-entropy value of $P(A_0 = A_1)$ leaps across most of the identified interval, which itself moves continuously. So bifurcations and phase-transition-like jumps of a selected law do not come from the geometry of the fibre, which is convex and Lipschitz. They come from a selection objective that is not concave, or from restricting to a model family that is not convex (latent classes, mixtures), where the same Maxwell mechanism applies. That is where a discriminant geometry of probability models would begin; we leave it open.

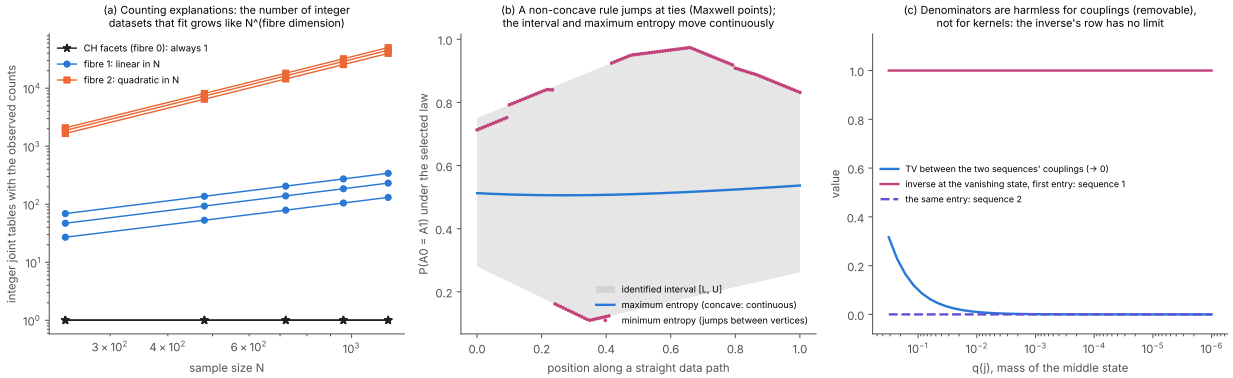


Figure 27: Algebra and selection that is not concave. (a) Integer joint tables with the observed counts against sample size, for the sparsity-matched faces of Study 18: constant on CH facets, linear for fibre 1, quadratic for fibre 2. (b) Along a straight data path, the minimum-entropy rule jumps between vertices at ties (Maxwell points), while maximum entropy and the identified interval move continuously. (c) Markov gluing's denominator is removable for couplings, but the Bayesian inverse's row at a vanishing state has no limit.

13.25 Trajectories of decision viability

A system that keeps learning moves its data, and with them the fibre, the selected law and the decision. This subsection asks what the trajectory of a decision can reveal, and gives the one quantity that tracks viability exactly whatever moves the data, whether drift, noise or poisoning.

Definition 96 (viability shares)**.** For a decision with $w \cdot s(d) > \tau$, the *viability shares* $q^*_{\text{some}}(d)$ and $q^*_{\text{all}}(d)$ are the least shares of the pooled data that one consistent batch (any joint law) must have so that, afterwards, some rule (resp. every rule) decides $w \cdot x \le \tau$.

Proposition 97 (viability under any disturbance)**.** *(1) After consistent batches of total share Q, whatever their number and laws, the data lie in $(1-Q)d + Q\,\mathcal{D}$, and every point of that set is reached by one batch. This is the reachable set of all disturbances of share Q, and the viability kernel of the decision, in the sense of viability theory [5], is $\{d : q^* > Q\}$.*

(2) q^*_{some} is the value of one linear program in (q, P, R): $\pi P = (1-q)d + \pi R$, $\mathbf{1} \cdot R = q$, $w \cdot P \le \tau$. q^*_{all} is the least of one linear program per dual vertex, as for U in proposition 79. Always $q^*_{\text{some}} \le q^*_{\text{all}}$, and $q^*_{\text{some}} = 0$ exactly when the decision is already rule-dependent.

Proof. (1) Pooling is averaging, so the pooled data are $(1-Q)d$ plus Q times a convex combination of points of $\mathcal{D}$. $\mathcal{D}$ is convex, so that combination is itself a point of $\mathcal{D}$, the data of a single batch. (2) Substitute $R = q \cdot x$ for the batch law x; the constraints are then linear in (q, P, R). VB1 reconstructs 20 random multi-batch histories by single batches and confirms q^*_{some} by bisection. □

Proposition 98 (open loop)**.** *Suppose the incoming batches do not depend on the decisions.*

(1) Forgetting. *Two streams that agree from step t_0 on coincide W steps later under a window of W batches, and differ by exactly N_{t_0}/N_t times their earlier difference under pooling. Disturbances therefore shrink, algebraically under pooling and with Lyapunov rate $\log(1-\eta) < 0$ under exponential forgetting.*

(2) Regimes. *While the batch law is fixed, the data move along a segment toward that law's data. The decision is eventually constant, unless the limit lies on the threshold.*

(3) Crossings. *Along a segment, L is convex and U concave, so each crosses τ at most twice. The difference of two actions' robust values is piecewise linear with at most $p_a + p_b - 1$ pieces, where p counts the dual vertices, so it changes sign at most that many times.*

(4) Ranking radius. *Rank the actions by robust value. No undetectable edit cheaper than $\min_{\text{adjacent } a,b}(v_a - v_b)/(\Lambda_a + \Lambda_b)$ changes the ranking, with Λ as in proposition 85.*

Proof. (1) The window mean forgets batches older than W, and the pooled mean weights the past by N_{t_0}/N_t. (2) $d_{t+1} - x = (1 - q_t)(d_t - x)$ for the regime's data x, so d_t moves toward x along a line. A threshold decision on a continuous function of d is eventually constant unless the limit is a boundary point. (3) A convex function along a line meets a level in an interval, so it crosses the level at most twice. The difference of two piecewise-linear functions has at most the sum of their pieces minus one. (4) Each robust value is Λ-Lipschitz (proposition 85(1)), so an adjacent gap closes only after an edit costing the gap divided by $\Lambda_a + \Lambda_b$. Checks VB2–VB4: along 150 segments the ends crossed τ at most twice. Random pairs of actions crossed at most once, against a bound of 127, which is valid but loose. No edit cheaper than the ranking radius changed the ranking. □

Proposition 99 (closed loop)**.** *Suppose the decision chooses the next batch's law from finitely many laws, and the data are a window of the last W batches. Then the state is the last W choices, so the decision sequence is eventually periodic with period at most $|laws|^W$.*

Proof. There are finitely many states, the update is deterministic, and the decision is a function of the state. Any orbit of such a system is eventually periodic. □

Example 100 (five regimes, and a feedback loop)**.** Take the act/abstain decision on $P(A_0 = A_1)$ with $\tau = 0.45$, and a window of six batches. The stream passes through five regimes of eight steps each (VB5, fig. 28(a,b)), with shares q^* measured as fractions of the window.

- *Clean* (the interior law 0.8 CH face + 0.2 uniform): the decision is rule-dependent ($q^*_{\text{some}} = 0$, $R = 0.107$), and $q^*_{\text{all}} = 0.246$.
- *Degeneration* (the rigid CH-facet law): the fibre collapses from dimension 7 to 0, R falls to 0, and q^*_{all} falls from 0.224 to 0.092.
- *Poisoning on the facet* (15% of each batch is the facet reading $(0, 1, 1, 0)$): the fibre stays a point and R stays 0, but q^*_{all} falls to 0.007 and q^*_{some} to 0.002 before the decision switches. Here the viability share is the only quantity that warns.
- *Poisoning off the facet*: the fibre reopens to dimension 7, the largest Hausdorff step (0.100) occurs at its start, R rises to 0.057, and the decision switches back.
- *Recovery*: everything returns to the clean values.

At most two decision switches occur per regime, as proposition 98(2,3) requires.

In the closed loop, acting brings batches that lower $P(A_0 = A_1)$ and abstaining brings batches that raise it (VB6, fig. 28(c,d)). The identified interval then straddles τ at every step, and the maximum-entropy decision chatters inside it. For windows $W = 3, \ldots, 8$ the decision sequence is periodic with period $W + 1$; the bound 2^W is proved, the pattern $W + 1$ only observed. The same law fed in open loop gives a decision that is constant from the first step.

Remark 101 (knots, braids and persistent topology)**.** Several caveats apply before these tools are used.

- *Knots.* Every closed curve in four or more dimensions can be unknotted, and our data move in eight. A knot type therefore appears only after a choice of three coordinates, and depends on that choice. We do not see a canonical one.
- *Braids.* Real-valued scores give a word of adjacent transpositions, not a braid. Unlike the decision margin, it does record swaps below the top two, and its robustness radius is the ranking radius of proposition 98(4). Genuine braids need two numbers per action, for example the centre and half-width of its robust interval, and braid entropy is meaningful only for recurrent regimes.
- *Persistent homology.* Sliding windows of an open-loop trajectory converge within each regime, so recurrent loops reflect a recurrent input or feedback, as in the closed loop above.
- *Pseudo-crossings.* In the language of probabilistic pseudo-knots [24], a decision switch is a classical crossing when it is identified on both sides, and a pseudo-crossing, resolved by the rule, while $R > 0$. Route A of example 86 is a single classical crossing, and route B a long pseudo-crossing. This restates the support of R; it is not a new invariant.

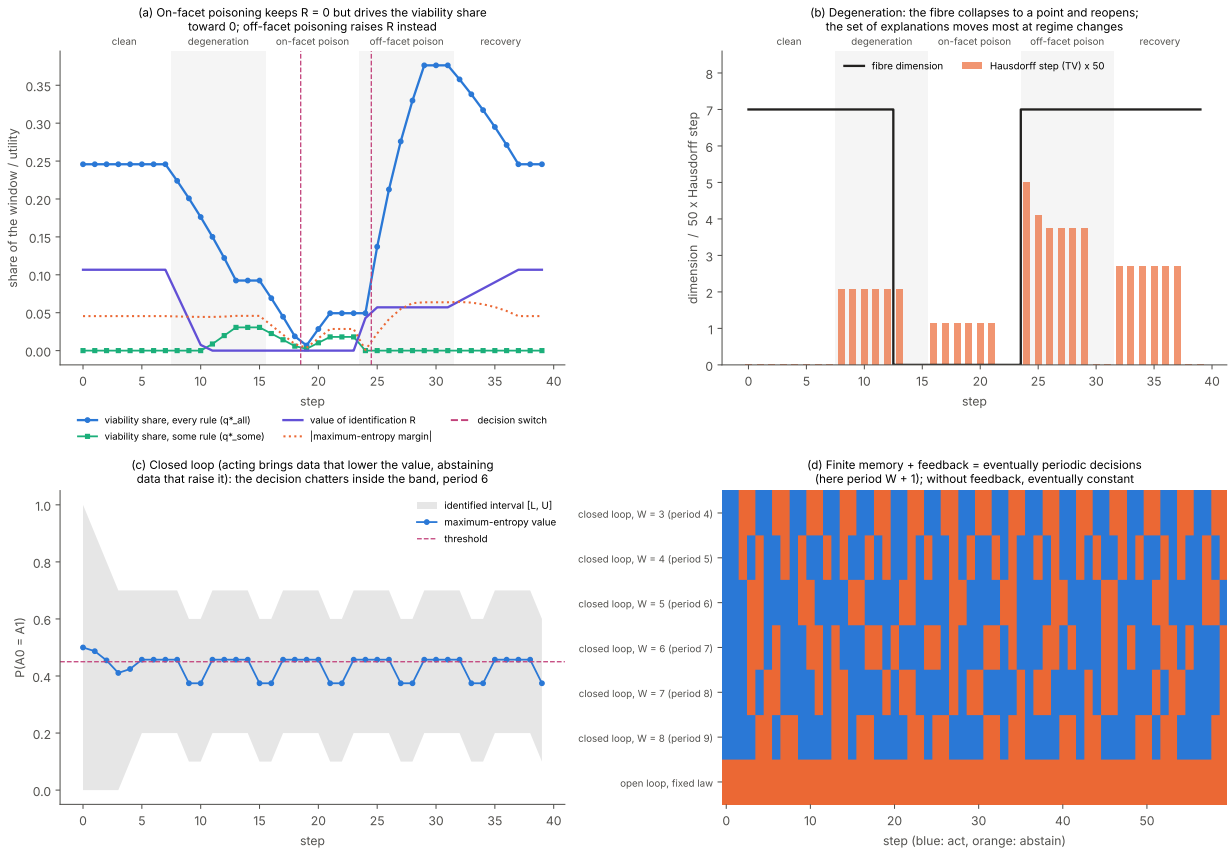


Figure 28: Trajectories of decision viability (example 100). (a) Five regimes in open loop: viability shares, value of identification and maximum-entropy margin; on-facet poisoning keeps $R = 0$ and is seen only by the viability share. (b) Fibre dimension and Hausdorff steps: degeneration collapses the fibre to a point, and off-facet poisoning reopens it. (c) Closed loop with a window of five batches: the decision chatters inside the identified band. (d) Closed-loop decision sequences are periodic (period $W+1$ here, at most 2^W in general); in open loop the decision is eventually constant.

Whether topology warns before the margin, R and q^* do is therefore a question about closed loops. Continuous-state feedback (forgetting with feedback, selective observation) can produce cycles and bifurcations of switching maps. That question is for a pre-registered study.

13.26 Histories, gluing and holonomy

So far a rule has seen only the current data. A system that keeps learning also has a history: the order in which contexts arrived, the tables it saw before they were revised, and the laws it selected along the way. Tohmé, Caterina and Gangle read the sheaf condition for local choices as independence from the path of previously solved problems. In their case-based example, two sequences of earlier problems lead to the same current problem and to different choices [79]. This subsection separates the two dependences that this raises,

dependence on the compatible law (the fibre) and dependence on the history,

the history being the path by which the data were reached, and measures the second geometrically.

Definition 102 (histories, updating rules, holonomy)**.** A *snapshot* is a set K of observed contexts with their tables, and its fibre J_K is the set of joint laws with those tables. A *history* $H = (K_1, \dots, K_n)$ is a sequence of snapshots, and $d(H) = K_n$ is its final snapshot. An *updating rule* carries the previously selected law to the current fibre, $\sigma_t = u(\sigma_{t-1}, K_t) \in J_{K_t}$, starting from the uniform law; write $\sigma(H) = \sigma_n$. The rule is *path independent* when $\sigma(H)$ depends only on $d(H)$, that is, when $\sigma = s \circ d$ for a section s. The *holonomy* of two histories with the same final snapshot is $\mathrm{hol}(H, H') = \mathrm{TV}(\sigma(H), \sigma(H'))$. A poison-then-remove loop is the special case $d(H) = d(H')$ in which H' stays at the start.

Three updating rules are compared: *KL updating* (the I-projection of σ_{t-1} onto J_{K_t}, minimum discrimination information), and *Euclidean* and *TV updating* (the nearest law of J_{K_t} in the Euclidean and total-variation distances, TV ties broken by a fixed linear order). The last two are minimal-change rules, in the spirit of belief revision. KL updating on a single context is Jeffrey's rule.

Proposition 103 (holonomy is bounded by the fibre)**.** *Let H, H' be histories with the same final snapshot K, and let u be any updating rule.*

(1) $\mathrm{hol}(H, H') \le \mathrm{diam}_{\mathrm{TV}} J_K$. *On a rigid face, where J_K is a point, every rule is path independent.*

(2) $|w \cdot \sigma(H) - w \cdot \sigma(H')| \le U - L$, *where $[L, U]$ is the identified interval of $w \cdot x$ over J_K. The threshold decision $w \cdot x \ge \tau$ can therefore depend on the history only if $L < \tau \le U$. An identified decision ($\tau < L$ or $\tau > U$) is path independent for every rule.*

(3) The value of information at $\sigma(H)$ lies in the envelope of theorem 91. When the criterion of that theorem holds, it is the same for every history and every rule.

Proof. Both $\sigma(H)$ and $\sigma(H')$ lie in J_K, and each statement is a property of all pairs of points of J_K: (1) by the definition of the diameter, (2) because $L \le w \cdot x \le U$ on J_K, and (3) by theorem 91. HY3 checks all three on 216 pairs of random histories. □

So history can change a decision only by moving the selection inside a fibre, and only when the fibre straddles the threshold. The decision-level sheaf condition is the static condition $R = 0$ of section 13.23. Which rules actually move?

Proposition 104 (flat and curved updating)**.** *Suppose the context sets only grow along the history ($K_1 \subseteq K_2 \subseteq \cdots$ as sets of contexts). Their tables may be revised, and every fibre is assumed to contain a strictly positive law.*

(1) KL updating is flat and memoryless: *σ_t is the maximum-entropy law of K_t at every t, whatever came before. Hence it is path independent, and poison-then-remove loops close.*

(2) Forgetting curves it. *If a context leaves the snapshot (a sliding window), KL updating keeps the statistics of the forgotten context in σ_t and is path dependent in general. On single contexts this is the non-commutativity of Jeffrey's rule [39].*

(3) Euclidean updating is flat in the interior. *As long as no positivity constraint binds, σ_t is the Euclidean projection of the uniform law onto the affine hull of J_{K_t}, so it is again memoryless. Holonomy appears only when a projection reaches the boundary of the simplex.*

(4) TV updating is curved *even in the interior: its nearest points are not projections onto the affine hulls.*

Proof. (1) Let $\mathcal{E}_K$ be the log-linear family of the contexts in K, the laws proportional to $\exp(\sum_{c\in K}\theta_c(x_c))$. The uniform law lies in $\mathcal{E}_{K_1}$. Suppose $\sigma_{t-1} \in \mathcal{E}_{K_{t-1}}$. Since $K_{t-1} \subseteq K_t$, also $\sigma_{t-1} \in \mathcal{E}_{K_t}$. The fibre J_{K_t} is the linear family defined by the same statistics, so the I-projection of any point of $\mathcal{E}_{K_t}$ onto it is the unique point of $\mathcal{E}_{K_t} \cap J_{K_t}$ (by the Pythagorean identity for I-divergence [19]). That point is the maximum-entropy law of K_t, which is strictly positive because the fibre contains a positive law. By induction, σ_t is the maximum-entropy law of K_t for every t. (2) When a context c leaves, σ_{t-1} need not lie in $\mathcal{E}_{K_t}$, so the induction fails. HY4 gives explicit path dependence. (3) Write $\operatorname{aff} J_{K_t} = a_t + V_t$. Growing contexts give $V_t \subseteq V_{t-1}$, and revised tables translate a_t. If $y = x - P_{V^\perp}(x-a)$ is the projection onto $a+V$ and $V' \subseteq V$, then $a' + P_{V'}(y-a') = a' + P_{V'}(x-a')$, because $P_{V'}P_{V^\perp} = 0$. So composing the projections gives the projection onto the last affine hull. Positivity enters only through the constraints $x \geq 0$. (4) This part is numerical: HY2 finds TV holonomy up to 0.26 between histories whose projections all stay strictly positive. □

HY1 confirms (1) on 90 random growing histories with revisions, to within 10^{-15}. HY2 confirms (3): the Euclidean holonomy is at most 2×10^{-16} when every projection stays positive, and nonzero in 32 of 40 cases at the boundary. HY6 confirms the loops: after 15% poisoning and its removal, KL and interior Euclidean updating return to within 10^{-15}, and TV updating fails to return in all 10 cases, by up to 0.047. Note that path independence of KL updating does not make it the right rule. It makes it a section, and proposition 103 still allows any section to sit on the wrong side of τ.

Example 105 (three snapshots, two histories)**.** Take the joint law with masses $k/20$, $k = (3,0,3,0,1,0,2,1,3,4,1,0,0,2,0,0)$ on the cells (a_0,a_1,b_0,b_1) in lexicographic order. Its context tables, with entries in the order $00,01,10,11$, are

$$(A_0,B_0):(0.20,0.30,0.45,0.05),\quad (A_0,B_1):(0.45,0.05,0.20,0.30),$$
$$(A_1,B_0):(0.50,0.20,0.15,0.15),\quad (A_1,B_1):(0.50,0.20,0.15,0.15).$$

The decision is to act when $P(A_0 = A_1) \geq \tau = 0.49$. A_0 and A_1 are never observed together, so this pair is never measured. Two histories reach the same final snapshot (HY5, fig. 29):

- *History A* sees $(A_0,B_0),(A_0,B_1)$ at t_1, adds (A_1,B_0) at t_2 and (A_1,B_1) at t_3.

- *History B* sees $(A_1, B_1), (A_1, B_0)$ at t_1, adds (A_0, B_1) at t_2 and (A_0, B_0) at t_3.

In the sheaf picture of [79], each snapshot is a family of local problems. At t_1 two tables share one variable, and at t_2 three tables form a path. On such an acyclic cover the local tables always glue once they agree on the shared variable [82]. The glued fibre is the intersection of the local fibres, of dimension 10 at t_1 and 8 at t_2 in both histories. The interval of $P(A_0 = A_1)$ is $[0, 1]$ at t_1 for both histories, and at t_2 it is $[0.2, 0.6]$ in history A and $[0.4, 0.8]$ in history B. At t_3 the cycle $A_0B_0A_1B_1$ closes. The fibre has dimension 7 and interval $[0.4, 0.6]$ for both histories, since the final snapshot is the same.

rule	$P(A_0 = A_1)$, A	$P(A_0 = A_1)$, B	action (A, B)	holonomy	VOI of B_0 (A, B)
KL	0.500	0.500	act, act	0	0.055, 0.055
Euclidean	0.4786	0.5125	abstain, act	0.053	0.052, 0.043
TV	0.450	0.600	abstain, act	0.225	0.107, 0

All of these are within the bounds of proposition 103. The TV diameter of the final fibre is 0.40, $U - L = 0.2$, and the envelope of the value of information is $[0, 0.157]$. The decision is history dependent for both minimal-change rules because τ lies inside $[0.4, 0.6]$, so $R > 0$. With τ outside it, no rule could split.

Contextual t_3. Replace the last table by $(A_1, B_1) : (0.65, 0.05, 0, 0.30)$, which has the same margins. Every three of the four tables still glue (each triple is a path), but the four do not. The fibre is empty, and the contextual fraction is 0.10. Moving $P(A_1 = 0, B_1 = 0)$ from 0.50 upward shrinks the interval of $P(A_0 = A_1)$ to a point at about 0.60, after which the fibre is empty (fig. 29(c)). This is the failure of the global gluing in the sheaf picture. It cannot occur before the cycle closes, and once it occurs no rule, path dependent or not, has a law to select.

Remark 106 (Tohmé's mechanisms, Das's square and order drift)**.** The two behavioural mechanisms of [79] sit in different places here.

- *Case-based choice* carries memory of earlier problems. It is history-indexed selection, the curved case of proposition 104: minimal-change updating, or I-projection with forgetting.
- *Framing*, as in prospect theory, changes the utility with the context. It changes the value functional rather than the fibre, so its effect on decisions and on the value of information is governed by proposition 89 and theorem 91.

Das's square, "select, then condition" against "condition, then select" (section 13.23), compares two paths to the same information, so its failure for re-selection is a holonomy of the same kind. The order drift of learned models (remark 88) is likewise a dependence of the selected law on the path through the variables. Proposition 103 bounds all three by the geometry of one fibre, and proposition 104(1) shows that KL updating over growing context sets has none of them.

Part IV
Deeper selection: 2×4 couplings

14 Three-dimensional fibres

A 2×3 fibre is a polygon, so selection has at most two levels. A 2×4 fibre $\mathrm{Cpl}(f, g)$ is a three-dimensional polytope, and the leximin recursion can cut it down in up to three steps: a sheet, then

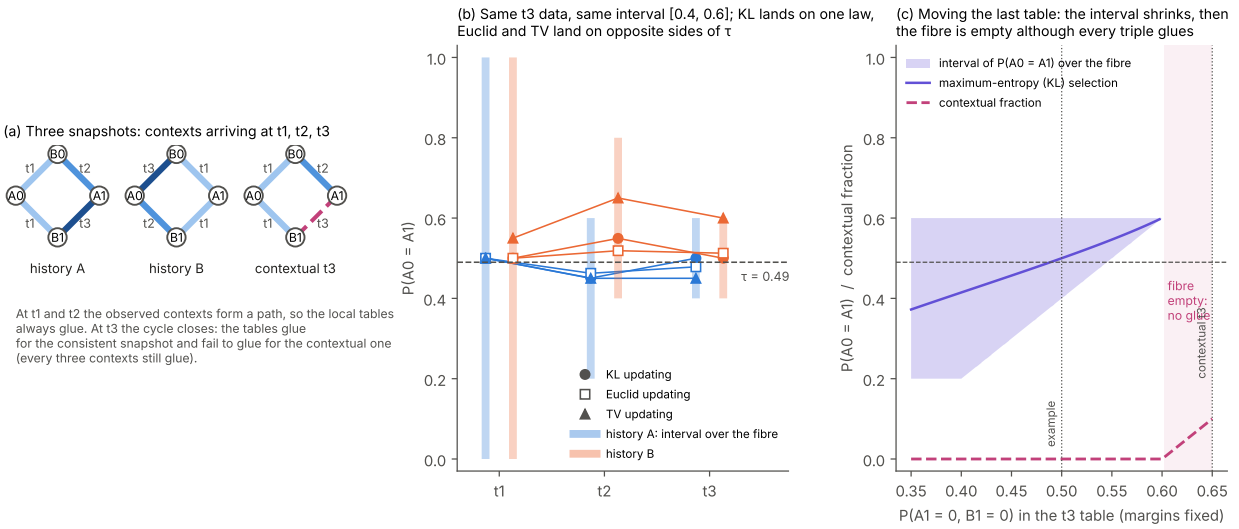


Figure 29: Three snapshots, two histories (example 105). (a) The context cycle with the time at which each context arrives. At t_1 and t_2 the observed contexts form a path and the tables glue; at t_3 the cycle closes, and the contextual variant does not glue. (b) The interval of $P(A_0 = A_1)$ over the fibre at each time (bars), with the laws selected by KL, Euclidean and TV updating (markers; blue history A, orange history B). At t_3 both histories have the same interval $[0.4, 0.6]$; KL lands on one law, and the minimal-change rules land on opposite sides of τ. (c) The t_3 snapshot as its last table varies with fixed margins: the interval shrinks to a point, then the fibre is empty, with contextual fraction rising from 0, although every three contexts still glue.

a line, then a point. This part asks what such a deeper hierarchy does on the other side, and it keeps three questions apart. Is the chamber Fano, so that the Floer layer of section 5 applies? How many dimensions does each level remove? And does the Clifford form of the positive object see each level? The answers are, in order: only 14 of the 148 chambers are Fano, and in those coupled three-step selection never occurs (theorem 108 and proposition 109); a level can remove more than one dimension; and in every example the Clifford form has one scale per removed dimension (Conjecture 115, proved at leading order in three examples).

14.1 Coordinates and the potential

Use $u = (a_{11}, a_{12}, a_{13})$. The eight cells are $a_{1j} = u_j$ and $a_{2j} = g_j - u_j$ for $j \le 3$, $a_{14} = f_1 - \sum_j u_j$, and $a_{24} = \sum_j u_j - (f_1 + g_1 + g_2 + g_3 - 1)$. Their normals are $\pm e_j$ and $\mp(1,1,1)$. Over the facet cells (a non-facet cell contributes no term), the potential is

$$W = \sum_{j \le 3} \Big([1j]\, y_j + [2j]\, \frac{T^{g_j}}{y_j} \Big) + [14]\, \frac{T^{f_1}}{P} + [24]\, T^{g_4 - f_1} P, \qquad P = y_1 y_2 y_3, \tag{2}$$

where $[ij] = 1$ if cell ij is a facet and 0 otherwise. For each example we keep three counts separate: the vertices of the fibre (equal to $\chi(X)$), the critical points of W, and the distinct fibres that carry them.

14.2 Ampleness, certified by curves

Whether $X_{(f,g)}$ is Fano is a statement about its fan, and so about the chamber of (f, g). We certify it with integers on the edges of the polytope, by a classical criterion, rather than by any probabilistic argument.

Proposition 107 (ampleness by curves)**.** *Let X be a smooth complete toric threefold with polytope Q.*

(1) (Nakai–Moishezon) A divisor D on a projective variety is ample if and only if $D^{\dim V} \cdot V > 0$ for every irreducible subvariety V of positive dimension [58].

(2) (toric Kleiman) On a complete toric variety it suffices to test the torus-invariant curves: D is ample if and only if $D \cdot C > 0$ for each of them, and nef if and only if $D \cdot C \geq 0$ [36, Theorem 6.3.13 and §6.3].

(3) The invariant curves are the edges of Q. If an edge lies on the facets i, j and joins the vertices whose third facets are k and l, the wall relation $v_k + v_l + a\, v_i + b\, v_j = 0$ has integer a, b, and $D_k \cdot C = D_l \cdot C = 1$, $D_i \cdot C = a$, $D_j \cdot C = b$, and $D_\rho \cdot C = 0$ otherwise [36, §6.4]. So $-K \cdot C = 2 + a + b$, and the Kähler class meets C in $-\lambda_k - \lambda_l - a\lambda_i - b\lambda_j$, which must equal the lattice length of the edge.

On a toric variety $-K$ is always big, since its polytope $\{m : \langle m, v_\rho \rangle \geq -1\}$ contains 0 in its interior. So nef already means weak Fano.

Part (3) contains its own test: the lattice length of every edge must come out of the same integers that give $-K \cdot C$. TF2 confirms it on all 2208 edges of all chambers. The curves with $-K \cdot C = 0$ matter twice. They are what makes a chamber weak Fano rather than Fano. And they span the anticanonically trivial directions of the Mori cone, so they are the geometric source of the classes that *may* contribute to the corrections of W (remark 113); a zero degree does not by itself give a nonzero contribution.

Theorem 108 (the 2×4 chambers)**.** *The nondegenerate 2×4 margins fall into 148 Kähler chambers, the sign patterns of $f_1 - g(J)$ over the 14 nonempty proper sets J of columns. Every chamber polytope is simple and smooth. By proposition 107 they are of six types:*

type	*chambers*	*facets*	χ	$(-K)^3$	*curves with* $-K \cdot C = 0$	W	*Floer reading*	*drop patterns seen*
$\mathbb{CP}^3$	2	4	4	64	0	*as is*	*Cho–Oh*	(3)
$\mathrm{Bl}_{pt}\mathbb{CP}^3$	8	5	6	56	0	*as is*	*Cho–Oh*	(3), (1,2)
$(\mathbb{CP}^1)^3$	4	6	8	48	0	*as is*	*Cho–Oh*	(1,1,1), *decoupled*
weak Fano	24	6	8	48	1	$(1 + \delta_k)$	*withheld*	(3), (1,2), (1,1,1)
weak Fano	64	7	10	40	3	$(1 + \delta_k)$	*withheld*	(3), (1,2), (1,1,1)
weak Fano	46	8	12	32	6	$(1 + \delta_k)$	*withheld*	(3), (1,2), (1,1,1)

Here χ is also the number of vertices, and "$(1+\delta_k)$" means the corrections of remark 113. In every chamber the normalised volume of $\mathrm{conv}\{v_c\}$ *over the facets equals the number of vertices. The* 148 *are chambers of the margin hyperplane arrangement only: leximin walls subdivide them further, so the drop pattern is not a chamber invariant, and the last column lists only the patterns seen in a census.*

Proof. An exact computation, TF1–TF3, with a machine-readable certificate, the file `chambers24.json` in `kahler/`, and an independent verifier (CC1–CC4). Each of the 148

chambers is certified by an exact rational interior point; that no other sign pattern is realised rests on linear-programming infeasibility. For each chamber, $-K \cdot C$ is computed on every edge from the integer wall relation, the type is read from the primitive relations, and an independent anticanonical vertex test agrees in all 148. The drop patterns are not invariants of a Kähler chamber, because leximin walls subdivide it; the last column lists those seen in a tie-free census (TF8). □

Proposition 109 (the box, and selection depth in the Fano chambers)**.** *(1)* $\mathrm{Cpl}(f,g)$ *is a box exactly when some column has* $g_j > \max(f_1, f_2)$*. Then column* j *never binds,* $X = (\mathbb{CP}^1)^3$*,* W *is a sum of three* $\mathbb{CP}^1$ *potentials, and the levels are the half-masses* $g_k/2$ *of the other three columns.*

(2) In a $\mathbb{CP}^3$ *chamber selection is immediate: pattern* (3)*. In a* $\mathrm{Bl}_{pt}\mathbb{CP}^3$ *chamber it is* (3) *or* $(1,2)$*.*

So coupled selection in three steps occurs only in weak Fano chambers.

Proof. (1) The column-j inequalities $0 \le f_1 - \sum_{k\neq j} a_{1k} \le g_j$ hold on the whole box $0 \le a_{1k} \le g_k$ exactly when $1 - g_j \le f_1 \le g_j$. Then the two cells of column j are not facets and their terms do not appear in (2). Each remaining column contributes $y + T^g/y$ in its own coordinate: y_k for a column $k \le 3$, and $z = T^{f_1}/P$ for column 4. These three coordinates are a unimodular basis. (2) By TF4 the facets of a $\mathbb{CP}^3$ chamber are one full row i, so the fibre is the simplex $\{a_{i\cdot} \ge 0,\ \sum_j a_{ij} = f_i\}$ and the equal split $f_i/4$ is selected at once. The facets of a $\mathrm{Bl}_{pt}\mathbb{CP}^3$ chamber are one full row i and one cell $i'j$, which cuts off a corner. If $f_i/4 < g_j/2$ the equal split is still selected at once. Otherwise column j binds first, at $g_j/2$, and the remaining three cells of row i share $f_i - g_j/2$ equally at the second level, a drop of two. □

A binding row finishes selection, since it leaves a candidate set of dimension $(n-2)(m-1) = 0$. The counting formula of theorem 23 gives only the first level, and the examples show that later levels can remove one or two dimensions. Write r_j for the number of dimensions removed at level j, the rank increment of the fixed normals; a level can also remove none, as the last level of example C does.

14.3 Three examples

Example 110 (A: the box)**.** Take $f = (2/5, 3/5)$ and $g = (1/20, 13/20, 1/10, 1/5)$. Column 2 never binds, so by proposition 109(1) the fibre is a box, $X = (\mathbb{CP}^1)^3$, and $W = (y_1 + T^{1/20}/y_1) + (y_3 + T^{1/10}/y_3) + (z + T^{1/5}/z)$ with $z = T^{2/5}/P$. The levels are 1/40, 1/20, 1/10, with drops $(1,1,1)$. The 8 critical points are $y_1 = \pm T^{1/40}$, $y_3 = \pm T^{1/20}$, $z = \pm T^{1/10}$, all over the leximin fibre $(1/40, 9/40, 1/20)$, with leading holonomies ± 1. Their critical values are $2(\pm x \pm x^2 \pm x^4)$ with $x = T^{1/40}$. For $T \in (0,1)$ exactly two coincide, $(+,-,-)$ and $(-,+,+)$, at the root of $x^3 + x - 1 = 0$, $T^* \approx 2.3 \times 10^{-7}$. There the m_0 argument for vanishing Hom does not apply (the holonomies still differ), and we do not compute that group. TF5 checks all of this exactly.

Proposition 111 (B: the blown-up projective space, exactly)**.** *Take* $f = (3/20, 17/20)$ *and* $g = (1/20, 1/5, 1/4, 1/2)$*. The facets are row* 1 *and cell* 21*, so* $X = \mathrm{Bl}_{pt}\mathbb{CP}^3$ *and* $W = y_1 + y_2 + y_3 + T^{3/20}/P + T^{1/20}/y_1$*. Put* $\tau = T^{1/20}$*.*

(1) Selection. *Column* 1 *binds at* $h_1 = 1/40$*, and row* 1 *then spreads* 1/8 *over three cells,* $h_2 = 1/24$*. The drops are* $(1,2)$*.*

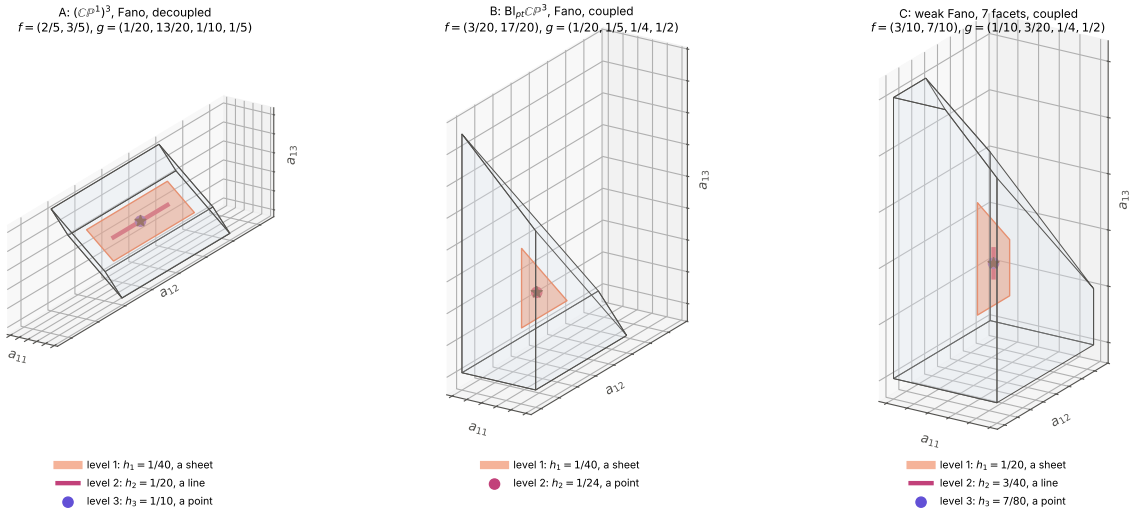


Figure 30: The three fibres and their candidate sets. Orange: the sheet left by the first level (a binding column). In A and C the second level leaves a line and the third a point; in B the second level is a binding row on what is left, and it fixes two dimensions at once. The star is the leximin point, the valuation of the positive critical point (theorem 27).

(2) Reduction. *At a critical point $y_2 = y_3 = c$ and $y_1 = \tau^3/c^3$, where $c^6 + \tau^2 c^4 - \tau^5 = 0$. As a cubic in c^2 this has discriminant $\tau^{10}(4\tau - 27)$, so there are 6 distinct critical points for $T \in (0, 1)$.*

(3) One fibre. *The Newton polygon of the cubic has a single edge. All six critical points lie over the leximin fibre $(1/40, 1/24, 1/24)$, with $c = \zeta T^{1/24}(1 + o(1))$ and $y_1 = \zeta^{-3} T^{1/40}(1 + o(1))$, $\zeta^6 = 1$.*

(4) Distinct values. *The critical values $w = 2y_1 + 2c$ are the roots of a sextic whose discriminant is $2^{36}\tau^{19}(4\tau - 27)^6(4\tau + 1)^2$. So they are distinct for every $T \in (0, 1)$.*

(5) Objects and morphisms *(Fano; cited, in the convention of section 13.7). The six are objects over one fibre, with distinct m_0, so every off-diagonal Hom vanishes and the category they generate is $\bigoplus_{c=1}^{6} \mathrm{Cl}(Q_c)$ over Λ, with no higher products. Generation is conditional.*

Proof. (1) By proposition 109(2), since $f_1/4 = 3/80 > g_1/2 = 1/40$. (2) The equations $y_j \partial_j W = 0$ give $y_2 = y_3 = T^{3/20}/P = c$ and $y_1 - \tau/y_1 = c$; eliminating gives the sextic, and the discriminant follows. (3) The lower hull of $(3, 0), (2, 2), (0, 5)$ in the plane of (power of c^2, power of τ) is the single segment from $(0, 5)$ to $(3, 0)$, of slope $-5/3$, so $c^2 \sim \tau^{5/3}$ and $c \sim \tau^{5/6} = T^{1/24}$. Then $y_1 = \tau^3/c^3$. (4) A resultant. (5) Proposition 31(2)–(3) and proposition 32, with (4). TF6 checks (2)–(4) symbolically. □

Example 112 (C: weak Fano, coupled in three steps)**.** Take $f = (3/10, 7/10)$ and $g = (1/10, 3/20, 1/4, 1/2)$. The facets are row 1 and cells 21, 22, 23, so seven facets, and the chamber is weak Fano with three curves of $-K \cdot C = 0$. Here $W = \sum_{j \le 3}(y_j + T^{g_j}/y_j) + T^{3/10}/P$. Columns 1 and 2 bind in turn, at $h_1 = 1/20$ and $h_2 = 3/40$. Row 1 then spreads $7/40$ over cells 13 and 14, at $h_3 = 7/80$. The last cell, 23, is fixed at $13/80$ without removing a dimension. The drops are $(1, 1, 1, 0)$. The critical equations reduce to one resultant of degree 10 in $c = T^{3/10}/P$, even in c.

Its Newton polygon in c^2 has an edge of slope $7/2$ with four roots and one of slope 1 with one root. So 8 critical points lie over the leximin fibre $(1/20, 3/40, 7/80)$, with leading holonomies of the form $(\pm1, \pm1, \pm1)$ or $(\pm1, \pm1, \pm i)$, and 2 lie over $(1/40, 1/40, 9/40)$, with holonomies $\pm(i, i, i)$. On that second fibre the leading set is $\{11, 12, 14, 23\}$, with leading coefficients $\rho = (i, i, i, -i)$ up to conjugation. (H1) holds and (H2) fails, yet $\operatorname{rank} M_c = 3 = \dim \operatorname{span}$: two more instances of the equality in theorem 38 without its sufficient condition. TF7 and TF9 check this.

Remark 113 (what changes in a weak Fano chamber)**.** For a semi-Fano toric manifold, that is, one with $-K$ nef, the potential of a fibre is $W^{\mathrm{LF}} = \sum_k (1 + \delta_k)\, T^{-\lambda_k} y^{v_k}$, where $\delta_k = \sum_{\alpha \neq 0} n_{\beta_k + \alpha} T^{\omega(\alpha)}$ runs over the classes α of rational curves with $-K \cdot \alpha = 0$ [35, Proposition 2.15]. The monomials are the same and only the coefficients change, by factors $1 + O(T^{\varepsilon})$. Three consequences, and one thing withheld:

(i) The valuations of the terms are unchanged, so the levels h_j, the leximin point and the leading coefficients are unchanged.

(ii) Theorem 27 needs real coefficients with uniform bounds. If the δ_k converge for small T, are real, and satisfy $0 < c \leq 1 + \delta_k(e^{-L}) \leq C$ for large L, then it applies to W^{LF} as it stands: the corrected positive critical point is selected at the leximin point, with error $O(1/L)$. If the δ_k are only formal Novikov series, the conclusion holds as a statement about valuations, by (i), and not about a real critical point.

(iii) The leading-order argument of proposition 116 uses only valuations and leading coefficients, so it applies to W^{LF} verbatim (argued, not checked).

(iv) Withheld: the exact critical points of W^{LF}, their number over each fibre, and the Floer reading of the ten objects. We do not compute the δ_k.

For each curve with $-K \cdot C = 0$ four statuses must be kept apart: zero anticanonical degree (computed here, exactly); an effective class α through it that contributes to some δ_k (not determined); a nonzero invariant $n_{\beta_k + \alpha}$ (not computed); the correction evaluated (not done).

14.4 One scale per removed dimension

Theorem 33 says that the reduction of Q/W mod T has rank r_1. The examples say more: the other directions of Q/W have exact scales, one for each dimension a later level removes. To state this without ordering eigenvalues, use determinantal divisors.

Definition 114 (scale valuations)**.** For a matrix M over Λ_0, let d_k be the least valuation of a $k \times k$ minor, and $e_k = d_k - d_{k-1}$, with $d_0 = 0$. The e_k are the valuations of the invariant factors of M. They do not change under $M \mapsto AMB$ with $A, B \in \mathrm{GL}_n(\Lambda_0)$, in particular under a change of integral basis of $H^1(L; \mathbb{Z})$.

For the positive object, Q/W is real symmetric and positive definite, and its scale valuations are the valuations of its eigenvalues. The sum of the $k \times k$ principal minors is the k-th elementary symmetric function of the eigenvalues; all terms are positive, so its valuation is that of the product of the k largest eigenvalues. And for a positive definite matrix $|\det M_{IJ}|^2 \leq \det M_{II} \det M_{JJ}$, so the least valuation over all $k \times k$ minors is attained on a principal one.

Conjecture 115 (one scale per removed dimension)**.** *For nondegenerate margins, the scale valuations of Q/W at the positive object are 0 with multiplicity r_1, and $h_j - h_1$ with multiplicity r_j for each later level j. A level that removes no dimension contributes nothing.*



Figure 31: Every critical point of the three potentials, grouped by fibre, with its leading holonomies drawn as dials (the arrow is the phase of y_j). A and B have one fibre each. C has two: eight points over the leximin fibre and two over $(1/40, 1/40, 9/40)$. In each case the number of critical points equals the normalised volume of the Newton polytope, which equals χ, and that equality certifies the list complete. For C these are the critical points of the uncorrected W (remark 113).

Proposition 116 (the conjecture in the three examples)**.** *At leading order, the scale valuations of Q/W are:*

(A) $(0,\ 1/40,\ 3/40)$ *for all* 8 *objects;*

(B) $(0,\ 1/60,\ 1/60)$ *for all* 6 *objects: the second level removes two dimensions and gives a double scale;*

(C) $(0,\ 1/40,\ 3/80)$ *for the* 8 *objects over the leximin fibre, with no scale from the fourth level; and* $(0,0,0)$ *for the two over the second fibre.*

So Conjecture 115 holds in A, B and C.

Proof. In every case $Q = \sum_k (\text{term } k)\, v_k v_k^\top$, and at a critical point $y_j - T^{g_j}/y_j = c$, where c is the value of the P-term. (A) In the unimodular basis $(\log y_1, \log y_3, \log z)$, $Q = \operatorname{diag}(W_1, W_3, W_z)$, with each factor's value $\pm 2T^h$ and $W \sim \pm 2T^{1/40}$. (B) $Q = \begin{pmatrix} 2y_1 & c & c \\ c & 2c & c \\ c & c & 2c \end{pmatrix}$ and $W = 2y_1 + 2c$, with $y_1 \sim T^{1/40}$ and $c \sim T^{1/24}$ (proposition 111). So $d_1 = 0$. The 2×2 minors have least valuation $1/40 + 1/24$, attained by $4y_1 c - c^2$, so $d_2 = 1/24 - 1/40 = 1/60$. Next, $\det Q = 6y_1 c^2 - 2c^3 \sim T^{1/40+1/12}$, so $d_3 = 1/30$, and $e = (0, 1/60, 1/60)$. (C) $Q = D + cJ$, with J the all-ones matrix and $D = \operatorname{diag}(2y_j - c)$. On the leximin fibre $y_1 \sim \pm T^{1/20}$, $y_2 \sim \pm T^{3/40}$ and $c \sim y_3 \sim T^{7/80}$, so $D \approx \operatorname{diag}(2y_1, 2y_2, y_3)$ and $W \approx 2y_1$. The 2×2 minors have least valuation $1/20 + 3/40$, so $d_2 = 1/40$. Next, $\det Q = D_1 D_2 D_3 + c\,(D_1 D_2 + D_1 D_3 + D_2 D_3) \approx 2y_3 D_1 D_2 \sim T^{17/80}$, with no cancellation, so $d_3 = 17/80 - 3/20 = 1/16$, and $e = (0, 1/40, 3/80)$. The second fibre has rank 3 by example 112. TF9 checks every case numerically: the eigenvalue slopes at the positive object, and the minors on every object over the leximin fibre. These finite-T checks corroborate the limiting valuations; the proof is the leading-order computation above. □

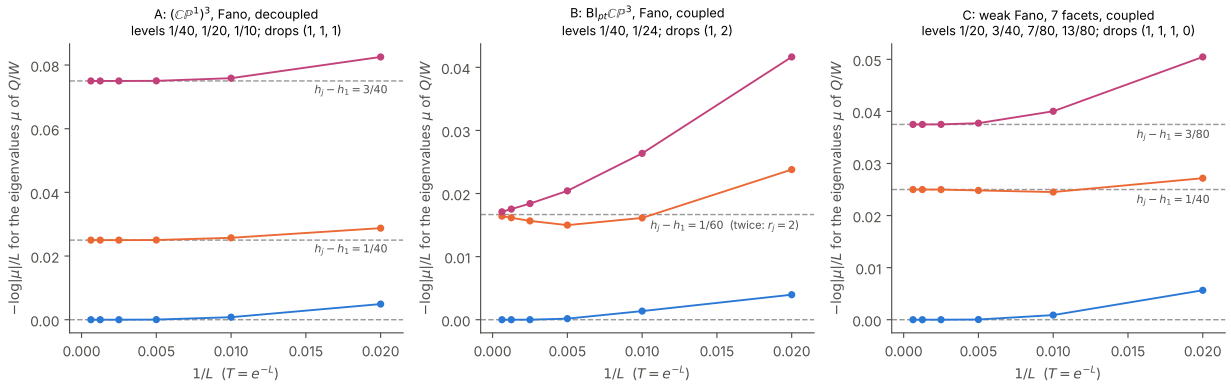


Figure 32: The eigenvalues of Q/W at the positive object, as $-\log|\mu|/L$ against $1/L$ for $L = 50, \ldots, 1600$. The finite-T offset is linear in $1/L$, so the curves run straight into their limits: 0 once, then $h_j - h_1$ for each removed dimension. In B the second level removes two dimensions and two eigenvalues share the limit $1/60$. In C the fourth level, $13/80$, removes none and leaves no trace.

In 2×3 the evidence was single scales: $(3\alpha - 1)/4$ on F_1, $1/200$ after one count, and $1/20$ in the column-binding example (section 13). The cube proves the conjecture for every box chamber, because the factors decouple. B is the first coupled case with a multiplicity, and C the first coupled case with three levels and a level that removes nothing. A proof in general would need the successive balances of theorem 27 to appear, one order at a time, in the minors of Q. That is the step we have not found.

14.5 Phases of the objects: Bloch spheres and holonomy groups

Each object in fig. 31 is drawn with three dials. Two pieces of information are attached to it. One is the torus fibre it lives on, whose position u in the polytope is a coupling, that is, a probability law on the cells of the table. The other is the holonomy of its flat connection, a point θ of a three-dimensional torus, which the dials show. Probabilities and phases appearing together suggest the picture physicists use for a two-level system, the Bloch sphere, and this subsection asks how far that picture applies.

Recall the Bloch sphere. A two-outcome law $(1-p, p)$ together with a phase θ defines the unit vector $\psi = \sqrt{1-p}\,|0\rangle + \sqrt{p}\,e^{i\theta}|1\rangle$ in $\mathbb{C}^2$, up to an overall phase. Such vectors are the points of $\mathbb{CP}^1$, which is a round sphere. The height on the sphere is $1-2p$, so latitude records the probability, and the longitude is θ. Explicitly, the point is

$$\big(2\sqrt{p(1-p)}\cos\theta,\ 2\sqrt{p(1-p)}\sin\theta,\ 1-2p\big).$$

The round metric splits accordingly. For $z = \sqrt{p}\,e^{i\theta}$ with any number of components, the Fubini–Study line element is

$$ds^2_{\mathrm{FS}} = \tfrac{1}{4}\sum_j \frac{dp_j^2}{p_j} + \sum_j p_j\,d\theta_j^2 - \Big(\sum_j p_j\,d\theta_j\Big)^2.$$

The first term is a quarter of the Fisher–Rao metric, and the rest measures relative phase. To see this, write $dz_j = (dp_j/2\sqrt{p_j} + i\sqrt{p_j}\,d\theta_j)e^{i\theta_j}$, use $\sum dp_j = 0$, and expand $ds^2 = \langle dz, dz\rangle - |\langle z, dz\rangle|^2$. Two remarks fix the setting. A point with three independent phases needs a three-dimensional torus, so the spaces here are three-dimensional, such as $(\mathbb{CP}^1)^3$ or $\mathrm{Bl}_{pt}\mathbb{CP}^3$, and not $\mathbb{CP}^2$, which carries only two relative phases. And the three dials are already relative phases, so turning all three together is not a change of overall phase. Indeed $\pm(i,i,i)$ are two distinct objects of example C.

Proposition 117 (example A is three Bloch spheres)**.** *In example A, $X = (\mathbb{CP}^1)^3$, and its three factors correspond to columns 1, 3 and 4 of the 2×4 table, with coordinates y_1, y_3 and $z = T^{f_1}/P$. On each factor the height is the conditional probability $P(\text{row } 2 \mid \text{column})$ and the longitude is the holonomy. The eight objects are the eight product states $|\pm\pm\pm\rangle$, where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$, and so they form an orthonormal basis of $(\mathbb{C}^2)^{\otimes 3}$. The positive object is $|{+}{+}{+}\rangle$.*

Proof. Column 2 never binds, so W is a sum of three potentials $y + T^a/y$, one for each remaining column (example 110). The two monomials of each are the two cells of that column, so the moment coordinate of a factor is the split of the column between the rows. The critical points of $y + T^a/y$ are $y = \pm T^{a/2}$. Both lie at the midpoint of the moment interval, which is the equator ($p = 1/2$), with longitude 0 or π. Two distinct sign patterns differ in some factor, where $\langle + | - \rangle = 0$. Each factor's circle is identified with its dual canonically up to a sign, and a sign does not move ± 1, so no choice enters. □

Examples B and C are not products, and the picture changes in two ways. First, the phases still have structure. Over each fibre, the holonomies of the objects form a coset of a finite subgroup of the torus, and over the leximin fibre they form a subgroup, whose identity is the positive object. In B the six holonomies are the powers of $(-1, \zeta, \zeta)$ with $\zeta = e^{i\pi/3}$, a cyclic group of order 6. In C the eight holonomies over the leximin fibre form $\mathbb{Z}/4 \times \mathbb{Z}/2$, generated by $(1, -1, i)$ and $(-1, -1, 1)$, and the two over the other fibre are $\pm(i, i, i)$, a coset that does not contain 1 (fig. 33). We found this in the three examples and have not proved it in general.

Second, the holonomy is a point of the torus dual to the fibre, while a point of X has its angles on the fibre torus itself. Turning objects into states therefore needs an identification of a lattice

with its dual. A product of spheres supplies one, factor by factor, and B and C do not. Different identifications place B's and C's groups at genuinely different points, so the overlaps of the resulting states are not invariants of the objects, and we do not compute them. Mixed states over the objects, $\sum_a q_a |z_a\rangle\langle z_a|$, would need weights q_a, and nothing here supplies them. In every case, all objects over one fibre carry the same coupling, so they agree on every expected utility, decision and value of information (proposition 94(1)).

Probabilistic pseudo knots. Diamantis and Kauffman attach a probability p_i to each undetermined crossing (pre-crossing) of a pseudo knot diagram, the probability that it resolves as a positive crossing. Assuming independence, they obtain a law on the 2^n complete resolutions and study invariants that are expectations over it, such as a probabilistic writhe and a probabilistic Kauffman bracket [24]. In the language above, $(p_1, \dots, p_n)$ is a point of the cube $[0,1]^n$, the moment polytope of $(\mathbb{CP}^1)^n$. Their law on resolutions is the outcome law of the product state with these heights, measured in the basis $|0\rangle, |1\rangle$ of each factor. The case $p_i = \frac{1}{2}$, the unweighted resolution set of pseudo knots introduced by Hanaki [21], is the equator. For $n = 3$ it is example A's positive object $|{+}{+}{+}\rangle$, whose outcome law is uniform on the eight resolutions.

Two consequences follow. Their invariants are expectations, so they do not see longitudes, for the same reason that decisions do not; a phase-sensitive version, which they list as future work, would need the Bloch longitude. And the same cube $\{0,1\}^n$ indexes the complete resolutions, the vertices of the box in example A, and the cube of smoothings on which Khovanov homology categorifies the Jones polynomial, graded by the number of 1-smoothings [22, 23]. A graded category built over the resolution cube, with the crossing probabilities as heights and the holonomies as longitudes, would join these. We do not construct it here.

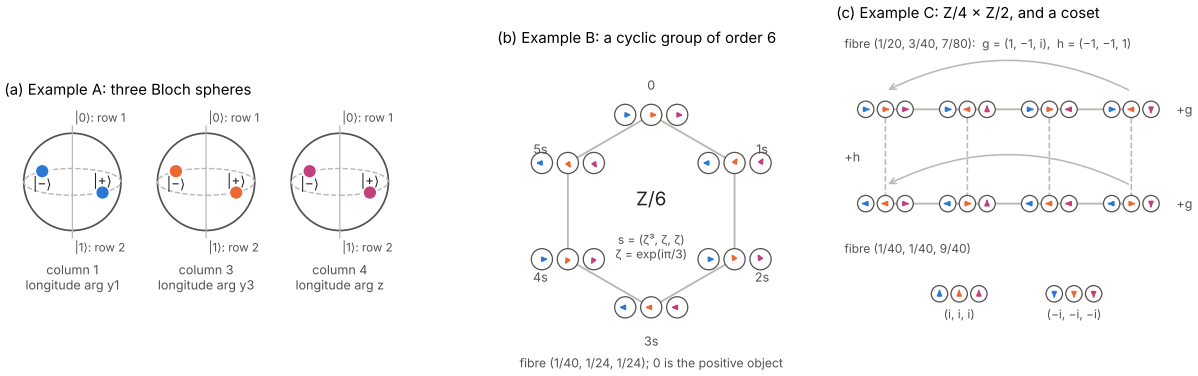


Figure 33: Phases of the objects of fig. 31. (a) Example A is $(\mathbb{CP}^1)^3$, with one Bloch sphere for each binding column. The height is $P(\text{row } 2 \mid \text{column})$, which is $1/2$ at the leximin fibre, and the longitude is the holonomy, so the eight objects are the product basis $|\pm\pm\pm\rangle$. (b) B's six holonomies form a cyclic group of order 6 (dials as in fig. 31; each edge adds the generator). (c) C's eight holonomies over the leximin fibre form $\mathbb{Z}/4 \times \mathbb{Z}/2$, and the two over the other fibre are $\pm(i, i, i)$, which differ by turning all three dials through π.

15 Outlook

Eight questions are left open.

- *Novikov weights and environments.* Section 7 makes one direction precise: soft evidence of size n on the scale N shifts λ by $-(n/N)\rho$. Whether the environment orders s^{r_e} of the layers paper act in the same way, and how hard evidence and forgetting act, is open.

- *The other walls.* Decision, contextuality and value-of-information walls (section 7), and the finite-n sharpness of the transition at an outcome wall, measured in the Fisher metric.

- *Beyond Fano.* For non-Fano toric manifolds, W acquires corrections [46]. Theorem 3 is about the leading potential only. In the weak Fano chambers of 2×4 the corrections only rescale coefficients (remark 113); computing the δ_k, and so the corrected objects of example C, is open.

- *Composition by selection.* By proposition 47, higher structure with invariant content needs morphism spaces that are not contractible; the natural candidate is the directed category of the critical points of W, whose Stokes data section 13.8 began to compute [74]. On the probabilistic side: continuity of selection across facet-set changes, a classification of the associativity locus for binary chains, and K_5.

- *One scale per removed dimension.* Conjecture 115 in general; then 3×3, where the fibre is four-dimensional, as a test of it and of the rank equality in theorem 38.

- *Signed branches and Stokes data.* The non-positive critical points have their own thimbles, which are not real and carry no probability law. Proposition 35 locates the Stokes walls on F_1; the Picard–Lefschetz decomposition itself, and the general window, are not computed. We do not know whether the signed fibre of example 6(a) has a probabilistic shadow. Section 13.8 crosses two leximin walls from the column-binding example; what the collision at the level-two wall does to the category, and an A_∞ functor along a path that avoids it in complexified parameters, are open.

- *The integral category.* Theorem 33 sees the outcome wall in the reduction mod T of one endomorphism algebra. Whether the full A_∞ category over Λ_0 changes there, and split generation in the non-monotone case [77], are open.

- *Wall-crossing.* The layers paper resolves degenerate directions by blow-ups off the torus orbits. Whether these correspond to wall-crossing in torus fibrations is a separate question, which needs geometric data that the finite setting does not supply.

A The ledger

`LEDGER.yaml` at the root of the repository is the single source for this table, which `kahler/render_ledger.py` generates. Status tags: [P] proved here, [U] used (classical), [C] cited and not proved, [E] exact computation, [N] numerical only, [D] a definition, [O] open. The category level is never filled by assumption. It records mathematical status, not novelty: [P] there marks a proved standard construction and says so, as for theorem 66. For each experiment the ledger also records the thresholds $s_{\text{prob}}, s_{\text{support}}, s_{\text{trop}}, s_{\text{crit}}, s_{\text{Stokes}}, s_{\text{context}}$, each as a number, "never" or "undefined". For F_1 along $\rho = e_1$ from $\alpha = 1/5$ they are: $s_{\text{trop}} = 1/5$; s_{crit} never; s_{support} never; s_{prob}, s_{Stokes} and s_{context} undefined until a dominance notion, a phase of $\hbar$, and a complex with an obstruction class are chosen.

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
soft-translation: Soft evidence along a translation, rho_k = ⟨v_k, t⟩ (an exponential tilt)	[D] reweighting by a log-linear factor: a reparametrisation of the same model; [P] Q translated by -s t; same Kahler class	[P] u* → u* - s t; h*, leading set and candidates unchanged	[C] the same fibres, translated; category [O] not addressed	none: rho is orthogonal to B - B (Corollary 12(2)); CHECKS DW7
soft-constant: Soft evidence along the constant, rho = 1 (normalisation)	[D] a common factor: removed by normalisation; [P] every facet moves out by s	[P] h* → h* + s; nothing else changes	[C] no change in which fibres carry Floer cohomology; category [O] not addressed	none: sum_k a_k = 0 for every affine dependence a; CHECKS DW7
soft-circuit: Soft evidence with a circuit component (lam_eff = lam - (n/N) rho)	[E] independence: a larger row or column survey (L1 = n1, L2 = n2); 2x3 needs n/2 more column observations to reach its wall; [P] moves the Kahler class; viability walls are the walls of the Kahler cone (F_1: alpha/L in 0, 1)	[P] outcome walls = normal fan of the balance polytope B; leading set, candidate dimension and face F jump there; h* concave piecewise linear	[C] F_1: two fibres with HF ≠ 0 below alpha/L = 1/3, one above (Cho-Oh); independence: one fibre on both sides; category [C] F_1: over Lambda the category of the four objects is unchanged up to A-infinity equivalence across the wall (on cited m_2); the positive object's Clifford form mod T drops rank 2 → 1; Stokes walls crowd the wall (section 12)	one circuit a: s* = -c(lam_0) / ⟨a, rho⟩; least s over rate range ≤ 1 is |c| / (||a||_1 / 2). F_1: a = (2,-1,2,-3), c = 3 alpha - L; independence: a = (c 1_r, -r 1_c)/gcd, c(lam) = (c L1 - r L2)/gcd; CHECKS DW2, DW4, DW5, DW6, DW7, CT2, CT6, MO1, MO6, MO7, MO8, MO9
margins: Observed counts (margins b)	[U] the MLE has the observed margins (Birch); independence: rows x cols / n^2; [U] the fibre over b; l_k(b) is the k-th margin (independence: exact)	[P] u* does not depend on b while lam is fixed	[D] compare L(b) with the Floer-selected L(u*), in Fisher-Rao distance within the model; category [O] not addressed	support: min_k l_k(b) = 0; centrality g(b)/h* = smallest margin / (n / max(r, c)) for independence with joint sampling; CHECKS CT4, CT5
margins-on-the-fibre: Margins (f, g) read on the fibre, the coupling polytope Cpl(f, g)	[E] the fibre's lattice points are the tables with these margins; Tier A is Fisher's noncentral hypergeometric family, odds ratios 1 at f(x)g; [U] Cpl(f, g) is Delzant; class = product coupling; 2x3: CP^2, CP^1 x CP^1, F_1, Bl_2 CP^2, dP_6 (coupling-polytopes paper)	[P] h* = min(f_i/m, g_j/n); leading cells = the binding row or column; candidate dimension (n-2)(m-1) or (n-1)(m-2); the selected coupling is the facet-leximin one (selection theorem), error O(1/log(1/T))	[C] example: three fibres with HF ≠ 0; one count moved: three become two (valuations by exact resultants); category [C] one count: objects 3 + 1 + 1 → 4 + 1 over the fibres; the positive object's Clifford form mod T drops rank 2 → 1 (section 12). Column-binding example: 5 critical points, 2 fibres, 5 objects (exact, all T); Hom = 0 off the diagonal, End = Cl(Q_c), all reduce to rank 1 (section 12.7; FOOO Lambda_0 convention)	the argmin of f_i/m, g_j/n changes. Example: f1/3 = 2/15 vs g3/2 = 27/200 (1/600 apart): one count crosses; the blow-up wall f1 = g2 is three counts away; CHECKS CF1, CF2, CF3, CF4, CF5, CF6, CF7, CF8, LX1, LX4, LX5, LX6, LX9, MO2, MO6, CB1, CB3, CB4, CB7, CB8

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
deeper-selection: Margins on a 2 x 4 fibre: a three-dimensional coupling polytope, selection in several levels	[E] the leximin coupling level by level: binding columns, then a residual row; a level can fix one dimension, two, or none (examples A, B, C); [E] 148 chambers, all smooth: 14 Fano (CP^3, Bl_pt CP^3, (CP^1)^3) and 134 weak Fano, certified by -K.C on every invariant curve (toric Kleiman); coupled three-step selection only in weak Fano chambers	[E] A: 8 critical points, 1 fibre; B: 6, 1 fibre, distinct values for all T; C: 10 over 2 fibres (8 + 2). Clifford scales 0 and h_j - h_1 with multiplicity r_j at leading order in A, B, C; in general a conjecture	[C] A, B Fano: Cho-Oh objects, Hom = 0 off the diagonal by m_0 (B for every T; A except at one T*). C weak Fano: W gets (1 + delta_k) corrections (Chan-Lau-Leung-Tseng); its Floer reading is withheld; category [C] B: the category of its six objects is a sum of Clifford algebras over Lambda (on cited theorems); weak Fano chambers not addressed	chamber: the sign vector of f1 - g(J); Fano: min over invariant curves of -K.C $\rangle$ 0; box: some g_j $\rangle$ max(f1, f2); CHECKS TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, CC1, CC2, CC3, CC4
path-across-walls: A path of margins inside one Kahler chamber (Stage IV): f = (2/5, 3/5), g = (7/10 - gamma, 3/10, gamma), gamma in [3/20, 11/40]	[E] the leximin coupling changes at gamma = 1/5 (level 2: cells 11, 12, 22 tie; the column-binding example lies there) and at 4/15 (level 1: row 1 starts to bind); [E] one Kahler chamber (the pentagon) along the whole path: no blow-up wall	[E] objects over fibres 4 + 1 below 1/5, 2 + 3 at 1/5, 4 + 1 between, 3 + 1 + 1 above 4/15 (Newton polygon); a collision of two real critical points at gamma_c(T) $\to$ 1/5 (numerical: one discriminant zero at each L tested), none at 4/15; the positive object's Clifford scale h2 - h1 bends at 1/5 and closes at 4/15	[C] away from gamma_c(T) the critical values are distinct, so Hom = 0 off the diagonal and End = Cl(Q_c) at every point (cited theorems); at gamma_c a critical point is degenerate and the cited theorems do not apply; category [O] pointwise the same category over Lambda on both sides of the level-1 wall; no A-infinity functor along the path is constructed; the effect of the collision is open	level 2: f1 - g3/2 = g2; level 1: g3/2 = f1/3; collision: a sign change of the discriminant of the quintic; CHECKS PP1, PP2, PP3, PP4, PP5, PP6, PP7, PP8, PP9

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
composition: Composing couplings over a shared marginal: Markov gluing (g1 D^-1 g2) against selection (the facet-leximin point of the multi-coupling fibre, then the end marginal)	[E] Markov gluing is associative and unital, and is the maximum-entropy point of the fibre; selection is unital, commutes with transposition, and is not associative (26 of 30 random chains); K_4 example: 11 distinct exact composites; binary chains: selection is Frechet-midpoint gluing, uniform margins compose correlations by a signed minimum (associative), and among 9^4 margin tuples only the uniform one is associative everywhere; full-dimensional cells of associativity certified exactly for two non-uniform tuples; every selection behind the K_4 examples certified exactly; [P] the Markov composite is the Birch point of the chain model's fibre; the fibres are convex polytopes, so fillers K_n → Pi(p0, pn) exist and carry no homotopy type	[N] at finite T the associator of the binary K_3 example is -41/960 at every L tested: no common leading term (valuation 0); composites converge faster than 1/L	[C] binary chains: the selected composite is the marginal of the unique Floer-nontrivial torus fibre of the reduced space prod CP^1 (Cho-Oh), equivalently the unique fibre with HF(R, L) $\neq 0$ for the real Lagrangian (Alston-Amorim, characteristic 2); Markov takes the Birch point; K_4 fibres beyond 3-way are not Fano (some orbifold), so only FOOO non-displaceability applies to the Delzant ones; category [O] no-go: any A-infinity structure on chains of coupling polytopes is quasi-equivalent to the trivial one, so the associator carries no invariant there; the filtered-m_3 conjecture in the form 'valuation = h_j - h_1' is refuted; a non-contractible candidate (the directed category of critical points) is not addressed	associator = difference of the two bracketed composites; Markov: zero identically; CHECKS AS1, AS2, AS3, AS4, AS5, AS6, AS7, SB1, SB2, SB3, SB4, SB5, SB6, SB7, SX1, SX2, SX3, SF1, SF2, SF3, SF4, SF5

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
probability-category: The dagger symmetric monoidal category of couplings under Markov gluing, and the symmetry profile of selection; what the data determine about a composite, and how little changes a composed decision	[E] Markov gluing is a dagger, monoidal and natural for end coarse-graining; selection is closed, unital, reversal- and relabelling-equivariant and commutes with copied factors, but fails the tensor and end coarse-graining squares (exact counterexamples); both fail middle coarse-graining; binary ends: the data determine [sum L_b, sum U_b], selection is its centre (least worst-case error); in the example Markov flips at 2.6% of records edited or with an unrecorded middle distinction, selection at 12.9%, the data force a warning at 16.7%; [P] the copied-factor identity: the fibre is a product of scaled copies and leximin factors over products; the tensor square for Markov: maximum entropy is additive over independent systems	[–] no finite-T statement	[–] no Floer statement; category [U] the underlying dagger symmetric monoidal category is standard (Markov kernels with Bayesian inversion); selection is not a second composition on it, and no higher structure is claimed	flip edits are exact roots of affine pieces, with the decision checked on both sides; CHECKS NC1, NC2, NC3, NC4, NC5, NC6, NC7, NC8, NC9

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
marginal-sections: Composition rules as sections of the projection from joint laws to marginal data; the 4-cycle (CHSH) cover; Simpson reversals	[E] 4-cycle: D has 16 vertices and 24 facets (16 local, 8 Clauser-Horne), so D = L cut by CH; 2721 faces with fibre dimension |V| - 1 - dim G: interior 7, positivity facets 4, CH facets and their faces 0; symmetric path in D iff S $\leq$ 2, fixed fibre a point so all equivariant sections agree; Simpson reversals by selection and by middle coarse-graining, exact; every n-cycle: noncontextuality facets are simplices with 2n vertices (all facets exact for n $\leq$ 5, listed facets for n $\leq$ 8), so the contextuality boundary is rigid; [P] the projection theorem (fibre dimension over faces; Lipschitz fibres); strictly concave sections are continuous; the entropy cocycle gives system independence only with c $\rangle$ 0 (with c $\langle$ 0 no section); maximum entropy is interior, equivariant, system independent and inner-face consistent; inner-face consistency implies associativity	[N] generic paths: the disagreement of maximum entropy and leximin vanishes at a CH facet and stays positive at a positivity facet and a 6-face; identified interval [lam/2, (1 - lam)/2] on the symmetric path; the power family phi_q: tensor-square defect 0 only at Shannon (q = 1), rising toward the leximin value	[–] no Floer statement; category [P] (standard categorical construction) Theorem sec2cat: an inner-face consistent section family gives an associative composition and a category; the maximum-entropy section gives Markov gluing, Cpl (proved); leximin fails inner-face consistency, with the selection associator and Theorem k4 as witness (proved); uniqueness of maximum entropy among associative section families is open (Problem assocsec); no new category is claimed	exact facet enumeration of D and exact face lattice with affine ranks; CHECKS MS1, MS2, MS3, MS4, MS5, MS6, MS7, MS8, MS9, MS10, MS11

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
poisoning: Undetectable poisoning of marginal data: the edits that flip a decision, where they can be placed, and removal over time	[E] circuits of V: 188 for n = 4 (supports 4, 6, 8, 9), 481 for n = 5 (supports 4 to 11), 5n of support 4 for n = 4..8; undetectable placements on all 2721 faces of the 4-cycle's D (vertex 13, CH facet 252, positivity facet 287, interior 376), monotone under inclusion; [P] two thresholds c_some $\leq$ c_s $\leq$ c_all; c_some is one linear program and c_all a minimum over dual vertices; a cheapest flip is a conformal sum of at most min(a, dim V) circuits (Rockafellar's elementary vectors); placement is monotone in the hidden support; dim G + fibre dim = |S| - 1; removal maps form a functor on the opposite of the poset of information states; forgetting contexts widens the interval; a one-batch flip lasts while k / (N_t + k) $\geq$ q*	[N] one support-4 circuit is a cheapest flip in 30/30, 30/30, 28/30, 28/30, 24/29 random cases for n = 4..8 (gap 0.01); budgeted placement at the Study 18 faces; a one-batch poisoning stream: every rule flipped 1 step, maximum entropy 6 steps, only the poisoned batch's removal reverts, first-order estimate within 6%	[–] no Floer statement; category [D] information states with removal maps: a presheaf on the poset of (batches, contexts); nothing beyond functoriality and commuting removal squares is claimed	dual-vertex enumeration of pi^T y + y0 1 $\geq$ w; exact tangent counts with rational facet normals and integer circuits; CHECKS PF1, PF2, PF3, PF4, PF5, PF6, PF7, PF8, PF9
poisoning-geometry: The geometry of poisoning: certified radii, fibre deformation, crossing the contextuality boundary, continuity of selection	[E] Lipschitz constants of L and U for P(A0 = A1): 1 and 1 (6 dual vertices each, 376 unit circuits); for P(A0 = 0, A1 = 0): 1/2 and 1/2; [P] certified radii c_some $\geq$ (L - tau)/Lam_L, c_all $\geq$ (U - tau)/Lam_U; width Lipschitz with Lam_L + Lam_U while fibre dimension jumps; closed-form distance to the contextuality boundary; continuous rules with unique optima are continuous (Berge with Lipschitz fibres)	[N] the bound is attained for some rule in 45 of 75 cases, never off by more than 2; staged example: route on the facet stays rigid (every rule flips at share 0.0924), route off it opens to dimension 7 (every rule at 0.4977); PR overreach contextual at once from a rigid point, and at the predicted share from inside	[–] no Floer statement; category [–] no categorical statement	dual vertices enumerated exactly; circuits exact; facet list exact (MS3); CHECKS PG1, PG2, PG3, PG4, PG5, PG6

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
bayes-inversion: Bayesian inversion of a coupling, against composition rules (sections of the chain cover)	[E] Markov gluing and selection commute with inversion on 40 random rational chains; a non-equivariant vertex rule commutes on 13/40; null middle states never reach a composite; a structural zero with full-support margins keeps unique inverses and a positive-width interval; [–] no toric statement	[N] order drift as Bayesian inconsistency: Study 18 plain 0.315 / 0.281 / 0.174, consistency-trained 0.011 / 0.007 / 0.010	[–] no Floer statement; category [P] (standard categorical construction) the dagger of Cpl is Bayesian inversion (Cho-Jacobs); composition by a reversal-equivariant section commutes with it; on binary-end chains every linear rule with a unique optimum does; inversion is a symmetry of the marginal projection, not a section	exact rational arithmetic on every composite; CHECKS IS1, IS2, IS3, IS4, IS5, IS6
value-of-information: Decisions over a fibre: utility image, value of identification (minimax regret), instance information, dilation, Das's commuting square	[E] act / abstain closed forms R = 2 min((tau - L)+, (U - tau)+) and randomisation gap R/2; Good's inequality as a triangle inequality; [P] utility image a polytope; identified iff R = 0; R $\leq$ width against an action optimal somewhere; R Lipschitz; more contexts never raise R; R = 0 on rigid faces; mixing: L convex, U and width concave; committed robust VOI $\geq$ 0, mixed policies via the minimax theorem	[N] staged routes: route A R = 0 while the decision flips; route B R $\rangle$ 0 on (1222, 11890) units, peak 0.258; pure-policy robust value leaves the envelope by exactly the value of randomising; dilation for some outcome 300/300, strict 9/300; Das square gap 0.084-0.274 (random), 0.220-0.459 (rigid faces)	[–] no Floer statement; category [D] commit-then-condition versus condition-then-select as a square; it fails for re-selection on marginal data	linear programs for every U_ab; exact fibre vertices for Hausdorff distances; CHECKS VG1, VG2, VG3, VG4, VG5, VG6, VG7, VG8
voi-determination: When local data determine the value of information; value-based distance between fibres	[E] exact envelope of VOI over the fibre by one LP per policy cell; four cases separate the three determinations (law, decision, VOI); [P] Theorem determination: VOI constant on J_d iff every full-dimensional cell's value functional is cover-additive on the fibre's support; sup over |u| $\leq$ 1 of the robust-value change = 2 x Hausdorff (TV) distance, attained by one action	[N] criterion = envelope collapse on 80/80 random problems; local VOI defined for a PR box (0.45); dual utility attains 2 dH to 1e-7 on 6 pairs	[–] no Floer statement; category [D] VOI glues from local tables exactly when the value functionals are sums over contexts (the decision-value counterpart of Tohme-Caterina-Gangle)	slack LPs for full-dimensional cells; exact fibre vertices for Hausdorff distances; CHECKS VD1, VD2, VD3, VD4

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
viability: Trajectories of decision viability: viability shares under any disturbance, open and closed loops, ranking radius	[E] closed loop with finite window: decisions eventually periodic, period $\leq$ 2^W (observed W + 1 for W = 3..8); [P] reachable set and viability kernel; forgetting identities; segments per regime; at most two threshold crossings per segment for L and U; ranking radius certified; closed-loop periodicity	[N] five-regime stream: on-facet poisoning keeps R = 0 and dimension 0 while q*_all falls 0.092 $\rightarrow$ 0.007; off-facet poisoning raises R to 0.057 and the Hausdorff step to 0.100; at most two switches per regime	[–] no Floer statement; category [–] no categorical statement	LPs for viability shares; dual vertices and circuits for Lipschitz constants; exact fibre vertices for Hausdorff steps; CHECKS VB1, VB2, VB3, VB4, VB5, VB6
history-holonomy: History-indexed selection: holonomy of updating rules along histories of snapshots; gluing on the context cycle	[N] three snapshots (law k/20): KL acts in both histories; Euclid (0.4786 vs 0.5125) and TV (0.45 vs 0.60) split at tau = 0.49; contextual t3: every triple glues, the four do not (contextual fraction 0.10); [P] holonomy bounds by the fibre; KL flat over growing contexts (log-linear family and Pythagorean identity); Euclidean flat in the interior (nested affine projections)	[N] TV updating curved in the interior (holonomy up to 0.26); Euclidean curved at the boundary (32/40); forgetting curves KL (Jeffrey non-commutativity)	[–] no Floer statement; category [D] path independence = factoring through the final snapshot (sigma = s o d), the selection-level sheaf condition of Tohme-Caterina-Gangle	LP feasibility for gluing; exact vertex enumeration for the TV diameter; exact active-set QP for Euclidean updating; CHECKS HY1, HY2, HY3, HY4, HY5, HY6
fisher-relevance: Fisher geometry against decision relevance: phases, Fisher-null directions, Fisher-Rao bound, volume coefficients	[N] two vertices of one fibre at Fisher-Rao distance 2.33 with equal VOI; laws at Fisher-Rao distance 0.002 across a wall with different actions; [P] phases decision-null; fibre = Fisher-null space of the data; one-sided Fisher-Rao Lipschitz bound; transitive isometries of the normal family	[E] normal-family volume coefficients in four coordinate systems and the isometry pullback (sympy)	[D] critical points over one fibre with different leading holonomies carry the same law: Floer separates what decisions cannot; category [–] no categorical statement	eigen-decomposition of the observed Fisher information; exact VOI envelope by LPs; sympy for the normal family; CHECKS FV1, FV2, FV3, FV4, FV5
bloch-phases: Phases of the Floer objects of Part IV: Bloch spheres, holonomy groups, lattice identifications	[P] the Bloch z-coordinate of each factor of A is a column-conditional row probability; all objects over a fibre carry one coupling; [P] A = (CP^1)^3: eight objects = the sigma_x product basis, canonical; Fubini-Study = Fisher/4 + phase	[N] holonomies form (Z/2)^3, Z/6, Z/2 x Z/4 over the leximin fibres and a Z/2-coset over C's second fibre	[N] Bloch placement needs a lattice identification: 37 (B) and 27 (C) distinct placements under 50 random unimodular choices; 1 for A; category [–] no categorical statement; mixed states over objects left open	exact Gram matrix for A; group closure and Smith normal form on rounded holonomies; CHECKS BL1, BL2, BL3, BL4

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
algebra-selection: Algebra of the probability operations, counting explanations, and selection that is not concave	[E] maximum entropy (1 + sqrt 3)/4 from rational data (sympy); integer explanations 1 / linear / quadratic on fibre-0 / 1 / 2 faces; [P] removable singularity of gluing and the dagger, not of kernels; leximin and Markov rational; Ehrhart degree = fibre dimension; concave-minimising rules jump only at vertex ties (Maxwell points)	[N] minimum-entropy jumps 0.82 / 0.44 / 0.69 in TV at ties within 1e-13; maximum entropy at most 0.009 per step	[–] no Floer statement; category [–] no categorical statement	sympy exact root; exact integer enumeration; exact fibre vertices; CHECKS AL1, AL2, AL3, AL4
refinement: Refinement: add a distinction (a facet normal, a toric blow-up)	[O] which table operation adds a facet is not yet identified; [U] CP^2 → F_1 is the blow-up at a point	[E] creates the circuit (2,-1,2,-3) and the wall alpha/L = 1/3; CP^2 had none	[C] F_1 gains a second fibre with HF ≠ 0 below the wall; category [O] not addressed	m - n - 1 increases by one; CHECKS DW4
structural-zero: Hard evidence: a structural zero (support restriction)	[E] 2x2 with one structural zero: quasi-independence on three cells; [E] the unit square becomes a unimodular triangle, not a face of it: CP^1 x CP^1 is replaced by CP^2	[E] the one circuit and the one wall disappear	[C] CP^2: one fibre (the Clifford torus); category [O] not addressed	face test fails; DW circuits of the triangle: none; CHECKS CT7
independent-product: Independent product of two models	[E] independence of rows and columns: Delta_(r-1) x Delta_(c-1); [U] the product manifold	[P] adds exactly one circuit, the relative scale (q1, -q2); u* = (u1*, u2*) never moves	[C] independence: exactly one fibre with HF ≠ 0 for every L1, L2, so this wall changes no Floer data; category [O] Kunneth-type statements not addressed	m - n - 1 = (m1 - n1 - 1) + (m2 - n2 - 1) + 1; CHECKS CT1, CT2, CT3
forgetting: Forgetting an environment / marginalisation	[P] the layers paper: the forgetting square, reversal walls, pooled orders; [O] the image configuration under a linear map of exponents: not yet worked out	[O] not yet worked out	[O] not addressed; category [O] not addressed	layers paper, theorem "contextual Bayes with an environment"; CHECKS EN1, EN2, EN3
coupling: Dependent coupling, convolution, intervention, Markov evolution	[O] not yet worked out; [O] convolution would be a Minkowski sum of supports	[O] not yet worked out	[O] not addressed; category [O] not addressed	-; CHECKS –

B Exact inputs

Every example can be rebuilt from these data; the checks named in each statement recompute everything else.

- *The column-binding example and its path* (sections 13.7 and 13.8): $f = (2/5, 3/5)$, $g = (7/10 - \gamma,\ 3/10,\ \gamma)$; the example is $\gamma = 1/5$, the path $\gamma \in [3/20, 11/40]$.
- *The* 2×4 *examples* (section 14): A: $f = (2/5, 3/5)$, $g = (1/20, 13/20, 1/10, 1/5)$; B: $f =$

$(3/20, 17/20)$, $g = (1/20, 1/5, 1/4, 1/2)$; C: $f = (3/10, 7/10)$, $g = (1/10, 3/20, 1/4, 1/2)$. The 148 chambers are in `kahler/chambers24.json`.

- *The binary chain of theorem 45*: $\gamma_i = \begin{pmatrix} t_i & p_i - t_i \\ p_{i+1} - t_i & 1 - p_i - p_{i+1} + t_i \end{pmatrix}$ with $(p_0, \ldots, p_4) = (2/5, 1/3, 3/7, 1/4, 5/11)$ (first entries) and $(t_1, \ldots, t_4) = (13/60, 7/30, 1/80, 3/80)$. The K_3 example of section 13.9 is $\gamma_1, \gamma_2, \gamma_3$. Example 56 and section 13.14 use γ_1, γ_2, with threshold $\tau = 1/2$ on $P(X_2 = 0 \mid X_0 = 0)$; the other inputs of proposition 58 are listed there.

- *The chain with ternary intermediates* (fig. 16), rows indexed by the first variable:

$$\gamma_1 = \begin{pmatrix} \frac{1}{150} & \frac{19}{150} & \frac{4}{15} \\ \frac{4}{25} & \frac{31}{150} & \frac{7}{30} \end{pmatrix}, \quad \gamma_2 = \begin{pmatrix} \frac{7}{50} & \frac{2}{75} \\ \frac{18}{175} & \frac{121}{525} \\ \frac{13}{70} & \frac{11}{35} \end{pmatrix},$$

$$\gamma_3 = \begin{pmatrix} \frac{3}{35} & \frac{81}{1400} & \frac{57}{200} \\ \frac{4}{35} & \frac{339}{1400} & \frac{43}{200} \end{pmatrix}, \quad \gamma_4 = \begin{pmatrix} \frac{9}{125} & \frac{23}{375} & \frac{1}{15} \\ \frac{209}{3000} & \frac{79}{750} & \frac{1}{8} \\ \frac{13}{120} & \frac{1}{6} & \frac{9}{40} \end{pmatrix}.$$

- *The cells of associativity* (proposition 50(2)): margins $(2/5, 1/3, 3/7, 1/4)$ at the point $(13/60, 7/30, 29/240)$, and margins $(3/20, 1/4, 3/4, 17/20)$ at the point $(1/40, 1/48, 51/80)$. The check SX1 prints the inequalities of both cells.

Checks

code	`engine/pce/kahlerbridge.py`, `engine/pce/datawalls.py`, `engine/pce/contingency_toric.py`, `engine/pce/couplingfibre.py`, `engine/pce/floermorph.py`, `engine/pce/colbind.py`, `engine/pce/cpl24.py`, `engine/pce/pentpath.py`, `engine/pce/selcomp.py`
checks	`engine/check_kahlerbridge.py` (KB, 8/8), `engine/check_datawalls.py` (DW, 7/7), `engine/check_contingency_toric.py` (CT, 7/7), `engine/check_couplingfibre.py` (CF, 8/8), `engine/check_leximin.py` (LX, 9/9), `engine/check_floermorph.py` (MO, 9/9), `engine/check_colbind.py` (CB, 8/8), `engine/check_cpl24.py` (TF, 9/9), `engine/check_pentpath.py` (PP, 9/9), `engine/check_selcomp.py` (AS, 7/7), `engine/check_selbin.py` (SB, 7/7), `engine/check_selcert.py` (SX, 3/3), `engine/check_selfloer.py` (SF, 5/5), `engine/check_chambers24_cert.py` (CC, 4/4), `engine/check_ledger.py` (LG, 3/3)
figures	`kahler/fig_kahler.py`, `kahler/fig_datawalls.py`, `kahler/fig_coupling.py`, `kahler/fig_leximin.py`, `kahler/fig_morph.py`, `kahler/fig_colbind.py`, `kahler/fig_cpl24.py`, `kahler/fig_pentpath.py`, `kahler/fig_selcomp.py`, `kahler/fig_selbin.py`
certificate	`kahler/chambers24.json`
ledger	`LEDGER.yaml`, rendered by `kahler/render_ledger.py`

The checks corroborate the statements numerically, or by exact rational arithmetic where marked; they do not prove them. The proofs stand on their own. The F_1 blow-up sizes used are $\alpha = 1/5$, $1/4$, $2/5$, $3/5$, $4/5$.

KB1	proposition 1(1)–(2): moment map $= \nabla\varphi$, Fisher information $= \mathrm{Hess}\,\varphi$, positive definite (30 supports)

KB2	proposition 1(4): $\psi = \sum q \log(q/c)$, $\nabla\psi = u(p)$, $\operatorname{Hess}\psi = \text{Fisher}^{-1}$ (20 supports)
KB3	proposition 1(5): $\int \det \operatorname{Hess}\varphi$ over boxes equals the area of the moment image
KB4	theorem 3(1)–(2): one positive critical point, mean-zero law, the valuations of examples 5 and 6, balanced leading terms
KB5	theorem 3(3): the segment of candidates for $\alpha = 3/5$, and its endpoints $(1/5, 1/5)$, $(3/5, 1/5)$, with leading sets $\{v_1, v_2, v_4\}$, $\{v_2, v_3, v_4\}$ and no positive balance; the barycentre alone for $\alpha = 1/5$ and for $\mathbb{CP}^2$ (exact rational points)
KB6	theorem 3(4): all four critical points of F_1 in 80-digit arithmetic; the signed critical point and its signed balance
KB7	proposition 7: Hessian $\operatorname{diag}(H, -H)$ at y_T; Laplace covariance $\hbar H^{-1}$ with relative error $O(\hbar/W_T(y_T))$ ($\mathbb{CP}^2$, F_1 with $\alpha = 3/5$)
KB8	theorem 8, example 9: moments against the uniform law on Q_h for $h < h^*$; concentration on $\operatorname{argmax} g$ at h^*; the uniform segment; the levels 3/10, 9/20; $\log T = -200, -800$
DW1	definition 10: evidence $e^{-n\rho}$ at $T = e^{-N}$ is the shift $\lambda - (n/N)\rho$
DW2	proposition 11(1)–(2): duality; optimal support = leading set at the centre of argmax; $\dim \operatorname{argmax} = n - \dim \sigma_0$ (40 random configurations)
DW3	proposition 11(3): the limit law lies in the optimal face (23 cases)
DW4	example 14(a): the circuit $(2, -1, 2, -3)$; the wall $\alpha/L = 1/3$ inside $0 < \alpha/L < 1$; the diagonal flip (exact rationals); $\mathbb{CP}^2$ one chamber
DW5	example 14(b),(c): candidates and valuations for $\alpha = 1/10, \ldots, 9/10$; the wall law from $x^4 = x + 1$ at $T = 10^{-20}, 10^{-60}$
DW6	example 14(d): the discriminant, symbolically; its terms tie only at $\alpha = 1/3$
DW7	corollary 12, example 14(e): $s^* = 1/5$ for $\rho = e_1$; no wall for e_4, $(0, -1, 0, 1)$, translations or constants; least evidence 1/10; $\mathbb{CP}^2$ never
CT1	proposition 15: $(r-1)(c-1)$ moves; one facet circuit ($r, c \in \{2, 3, 4\}$)
CT2	theorem 17(1): h^*, the leading sets and the wall $cL_1 = rL_2$ (exact rationals)
CT3	theorem 17(2)–(3): the rc critical points and their common valuation; the exact positive law; candidate dimensions
CT4	proposition 18(1): $\ell_k(b)$ = margins; MLE margins and positivity (200 tables)
CT5	proposition 18(3): the block-diagonal Fisher information; properties of d_Δ
CT6	example 19: $h^* = n/3$, the candidate segment, u^*, and $n/2$ more column observations
CT7	example 20: the triangle is not a face; no circuit
CF1–CF2	example 21: the pentagon, its facets, Delzant; 715 tables (Pick), exact
CF3	proposition 22: odds-ratio identity; exact hypergeometric mean $(12, 86/5)$; Birch at the leximin
CF4	theorem 23(1)–(3): the closed form against the facet programme; binding cells are facets; candidate dimensions
CF5	example 24: five critical points by exact resultants; positions, signs, leading sets, signed balances
CF6	theorem 23(4): positive valuation = leximin coupling (20 random margins)
CF7	fig. 5: the thimble law against uniform on Q_h; concentration at the leximin
CF8	example 25: one count; the walls' distances; three valuations become two
LX1–LX3	theorem 27: $L\lvert u_L - u^{\text{lex}}\rvert$ bounded on 30 random configurations (multiprecision Newton); the projected equation; $\mathbb{CP}^2$, F_1
LX4	corollary 29: facet-leximin = all-cell leximin (292 margins); convergence
LX5–LX9	example 30: the fibre and its two levels (exact); the $\ln 2/2$ correction; the five critical points; the thimble; the one-count levels
MO1–MO2	the objects of F_1 and of the 2×3 fibres: counts, fibres, types, holonomies
MO3–MO5	proposition 31: disjoint fibres, distinct holonomies and values; $Q = W \operatorname{Cov}_w(v)$; Clifford generators spanning M_2 (18 objects)
MO6	theorem 33: $Q/W \to \operatorname{Cov}_{q^*}(v)$; reduced ranks $2, 2, 1, 1$ on F_1; $2 \to 1$ after one count; valuations of the small eigenvalue
MO7	theorem 34: no collision, Morse along $\alpha \in [1/5, 3/5]$; explicit real isomorphisms
MO8–MO9	proposition 35: Stokes walls within $4/\log(1/T)$ of the wall, none outside, rate $1/\log(1/T)$

CB1–CB4	proposition 36: the reduction to the quintic; the Newton polygon; distinct and nondegenerate for all $T > 0$; distinct critical values on $(0, 1)$ (exact, symbolic)
CB5–CB6	the table of section 13.7: leading coefficients, offsets $-\log\lvert\rho\rvert/L$, leading values; leading Clifford forms and determinants
CB7	proposition 39: 5 objects, 2 fibres; distinct values and holonomies; $\mathrm{Cl}(Q_c) \cong M_2$
CB8	theorem 38: (H1), convergence to M_c, rank = span on 23 objects
PP1–PP2	proposition 40(1): one Kähler chamber along the path; the leximin levels and the two walls (exact)
PP3–PP4	proposition 40(2)–(3): the reduction to (1); Newton polygons at 10 rational γ (exact)
PP5	proposition 40(3): valuations, real and complex types, leading phases at $L = 200, 400$
PP6	proposition 41(1)–(2): one discriminant zero, at the level-two wall, for $L = 50, \ldots, 400$; none at the level-one wall
PP7	proposition 41(3): no coincidence of critical values away from the collision
PP8	proposition 41(4): the positive object's scale against $h_2 - h_1$ along the path
PP9	Stokes alignments at three phases and three L, within $1.5/L$ of a wall
TF1	theorem 108: 148 chambers, each with an exact interior point; all simple and smooth
TF2	proposition 107(3): integer wall relations on all 2208 edges; the Kähler degree equals the lattice length (exact)
TF3	theorem 108: the six types by toric Kleiman; the vertex test agrees; volume = χ; $(-K)^3 = 64, 56, 48, 48, 40, 32$
TF4	proposition 109: the box criterion on all chambers; the facets of the $\mathbb{CP}^3$ and $\mathrm{Bl}_{pt}\mathbb{CP}^3$ chambers
TF5	example 110: 8 critical points over one fibre; the one coincidence of critical values at $x^3 + x - 1 = 0$ (exact)
TF6	proposition 111(2)–(4): the sextic, its discriminants, one fibre, holonomies $(\zeta^{-3}, \zeta, \zeta)$ (exact)
TF7	example 112: degree-10 resultant, its Newton polygon, 8+2 critical points over two fibres; three curves with $-K \cdot C = 0$
TF8	propositions 109 and 111 and examples 110 and 112: the leximin levels and drops exactly; a tie-free census of drop patterns by chamber type
TF9	proposition 116: eigenvalue slopes at the positive object ($L = 800 \to 1600$); minors on all 22 leximin-fibre objects; rank 3 on C's second fibre
CC1–CC4	theorem 108: the certificate `kahler/chambers24.json`, verified independently (standard library, exact rationals)
AS1–AS2	proposition 43: Markov gluing associative and unital, one composite per face, maximal entropy (exact); face counts of K_3, K_4, K_5
AS3–AS4	theorem 45: the eleven composites, binary and with ternary intermediates (exact rationals)
AS5–AS6	proposition 44: unit and transposition (exact); the non-associativity census
AS7	finite T: the associator $-41/960$ at every L; convergence of the composites
SB1–SB2	theorem 48 and corollary 49: Fréchet-midpoint gluing on 60 chains; signed-minimum correlations and associativity on uniform margins (exact)
SB3–SB4	continuity: affine on a fixed type (exact); no jump across ≥ 10 facet-set changes (numerical)
SB5–SB6	proposition 50: only uniform margins associative among 9^4 tuples; the locus has positive measure (exact grids)
SB7	example 51: the one-parameter associator, its kinks and zeros (exact)
SX1	proposition 50(2): two full-dimensional cells where $A = 0$ (exact)
SX2	theorem 45 and the ternary example: exact certificates for all 57 selections
SX3	the ternary configuration: hull, Markov point and crossings by exact orientation tests
SF1–SF3	proposition 52: the fibre is $\prod \mathbb{CP}^1$ (exact traversal); Cho–Oh and Alston–Amorim conditions at the midpoints (exact)
SF4–SF5	Birch against Floer points; the K_4 fibres: Delzant, orbifold and Fano tests (exact traversal)

NC1–NC3	propositions 55 and 57: the dagger symmetric monoidal structure of Markov gluing; units, reversal, relabelling and copied factors for selection (exact)
NC4–NC6	proposition 58: the tensor, end and middle counterexamples, with all inputs (exact)
NC7	proposition 60: the identified interval against linear programs over the fibre; selection at its centre
NC8–NC9	example 62: the interval, worst-case errors, flipping edits and zero-cost changes (exact)
MS1–MS2	theorems 65 and 66: both rules are sections, equivariant; maximum entropy is the inner-face consistent Markov chain, leximin is not
MS3–MS5	propositions 72 and 73: the 24 facets of $\mathcal{D}$; the 2721 faces and their fibre dimensions; contextual data on the symmetric path (exact)
MS6–MS7	proposition 75 and the path table: symmetric path (exact rank; interval numerical); generic paths (numerical)
MS8–MS9	proposition 76: the two Simpson reversals (exact)
MS10	theorem 77: all facets of $\mathcal{D}$ for the n-cycle, $n = 3, 4, 5$; the listed facets and vertex counts for $n = 6, 7, 8$ (exact)
MS11	proposition 71: the $c < 0$ example (exact); the tensor-square defect along the power family (numerical)
LG1–LG3	appendix A: the schema; every cited check passes in the log; the rendered table matches `LEDGER.yaml`

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